

# Mixture Modeling with `MplusAutomation`

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# Contents



# Mixture Modeling with MplusAutomation

Welcome! This will be a collection of resources that will teach you how to apply mixture modeling using Mplus(?) and **MplusAutomation**(?)! These resources will serve as a comprehensive guide to understanding and applying mixture models using Mplus and its automation capabilities with **MplusAutomation**. Here, you will learn from start to finish how to apply a range of mixture modeling using Mplus with the **MplusAutomation** package.

Note: This book is a continuous work in progress. The code presented may be updated and/or expanded as research progresses. Please treat the material as a living document rather than a final product.

## Stay in touch!

- Please visit our website to learn more about the IMMERSE fellowship.
  - For all code and materials found in this Bookdown, see [here](#).
- Visit our GitHub account to access all the IMMERSE training materials.
- Follow us on BlueSky and X to stay-up-to date on our fellowship!

## Acknowledgements

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We also thank all the IMMERSE fellows who provided feedback during the development of these materials.

## **Part I**

# **Introduction**





# Chapter 1

## R and RStudio

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This walkthrough is presented by the IMMERSE team and will go through some common tasks carried out in R. There are many free resources available to get started with R and RStudio. One of our favorites is *R for Data Science*.

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- *R*(?) is a free, open-source programming language and environment widely used for statistical computing, data analysis, and data visualization.
  - *RStudio*(?) is an integrated development environment (IDE) for R, providing an intuitive interface that makes coding, visualization, and project management more accessible.
  - *Mplus*(?) is a statistical modeling program used for analyzing complex data, such as latent variable models, structural equation modeling, and growth modeling. This book uses an R package called **MplusAutomation** to automate the process of running models, extracting results, and generating data visualizations.
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### 1.1 Installation

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### 1.1.1 Step 0: Install R, RStudio, and Mplus

Here you will find a guide to installing both R and R Studio. You can also install Mplus here.

*Note:* The installation of Mplus requires a paid license with the mixture add-on. IMMERSE fellows will be given their own copy of Mplus for use during the one year training.

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## 1.2 Set-up

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### 1.2.1 Step 1: Create a new R-project in RStudio

R-projects help us organize our folders , filepaths, and scripts. To create a new R project:

- File -> New Project...

Click “New Directory” -> New Project -> Name your project

### 1.2.2 Step 2: Create an R-markdown document

An R-markdown file provides an authoring framework for data science that allows us to organize our reports using texts and code chunks. This document you are reading was made using R-markdown!

To create an R-markdown:

- File -> New File -> R Markdown...

In the window that pops up, give the R-markdown a title such as “**Introduction to R and RStudio**” Click “OK.” You should see a new markdown with some example text and code chunks. We want a clean document to start off with so delete everything from line 10 down. Go ahead and save this document in your R Project folder.

### 1.2.3 Step 3: Load packages

Your first code chunk in any given markdown should be the packages you will be using. To insert a code chunk, either use the keyboard shortcut `ctrl + alt + i` or Code  $\rightarrow$  Insert Chunk or click the green box with the letter C on it. There are a few packages we want our markdown to read in:

```
library(psych) # describe()
library(here) # helps with filepaths
library(gt) # create tables
library(tidyverse) # collection of R packages designed for data science
```

As a reminder, if a function does not work and you receive an error like this: `could not find function "random_function"`; or if you try to load a package and you receive an error like this: `there is no package called 'random_package'`, then you will need to install the package using `install.packages("random_package")` in the console (the bottom-left window in R studio). Once you have installed the package you will *never* need to install it again, however you must *always* load in the packages at the beginning of your R markdown using `library(random_package)`, as shown in this document.

The style of code and package we will be using is called `tidyverse(?)`. Most functions are within the `tidyverse` package and if not, I've indicated the packages used in the code chunk above.

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## 1.3 Explore the data

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### 1.3.1 Step 4: Read in data

To demonstrate mixture modeling in the training program and online resource components of the IES grant we utilize the *Civil Rights Data Collection (CRDC)* (CRDC) data repository. The CRDC is a federally mandated school-level data collection effort that occurs every other year. This public data is currently available for selected latent class indicators across 4 years (2011, 2013, 2015, 2017) and all US states. In this example, we use the Arizona state sample. We utilize six focal indicators which constitute the latent class model in our example; three variables which report on harassment/bullying in schools based on disability, race, or sex, and three variables on full-time equivalent school staff

LCA indicators<sup>1</sup>

| Name           | Label                                                                    |
|----------------|--------------------------------------------------------------------------|
| leaid          | District Identification Code                                             |
| ncessch        | School Identification Code                                               |
| report_dis     | Number of students harassed or bullied on the basis of disability        |
| report_race    | Number of students harassed or bullied on the basis of race, color, or n |
| report_sex     | Number of students harassed or bullied on the basis of sex               |
| counselors_fte | Number of full time equivalent counselors hired as school staff          |
| report_sex     | Number of full time equivalent psychologists hired as school staff       |
| counselors_fte | Number of full time equivalent law enforcement officers hired as school  |

<sup>1</sup>Civil Rights Data Collection (CRDC)

hires (counselor, psychologist, law enforcement). This data source also includes covariates on a variety of subjects and distal outcomes reported in 2018 such as math/reading assessments and graduation rates.

**To read in data in R:**

```
data <- read_csv(here("data", "crdc_lca_data.csv"))
```

**Ways to view data in R:**

1. click on the data in your Global Environment (upper right pane) or use...

```
View(data)
```

2. `summary()` gives basic summary statistics & shows number of NA values (great for checking that data has been read in correctly)

```
summary(data)
#>      leaid          ncessch      report_dis
#> Length:2027      Length:2027      Min.    :0.0000
#> Class :character  Class :character  1st Qu.:0.0000
#> Mode  :character  Mode  :character  Median :0.0000
#>                                     Mean  :0.0425
#>                                     3rd Qu.:0.0000
#>                                     Max.   :1.0000
#>                                     NA's   :27
#>      report_race      report_sex      counselors_fte
```

```
#> Min. :0.000 Min. :0.00 Min. :0.0000
#> 1st Qu.:0.000 1st Qu.:0.00 1st Qu.:0.0000
#> Median :0.000 Median :0.00 Median :0.0000
#> Mean :0.103 Mean :0.17 Mean :0.4595
#> 3rd Qu.:0.000 3rd Qu.:0.00 3rd Qu.:1.0000
#> Max. :1.000 Max. :1.00 Max. :1.0000
#> NA's :27 NA's :27 NA's :27
#> psych_fte law_fte
#> Min. :0.0000 Min. :0.0000
#> 1st Qu.:0.0000 1st Qu.:0.0000
#> Median :0.0000 Median :0.0000
#> Mean :0.4742 Mean :0.1255
#> 3rd Qu.:1.0000 3rd Qu.:0.0000
#> Max. :1.0000 Max. :1.0000
#> NA's :30 NA's :27
```

3. `names()` provides a list of column names. Very useful if you don't have them memorized!

```
names(data)
#> [1] "leaid" "necessch" "report_dis"
#> [4] "report_race" "report_sex" "counselors_fte"
#> [7] "psych_fte" "law_fte"
```

4. `head()` prints the top 6 rows of the dataframe

```
head(data)
#> # A tibble: 6 x 8
#>   leaid ncessch report_dis report_race report_sex
#>   <chr> <chr> <dbl> <dbl> <dbl>
#> 1 0400001 040000100120 0 0 0
#> 2 0400001 040000100616 0 0 1
#> 3 0400001 040000101204 0 0 1
#> 4 0400001 040000101871 0 1 1
#> 5 0400001 040000101872 0 0 0
#> 6 0400001 040000102344 0 0 0
#> # i 3 more variables: counselors_fte <dbl>,
#> # psych_fte <dbl>, law_fte <dbl>
```

### 1.3.2 Step 5: Descriptive Statistics

Let's look at descriptive statistics for each variable. Because looking at the ID variables' (`leaid`) and (`necessch`) descriptives is unnecessary, we use `select()` to remove the variable by using the minus (-) sign:

```
data %>%
  select(-leaid, -ncessch) %>%
  summary()
```

| #>          | report_dis | report_race   | report_sex   |
|-------------|------------|---------------|--------------|
| #> Min.     | :0.0000    | Min. :0.000   | Min. :0.00   |
| #> 1st Qu.: | 0.0000     | 1st Qu.:0.000 | 1st Qu.:0.00 |
| #> Median : | 0.0000     | Median :0.000 | Median :0.00 |
| #> Mean :   | 0.0425     | Mean :0.103   | Mean :0.17   |
| #> 3rd Qu.: | 0.0000     | 3rd Qu.:0.000 | 3rd Qu.:0.00 |
| #> Max. :   | 1.0000     | Max. :1.000   | Max. :1.00   |
| #> NA's :   | 27         | NA's :27      | NA's :27     |

| #>          | counselors_fte | psych_fte      | law_fte        |
|-------------|----------------|----------------|----------------|
| #> Min.     | :0.0000        | Min. :0.0000   | Min. :0.0000   |
| #> 1st Qu.: | 0.0000         | 1st Qu.:0.0000 | 1st Qu.:0.0000 |
| #> Median : | 0.0000         | Median :0.0000 | Median :0.0000 |
| #> Mean :   | 0.4595         | Mean :0.4742   | Mean :0.1255   |
| #> 3rd Qu.: | 1.0000         | 3rd Qu.:1.0000 | 3rd Qu.:0.0000 |
| #> Max. :   | 1.0000         | Max. :1.0000   | Max. :1.0000   |
| #> NA's :   | 27             | NA's :30       | NA's :27       |

Alternatively, we can use the `psych::describe()` function to give more information:

```
data %>%
  select(-leaid, -ncessch) %>%
  describe()
```

| #>                | vars | n    | mean | sd   | median | trimmed | mad | min |
|-------------------|------|------|------|------|--------|---------|-----|-----|
| #> report_dis     | 1    | 2000 | 0.04 | 0.20 | 0      | 0.00    | 0   | 0   |
| #> report_race    | 2    | 2000 | 0.10 | 0.30 | 0      | 0.00    | 0   | 0   |
| #> report_sex     | 3    | 2000 | 0.17 | 0.38 | 0      | 0.09    | 0   | 0   |
| #> counselors_fte | 4    | 2000 | 0.46 | 0.50 | 0      | 0.45    | 0   | 0   |
| #> psych_fte      | 5    | 1997 | 0.47 | 0.50 | 0      | 0.47    | 0   | 0   |
| #> law_fte        | 6    | 2000 | 0.13 | 0.33 | 0      | 0.03    | 0   | 0   |

| #>                | max | range | skew | kurtosis | se   |
|-------------------|-----|-------|------|----------|------|
| #> report_dis     | 1   | 1     | 4.53 | 18.55    | 0.00 |
| #> report_race    | 1   | 1     | 2.61 | 4.82     | 0.01 |
| #> report_sex     | 1   | 1     | 1.76 | 1.08     | 0.01 |
| #> counselors_fte | 1   | 1     | 0.16 | -1.97    | 0.01 |
| #> psych_fte      | 1   | 1     | 0.10 | -1.99    | 0.01 |
| #> law_fte        | 1   | 1     | 2.26 | 3.11     | 0.01 |

What if we want to look at a subset of the data? For example, what if we want to subset the data to observe a specific school district? (`leaid`) We can use `tidyverse::filter()` to subset the data using certain criteria.

```
data %>%
  filter(leaid == "0408800") %>%
  describe()
#>           vars  n mean   sd median trimmed  mad min
#> leaid*         1 86 1.00 0.00   1.0   1.00 0.00  1
#> ncessch*       2 86 43.50 24.97  43.5  43.50 31.88  1
#> report_dis     3 86 0.05 0.21   0.0   0.00 0.00  0
#> report_race    4 86 0.15 0.36   0.0   0.07 0.00  0
#> report_sex     5 86 0.19 0.39   0.0   0.11 0.00  0
#> counselors_fte 6 86 0.95 0.21   1.0   1.00 0.00  0
#> psych_fte      7 86 0.19 0.39   0.0   0.11 0.00  0
#> law_fte        8 86 0.14 0.35   0.0   0.06 0.00  0
#>           max range skew kurtosis   se
#> leaid*         1    0  NaN      NaN 0.00
#> ncessch*       86   85 0.00   -1.24 2.69
#> report_dis     1    1  4.23   16.10 0.02
#> report_race    1    1  1.91    1.68 0.04
#> report_sex     1    1  1.59    0.52 0.04
#> counselors_fte 1    1 -4.23   16.10 0.02
#> psych_fte      1    1  1.59    0.52 0.04
#> law_fte        1    1  2.04    2.21 0.04

#You can use any operator to filter: >, <, ==, >=, etc.
```

Since we have binary data (0,1), it would be helpful to look at the proportions:

```
data %>%
  drop_na() %>%
  pivot_longer(report_dis:law_fte, names_to = "variable") %>%
  group_by(variable) %>%
  summarise(prop = sum(value)/n(),
            n = n()) %>%
  arrange(desc(prop))
#> # A tibble: 6 x 3
#>   variable      prop      n
#>   <chr>      <dbl> <int>
#> 1 psych_fte    0.481  1970
#> 2 counselors_fte 0.459  1970
#> 3 report_sex   0.173  1970
#> 4 law_fte      0.127  1970
#> 5 report_race  0.105  1970
#> 6 report_dis   0.0431 1970
```





## Chapter 2

# MplusAutomation

---

**MplusAutomation(?)** is designed to streamline the use of Mplus, a powerful statistical software for modeling complex data developed by Muthen and Muten (<https://www.statmodel.com>). With **MplusAutomation**, researchers can automate the process of estimating latent variable models, running batches of models, extracting results, and generating data visualizations - all within the R environment.

---

### WHAT?

- **MplusAutomation** is an R package
  - It “wraps around” the **Mplus** program
  - Requires both R & **Mplus** software
  - Requires learning some basics of 2 programming languages
  - Car metaphor: R/Rstudio is the *steering wheel or dashboard* & Mplus is the *engine*
- 

### WHY?

- **MplusAutomation** can provide clearly organized work procedures in which every research decision can be documented in a single place
- Increase reproducibility, organization, efficiency, and transparency

---

## HOW?

- The interface for MplusAutomation is entirely within R-Studio. You do not need to open Mplus
  - The code presented will be very repetitive by design
- 

Below is a template for `mplusObject()` & `mplusModeler()` functions. Use this template to run statistical models with Mplus.

```
m_template <- mplusObject(

  TITLE =
    "",

  VARIABLE =
    "",

  ANALYSIS =
    "",

  PLOT =
    "",

  OUTPUT =
    "",

  usevariables = colnames(),
  rdata = )

m_template_fit <- mplusModeler(m_template,
  dataout=here("", ".dat"),
  modelout=here("", ".inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

## Part II

# Latent Class Analysis



## Chapter 3

# Enumeration

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Example: Bullying in Schools

---

To demonstrate mixture modeling in the training program and online resource components of the IES grant we utilize the *Civil Rights Data Collection (CRDC)*(?) data repository. The CRDC is a federally mandated school-level data collection effort that occurs every other year. This public data is currently available for selected latent class indicators across 4 years (2011, 2013, 2015, 2017) and all US states. In this example, we use the Arizona state sample. We utilize six focal indicators which constitute the latent class model in our example; three variables which report on harassment/bullying in schools based on disability, race, or sex, and three variables on full-time equivalent school staff hires (counselor, psychologist, law enforcement). This data source also includes covariates on a variety of subjects and distal outcomes reported in 2018 such as math/reading assessments and graduation rates.

---

Load packages

```
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
```

LCA indicators<sup>1</sup>

| Name           | Label                                                                    |
|----------------|--------------------------------------------------------------------------|
| leaid          | District Identification Code                                             |
| ncessch        | School Identification Code                                               |
| report_dis     | Number of students harassed or bullied on the basis of disability        |
| report_race    | Number of students harassed or bullied on the basis of race, color, or n |
| report_sex     | Number of students harassed or bullied on the basis of sex               |
| counselors_fte | Number of full time equivalent counselors hired as school staff          |
| psych_fte      | Number of full time equivalent psychologists hired as school staff       |
| law_fte        | Number of full time equivalent law enforcement officers hired as school  |

<sup>1</sup>Civil Rights Data Collection (CRDC)

```
library(janitor)
library(gt)
library(cowplot)
library(DiagrammeR)
```

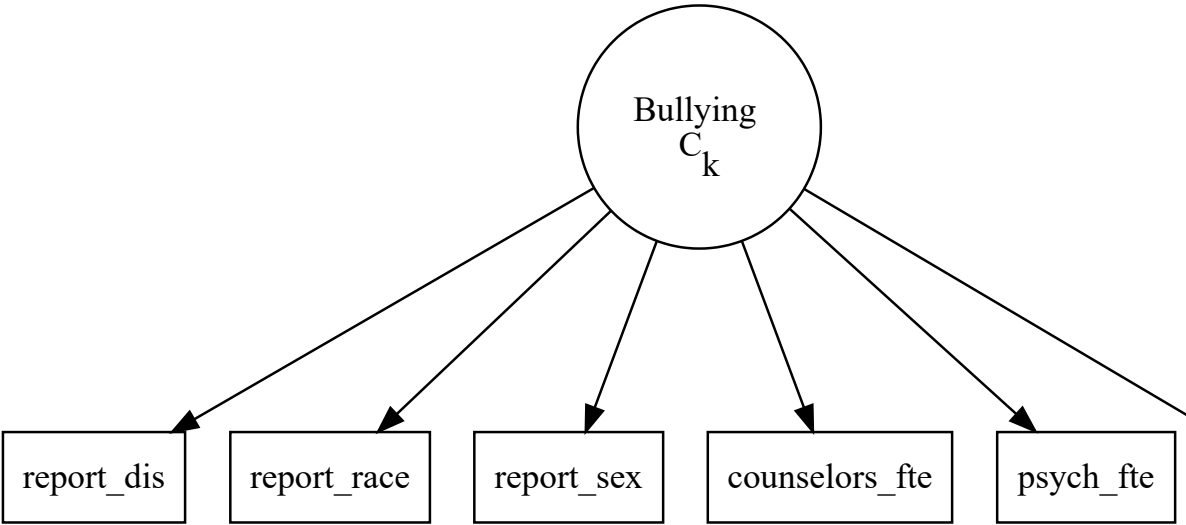
### 3.1 Variable Description

---

#### Variables have been transformed to be dichotomous indicators using the following coding strategy

Harassment and bullying count variables are recoded 1 if the school reported at least one incident of harassment (0 indicates no reported incidents). On the original scale reported by the CDRC staff variables for full time equivalent employees (FTE) are represented as 1 and part time employees are represented by values between 1 and 0. Schools with greater than one staff of the designated type are represented by values greater than 1. All values greater than zero were recorded as 1s (e.g., .5, 1,3) indicating that the school has a staff present on campus at least part time. Schools with no staff of the designated type are indicated as 0 for the dichotomous variable.

---



## 3.2 Prepare Data

---

```
df_bully <- read_csv(here("data", "crdc_lca_data.csv")) %>%
  clean_names() %>%
  dplyr::select(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_ft
```

---

## 3.3 Descriptive Statistics

```
dframe <- df_bully %>%
  pivot_longer(
    c(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_fte),
    names_to = "Variable"
  ) %>%
  group_by(Variable) %>%
  summarise(
    Count = sum(value == 1, na.rm = TRUE),
    Total = n(),
    .groups = "drop"
  ) %>%
  mutate(`Proportion Endorsed` = round(Count / Total, 3)) %>%
  select(Variable, `Proportion Endorsed`, Count)

gt(dframe) %>%
  tab_header(
    title = md("**LCA Indicator Endorsement**"),
    subtitle = md("&nbsp;")
  ) %>%
  tab_options(
    column_labels.font.weight = "bold",
    row_group.font.weight = "bold"
  )
```

Save as image



## LCA Indicator Endorsement

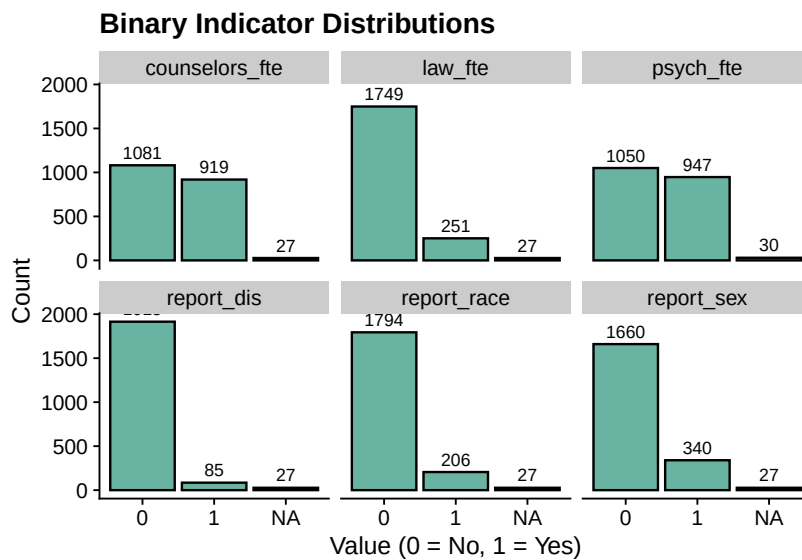
| Variable       | Proportion Endorsed | Count |
|----------------|---------------------|-------|
| counselors_fte | 0.453               | 919   |
| law_fte        | 0.124               | 251   |
| psych_fte      | 0.467               | 947   |
| report_dis     | 0.042               | 85    |
| report_race    | 0.102               | 206   |
| report_sex     | 0.168               | 340   |

```
gtsave(prop_table, here("figures", "prop_table.png"))
```

## Frequency Plot

```
data_long <- df_bully %>%
  pivot_longer(c(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_fte), names_to = "variable", values_to = "value")

# Bar plot for 0/1 indicators
ggplot(data_long, aes(x = factor(value))) +
  geom_bar(fill = "#69b3a2", color = "black") +
  geom_text(stat = "count", aes(label = after_stat(count)),
            vjust = -0.5, size = 3.5) +
  facet_wrap(~ variable) +
  labs(
    title = "Binary Indicator Distributions",
    x = "Value (0 = No, 1 = Yes)",
    y = "Count"
  ) +
  theme_cowplot()
```



### 3.4 Enumeration

This code uses the `mplusObject` function in the `MplusAutomation` package and saves all model runs in the `enum` folder.

```
lca_6 <- lapply(1:6, function(k) {
  lca_enum <- mplusObject(

    TITLE = glue("{k}-Class"),

    VARIABLE = glue(
      "categorical = report_dis-law_fte;
      usevar = report_dis-law_fte;
      classes = c({k}); "),

    ANALYSIS =
      "estimator = mlr;
      type = mixture;
      starts = 500 100;
      processors = 10;",

    OUTPUT = "sampstat residual tech11 tech14;",
```

```

PLOT =
  "type = plot3;
  series = report_dis-law_fte(*)";

usevariables = colnames(df_bully),
rdata = df_bully)

lca_enum_fit <- mplusModeler(lca_enum,
                             dataout=glue(here("enum", "bully.dat")),
                             modelout=glue(here("enum", "c{k}_bully.inp")),
                             check=TRUE, run = TRUE, hashfilename = FALSE)
})

```

**IMPORTANT:** Before moving forward, make sure to open each output document to ensure models were estimated normally.

---

## 3.5 Examine and extract Mplus files

Code by Delwin Carter (2025)

Check all Models for:

1. Warnings
2. Errors
3. Convergence and Loglikelihood Replication Information

```

source(here("functions", "extract_mplus_info.R"))

# Define the directory where all of the .out files are located.
output_dir <- here("enum")

# Get all .out files
output_files <- list.files(output_dir, pattern = "\\\\.out$", full.names = TRUE)

# Process all .out files into one dataframe
final_data <- map_dfr(output_files, extract_mplus_info_extended)

# Extract Sample_Size from final_data
sample_size <- unique(final_data$Sample_Size)

```

### 3.5.1 Examine Mplus Warnings

```
source(here("functions", "extract_warnings.R"))

warnings_table <- extract_warnings(final_data)
warnings_table
```

```
# Save the warnings table
#gtsave(warnings_table, here("figures", "warnings_table.png"))
```

### 3.5.2 Examine Mplus Errors

```
source(here("functions", "error_visualization.R"))

# Process errors
error_table_data <- process_error_data(final_data)
error_table_data
```

```
# Save the errors table
#gtsave(error_table, here("figures", "error_table.png"))
```

### 3.5.3 Examine Convergence and Loglikelihood Replications

```
source(here("functions", "summary_table.R"))

# Print Table with Superheader & Heatmap
summary_table <- create_flexitable(final_data, sample_size)
summary_table
```

| <i>N</i> = 2027 |            |      | Random Starts |       | Final starting value sets converging |      | LL Replication |        | Smallest Class |        |
|-----------------|------------|------|---------------|-------|--------------------------------------|------|----------------|--------|----------------|--------|
| Model           | Best LL    | npar | Initial       | Final | <i>f</i>                             | %    | <i>f</i>       | %      | <i>f</i>       | %      |
| 1-Class         | -5,443.409 | 6    | 500           | 100   | 100                                  | 100% | 100            | 100.0% | 2,027          | 100.0% |
| 2-Class         | -5,194.136 | 13   | 500           | 100   | 86                                   | 86%  | 86             | 100.0% | 444            | 21.9%  |
| 3-Class         | -5,122.478 | 20   | 500           | 100   | 94                                   | 94%  | 94             | 100.0% | 216            | 10.6%  |
| 4-Class         | -5,111.757 | 27   | 500           | 100   | 54                                   | 54%  | 19             | 35.2%  | 212            | 10.5%  |
| 5-Class         | -5,105.589 | 34   | 500           | 100   | 38                                   | 38%  | 6              | 15.8%  | 43             | 2.1%   |
| 6-Class         | -5,099.881 | 41   | 500           | 100   | 36                                   | 36%  | 6              | 16.7%  | 36             | 1.8%   |

```
# Save the flextable as a PNG image
#invisible(save_as_image(summary_table, path = here("figures", "housekeeping.png")))
```

### 3.5.4 Check for Loglikelihood Replication

Visualize and examine loglikelihood replication values for each output file individually

```
# Load the function for separate plots
source(here("functions", "ll_replication_plots.R"))

# Generate individual log-likelihood replication tables
ll_replication_tables <- generate_ll_replication_plots(final_data)
ll_replication_tables
```

\$c1\_bully.out

Log Likelihood Replications: 1-Class

Source File: c1\_bully.out

Log Likelihood

Replication Count

% of Valid Replications

−5,443.409

100.000

100.00%

\$c2\_bully.out

Log Likelihood Replications: 2-Class

Source File: c2\_bully.out

Log Likelihood

Replication Count

% of Valid Replications

−5,194.136

86.000

100.00%

\$c3\_bully.out

Log Likelihood Replications: 3-Class

Source File: c3\_bully.out

Log Likelihood

Replication Count

% of Valid Replications

−5,122.478

94.000

100.00%

\$c4\_bully.out

Log Likelihood Replications: 4-Class

Source File: c4\_bully.out

Log Likelihood

Replication Count

% of Valid Replications

−5,111.757

19.000

35.19%

−5,111.759

4.000

7.41%

−5,112.253

5.000

9.26%

−5,113.910

1.000

1.85%

−5,115.532

17.000

31.48%

−5,115.538

1.000

1.85%

−5,116.981

3.000

5.56%

−5,117.829

2.000

3.70%

−5,117.837

2.000

3.70%

\$c5\_bully.out

Log Likelihood Replications: 5-Class

Source File: c5\_bully.out

Log Likelihood

Replication Count

% of Valid Replications

−5,105.589

6.000

15.79%

−5,105.661

1.000

2.63%  
−5,105.791  
3.000  
7.89%  
−5,105.799  
3.000  
7.89%  
−5,106.628  
1.000  
2.63%  
−5,106.748  
1.000  
2.63%  
−5,106.864  
1.000  
2.63%  
−5,106.975  
1.000  
2.63%  
−5,106.983  
1.000  
2.63%  
−5,107.172  
4.000  
10.53%  
−5,107.449  
1.000  
2.63%  
−5,107.450  
2.000  
5.26%



−5,107.458

1.000

2.63%

−5,107.728

1.000

2.63%

−5,107.958

1.000

2.63%

−5,108.003

1.000

2.63%

−5,108.058

1.000

2.63%

−5,108.860

3.000

7.89%

−5,109.002

1.000

2.63%

−5,110.373

1.000

2.63%

−5,110.474

1.000

2.63%

−5,111.532

1.000

2.63%

−5,112.695

1.000  
2.63%  
\$c6\_bully.out  
Log Likelihood Replications: 6-Class  
Source File: c6\_bully.out  
Log Likelihood  
Replication Count  
% of Valid Replications  
−5,099.881  
6.000  
16.67%  
−5,100.272  
1.000  
2.78%  
−5,100.780  
1.000  
2.78%  
−5,100.874  
1.000  
2.78%  
−5,100.928  
2.000  
5.56%  
−5,101.017  
1.000  
2.78%  
−5,101.071  
3.000  
8.33%  
−5,101.089  
1.000

2.78%  
-5,101.332  
1.000  
2.78%  
-5,101.448  
1.000  
2.78%  
-5,101.494  
1.000  
2.78%  
-5,101.502  
2.000  
5.56%  
-5,101.512  
1.000  
2.78%  
-5,101.579  
2.000  
5.56%  
-5,101.859  
1.000  
2.78%  
-5,101.923  
1.000  
2.78%  
-5,101.964  
1.000  
2.78%  
-5,102.075  
1.000  
2.78%

−5,102.275

1.000

2.78%

−5,102.613

1.000

2.78%

−5,102.616

1.000

2.78%

−5,103.084

1.000

2.78%

−5,104.611

1.000

2.78%

−5,106.123

1.000

2.78%

−5,106.486

1.000

2.78%

−5,107.624

1.000

2.78%

Visualize and examine loglikelihood replication for each output file together

```
ll_replication_table_all <- source(here("functions", "ll_replication_processing.R"), 1)
ll_replication_table_all
```





**Are you confident with the results?**

- Errors?
- Warnings?
- Loglikelihood was Replicated?

If yes, Continue →

### 3.6 Table of Fit

First, extract data:

```
output_enum <- readModels(here("enum"), filefilter = "bully", quiet = TRUE)

# Extract fit indices
enum_extract <- LatexSummaryTable(
  output_enum,
  keepCols = c(
    "Title",
    "Parameters",
    "LL",
    "BIC",
    "aBIC",
    "BLRT_PValue",
    "T11_VLMR_PValue",
    "Observations"
  ),
  sortBy = "Title"
)

# Calculate additional fit indices
allFit <- enum_extract %>%
  mutate(CAIC = -2 * LL + Parameters * (log(Observations) + 1)) %>%
  mutate(AWE = -2 * LL + 2 * Parameters * (log(Observations) + 1.5)) %>%
  mutate(SIC = -.5 * BIC) %>%
```

```

mutate(expSIC = exp(SIC - max(SIC))) %>%
mutate(BF = exp(SIC - lead(SIC))) %>%
mutate(cmPk = expSIC / sum(expSIC)) %>%
dplyr::select(Title, Parameters, LL, BIC, aBIC, CAIC, AWE, BLRT_PValue, T11_VLMR_PValue, BF, cm
arrange(Parameters)

# Merge columns with LL replications and class size from `final_data`
merged_table <- allFit %>%
  mutate(Title = str_trim(Title)) %>%
  left_join(
    final_data %>%
      select(
        Class_Model,
        Perc_Convergence,
        Replicated_LL_Perc,
        Smallest_Class,
        Smallest_Class_Perc
      ),
    by = c("Title" = "Class_Model")
  ) %>%
  mutate(Smallest_Class = coalesce(Smallest_Class, final_data$Smallest_Class[match(Title, final_d
  relocate(Perc_Convergence, Replicated_LL_Perc, .after = LL) %>%
  mutate(Smallest_Class_Combined = paste0(Smallest_Class, "\u00A0", Smallest_Class_Perc, "%"))
  select(
    Title,
    Parameters,
    LL,
    Perc_Convergence,
    Replicated_LL_Perc,
    BIC,
    aBIC,
    CAIC,
    AWE,
    T11_VLMR_PValue,
    BLRT_PValue,
    Smallest_Class_Combined,
    BF,
    cmPk
  )

```

Then, create table:

```

fit_table1 <- merged_table %>%
  select(Title, Parameters, LL, Perc_Convergence, Replicated_LL_Perc,
    BIC, aBIC, CAIC, AWE,

```

```

        T11_VLMR_PValue, BLRT_PValue,
        Smallest_Class_Combined) %>%
gt() %>%
tab_header(title = md("***Model Fit Summary Table**")) %>%
tab_spanner(label = "Model Fit Indices", columns = c(BIC, aBIC, CAIC, AWE)) %>%
tab_spanner(label = "LRTs", columns = c(T11_VLMR_PValue, BLRT_PValue)) %>%
tab_spanner(label = md("Smallest\u00A0Class"), columns = c(Smallest_Class_Combined))
cols_label(
  Title = "Classes",
  Parameters = md("Par"),
  LL = md("*LL*"),
  Perc_Convergence = "% Converged",
  Replicated_LL_Perc = "% Replicated",
  BIC = "BIC",
  aBIC = "aBIC",
  CAIC = "CAIC",
  AWE = "AWE",
  T11_VLMR_PValue = "VLMR",
  BLRT_PValue = "BLRT",
  Smallest_Class_Combined = "n (%)"
) %>%
tab_footnote(
  footnote = md(
    "*Note.* Par = Parameters; *LL* = model log likelihood;
BIC = Bayesian information criterion;
aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion;
AWE = approximate weight of evidence criterion;
BLRT = bootstrapped likelihood ratio test p-value;
VLMR = Vuong-Lo-Mendell-Rubin adjusted likelihood ratio test p-value;
*cmPk* = approximate correct model probability."
  ),
locations = cells_title()
) %>%
tab_options(column_labels.font.weight = "bold") %>%
fmt_number(
  columns = c(3, 6:9),
  decimals = 2
) %>%
sub_missing(1:11,
  missing_text = "--") %>%
fmt(
  c(T11_VLMR_PValue, BLRT_PValue),
  fns = function(x)
    ifelse(x < 0.001, "<.001",
  scales::number(x, accuracy = .01))

```



```

) %>%
fmt_percent(
  columns = c(Perc_Convergence, Replicated_LL_Perc),
  decimals = 0,
  scale_values = FALSE
) %>%

cols_align(align = "center", columns = everything()) %>%
tab_style(
  style = list(cell_text(weight = "bold")),
  locations = list(
    cells_body(columns = BIC, row = BIC == min(BIC)),
    cells_body(columns = aBIC, row = aBIC == min(aBIC)),
    cells_body(columns = CAIC, row = CAIC == min(CAIC)),
    cells_body(columns = AWE, row = AWE == min(AWE)),
    cells_body(columns = T11_VLMR_PValue,
      row = ifelse(T11_VLMR_PValue < .05 & lead(T11_VLMR_PValue) > .05, T11_VLMR_PValue, NA)),
    cells_body(columns = BLRT_PValue,
      row = ifelse(BLRT_PValue < .05 & lead(BLRT_PValue) > .05, BLRT_PValue < .05, NA))
  )
)
fit_table1

```

---

Save table

```
gtsave(fit_table1, here("figures", "fit_table.png"))
```

---

### 3.7 Information Criteria Plot

```

allFit %>%
dplyr::select(2:7) %>%
rowid_to_column() %>%
pivot_longer(`BIC`:`AWE`,
  names_to = "Index",
  values_to = "ic_value") %>%

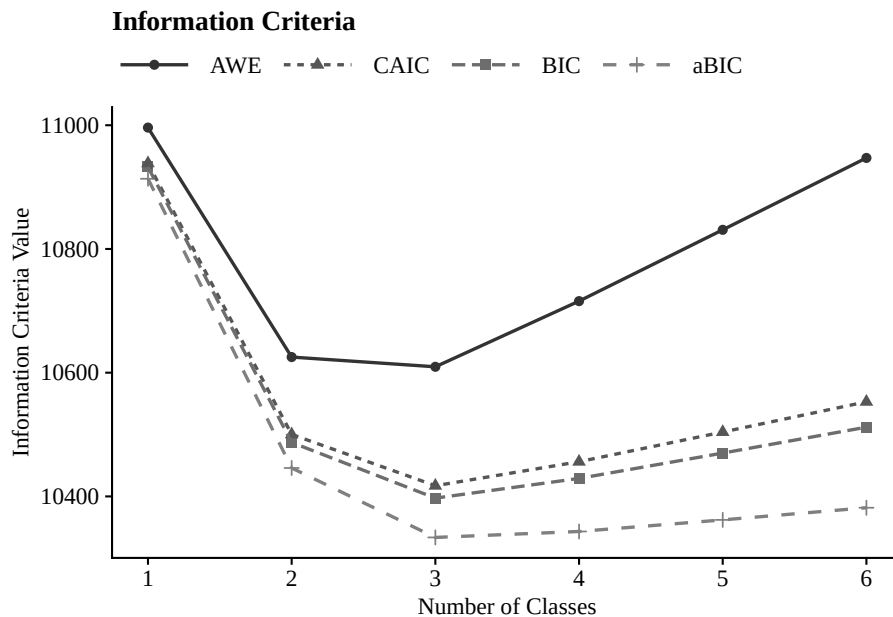
```

```

mutate(Index = factor(Index,
                      levels = c ("AWE", "CAIC", "BIC", "aBIC"))) %>%

ggplot(aes(
  x = rowid,
  y = ic_value,
  color = Index,
  shape = Index,
  group = Index,
  lty = Index
)) +
geom_point(size = 2.0) + geom_line(linewidth = .8) +
scale_x_continuous(breaks = 1:nrow(allFit)) +
scale_colour_grey(end = .5) +
theme_cowplot() +
labs(x = "Number of Classes", y = "Information Criteria Value", title = "Information
theme(
  text = element_text(family = "serif", size = 12),
  legend.text = element_text(family="serif", size=12),
  legend.key.width = unit(3, "line"),
  legend.title = element_blank(),
  legend.position = "top"
)

```



Save figure

```
ggsave(here("figures", "info_criteria.png"), dpi=300, height=5, width=7, units="in")
```

## 3.8 Compare Class Solutions

Compare probability plots for  $K = 1 : 6$  class solutions

```
model_results <- data.frame()

for (i in 1:length(output_enum)) {
  temp <- output_enum[[i]]$parameters$probability.scale %>%
    mutate(model = paste(i, "-Class Model"))

  model_results <- rbind(model_results, temp)
}

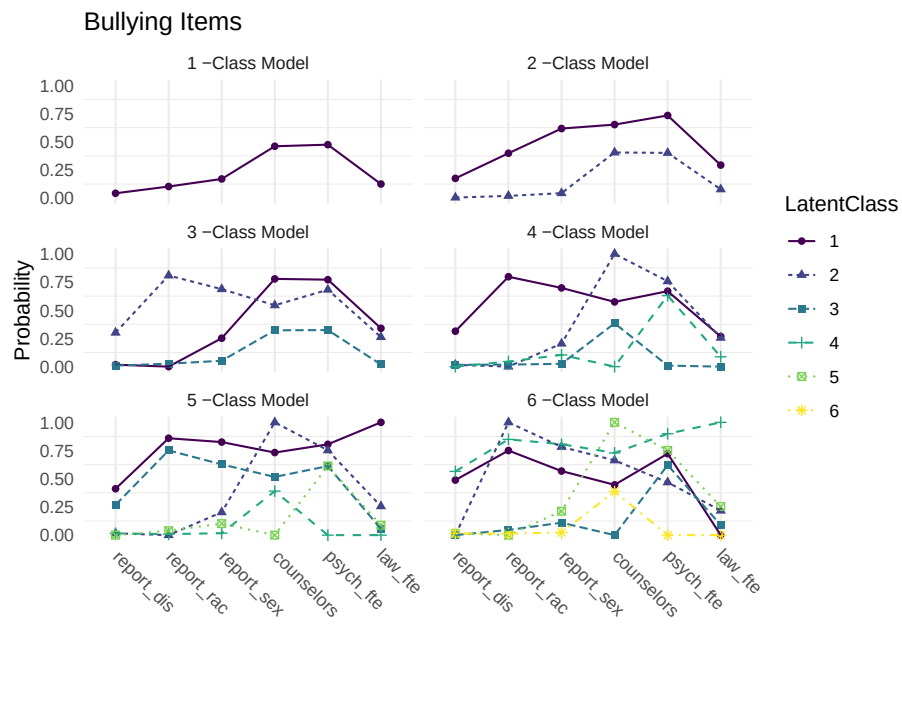
rm(temp)

compare_plot <- model_results %>%
  filter(category == 2) %>%
  dplyr::select(est, model, LatentClass, param) %>%
  mutate(param = as.factor(str_to_lower(param)),
         est = as.numeric(est))

compare_plot$param <- fct_inorder(compare_plot$param)

ggplot(
  compare_plot,
  aes(
    x = param,
    y = est,
    color = LatentClass,
    shape = LatentClass,
    group = LatentClass,
    lty = LatentClass
  )
) +
  geom_point() +
  geom_line() +
  scale_colour_viridis_d() +
```

```
ylim(0,1) +
facet_wrap( ~ model, ncol = 2) +
labs(title = "Bullying Items", x = NULL, y = "Probability") +
theme_minimal() +
theme(panel.grid.major.y = element_blank(),
      axis.text.x = element_text(angle = -45, hjust = -.1))
```



Save figure:

```
ggsave(here("figures", "compare_kclass_plot.png"), dpi=300, height=5, width=7, units="in")
```

### Model Warnings

| Output File  | # of Warnings                            | Warning Message(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c1_bully.out | There are 5 warnings in the output file. | <p>*** WARNING in VARIABLE command Note that only the first 8 characters of variable names are used in the output. Shorten variable names to avoid any confusion.</p> <p>*** WARNING in PLOT command Note that only the first 8 characters of variable names are used in plots. If variable names are not unique within the first 8 characters, problems may occur.</p> <p>*** WARNING in OUTPUT command SAMPSTAT option is not available when all outcomes are censored, ordered categorical, unordered categorical (nominal), count or continuous-time survival variables. Request for SAMPSTAT is ignored.</p> <p>*** WARNING in OUTPUT command TECH11 option is not available for TYPE=MIXTURE with only one class. Request for TECH11 is ignored.</p> <p>*** WARNING in OUTPUT command TECH14 option is not available for TYPE=MIXTURE with only one class. Request for TECH14 is ignored.</p> |
| c2_bully.out | There are 3 warnings in the output file. | <p>*** WARNING in VARIABLE command Note that only the first 8 characters of variable names are used in the output. Shorten variable names to avoid any confusion.</p> <p>*** WARNING in PLOT command Note that only the first 8 characters of variable names are used in plots. If variable names are not unique within the first 8 characters, problems may occur.</p> <p>*** WARNING in OUTPUT command SAMPSTAT option is not available when all outcomes are censored, ordered categorical, unordered categorical (nominal), count or continuous-time survival variables. Request for SAMPSTAT is ignored.</p>                                                                                                                                                                                                                                                                                   |
| c3_bully.out | There are 3 warnings in the output file. | <p>*** WARNING in VARIABLE command Note that only the first 8 characters of variable names are used in the output. Shorten variable names to avoid any confusion.</p> <p>*** WARNING in PLOT command Note that only the first 8 characters of variable names are used in plots. If variable names are not unique within the first 8 characters, problems may occur.</p> <p>*** WARNING in OUTPUT command SAMPSTAT option is not available when all outcomes are censored, ordered categorical, unordered categorical (nominal), count or continuous-time survival variables. Request for</p>                                                                                                                                                                                                                                                                                                        |

**Model Estimation Errors**

| Output File  | Model Type | Error Message                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| c2_bully.out | 2-Class    | THE BEST LOGLIKELIHOOD VALUE HAS BEEN REPLICATED. RERUN WITH AT LEAST TWO THE RANDOM STARTS TO CHECK THAT THE BEST LOGLIKELIHOOD IS STILL OBTAINED A REPLICATED.                                                                                                                                                                                                                                                                                                                |
| c3_bully.out | 3-Class    | THE BEST LOGLIKELIHOOD VALUE HAS BEEN REPLICATED. RERUN WITH AT LEAST TWO THE RANDOM STARTS TO CHECK THAT THE BEST LOGLIKELIHOOD IS STILL OBTAINED A REPLICATED. IN THE OPTIMIZATION, ONE OR MORE LOGIT THRESHOLDS APPROACHED EXTREME VALUES OF -15.000 AND 15.000 AND WERE FIXED TO STABILIZE MODEL ESTIMATION. THESE VALUES IMPLY PROBABILITIES OF 0 AND 1. IN THE MODEL RESULTS SECTION, THESE PARAMETERS HAVE 0 STANDARD ERRORS AND 999 IN THE Z-SCORE AND P-VALUE COLUMNS. |
| c4_bully.out | 4-Class    | THE BEST LOGLIKELIHOOD VALUE HAS BEEN REPLICATED. RERUN WITH AT LEAST TWO THE RANDOM STARTS TO CHECK THAT THE BEST LOGLIKELIHOOD IS STILL OBTAINED A REPLICATED. IN THE OPTIMIZATION, ONE OR MORE LOGIT THRESHOLDS APPROACHED EXTREME VALUES OF -15.000 AND 15.000 AND WERE FIXED TO STABILIZE MODEL ESTIMATION. THESE VALUES IMPLY PROBABILITIES OF 0 AND 1. IN THE MODEL RESULTS SECTION, THESE PARAMETERS HAVE 0 STANDARD ERRORS AND 999 IN THE Z-SCORE AND P-VALUE COLUMNS. |
| c5_bully.out | 5-Class    | THE BEST LOGLIKELIHOOD VALUE HAS BEEN REPLICATED. RERUN WITH AT LEAST TWO THE RANDOM STARTS TO CHECK THAT THE BEST LOGLIKELIHOOD IS STILL OBTAINED A REPLICATED. IN THE OPTIMIZATION, ONE OR MORE LOGIT THRESHOLDS APPROACHED EXTREME VALUES OF -15.000 AND 15.000 AND WERE FIXED TO STABILIZE MODEL ESTIMATION. THESE VALUES IMPLY PROBABILITIES OF 0 AND 1. IN THE MODEL RESULTS SECTION, THESE PARAMETERS HAVE 0 STANDARD ERRORS AND 999 IN THE Z-SCORE AND P-VALUE COLUMNS. |

Model Fit Summary Table<sup>1</sup>

| Classes | Par | <i>LL</i> | % Converged | % Replicated | Model Fit Indices |                  |                  |           |
|---------|-----|-----------|-------------|--------------|-------------------|------------------|------------------|-----------|
|         |     |           |             |              | BIC               | aBIC             | CAIC             |           |
| 1-Class | 6   | -5,443.41 | 100%        | 100%         | 10,932.50         | 10,913.44        | 10,938.50        | 10        |
| 2-Class | 13  | -5,194.14 | 86%         | 100%         | 10,487.26         | 10,445.96        | 10,500.26        | 10        |
| 3-Class | 20  | -5,122.48 | 94%         | 100%         | <b>10,397.24</b>  | <b>10,333.70</b> | <b>10,417.24</b> | <b>10</b> |
| 4-Class | 27  | -5,111.76 | 54%         | 35%          | 10,429.10         | 10,343.32        | 10,456.10        | 10        |
| 5-Class | 34  | -5,105.59 | 38%         | 16%          | 10,470.07         | 10,362.04        | 10,504.06        | 10        |
| 6-Class | 41  | -5,099.88 | 36%         | 17%          | 10,511.95         | 10,381.69        | 10,552.95        | 10        |

<sup>1</sup>*Note.* Par = Parameters; *LL* = model log likelihood; BIC = Bayesian information criterion; aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion; AWE = approximate weight of evidence criterion; BLRT = bootstrapped likelihood ratio test p-value; VLMR = Vuong-Lo-Mendell-Rubin adjusted likelihood ratio test p-value; *cmPk* = approximate correct model probability.





## Chapter 4

# Item Probability Plots

---

Example: Bullying in Schools

---

To demonstrate mixture modeling in the training program and online resource components of the IES grant we utilize the *Civil Rights Data Collection (CRDC)*(?) data repository. The CRDC is a federally mandated school-level data collection effort that occurs every other year. This public data is currently available for selected latent class indicators across 4 years (2011, 2013, 2015, 2017) and all US states. In this example, we use the Arizona state sample. We utilize six focal indicators which constitute the latent class model in our example; three variables which report on harassment/bullying in schools based on disability, race, or sex, and three variables on full-time equivalent school staff hires (counselor, psychologist, law enforcement). This data source also includes covariates on a variety of subjects and distal outcomes reported in 2018 such as math/reading assessments and graduation rates.

---

Load packages

```
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
```

LCA indicators<sup>1</sup>

| Name           | Label                                                                    |
|----------------|--------------------------------------------------------------------------|
| leaid          | District Identification Code                                             |
| ncessch        | School Identification Code                                               |
| report_dis     | Number of students harassed or bullied on the basis of disability        |
| report_race    | Number of students harassed or bullied on the basis of race, color, or n |
| report_sex     | Number of students harassed or bullied on the basis of sex               |
| counselors_fte | Number of full time equivalent counselors hired as school staff          |
| psych_fte      | Number of full time equivalent psychologists hired as school staff       |
| law_fte        | Number of full time equivalent law enforcement officers hired as school  |

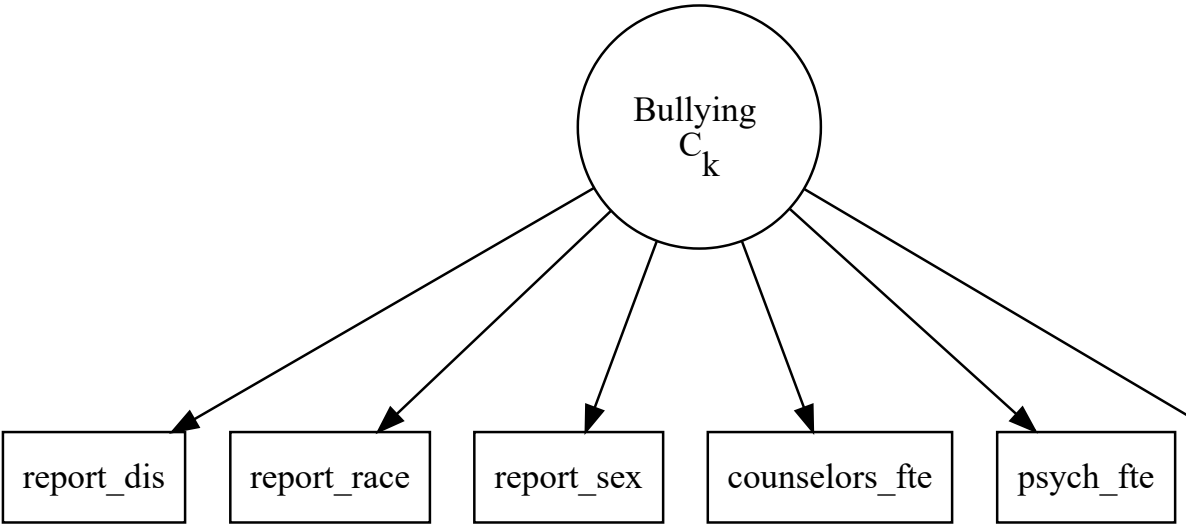
<sup>1</sup>Civil Rights Data Collection (CRDC)

```
library(janitor)
library(gt)
library(cowplot)
library(DiagrammeR)
```

## 4.1 Variable Description

### Variables have been transformed to be dichotomous indicators using the following coding strategy

Harassment and bullying count variables are recoded 1 if the school reported at least one incident of harassment (0 indicates no reported incidents). On the original scale reported by the CDRC staff variables for full time equivalent employees (FTE) are represented as 1 and part time employees are represented by values between 1 and 0. Schools with greater than one staff of the designated type are represented by values greater than 1. All values greater than zero were recorded as 1s (e.g., .5, 1,3) indicating that the school has a staff present on campus at least part time. Schools with no staff of the designated type are indicated as 0 for the dichotomous variable.



## 4.2 Prepare Data

---

```
df_bully <- read_csv(here("data", "crdc_lca_data.csv")) %>%
  clean_names() %>%
  dplyr::select(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_ft
```

---

## 4.3 Descriptive Statistics

```
dframe <- df_bully %>%
  pivot_longer(
    c(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_fte),
    names_to = "Variable"
  ) %>%
  group_by(Variable) %>%
  summarise(
    Count = sum(value == 1, na.rm = TRUE),
    Total = n(),
    .groups = "drop"
  ) %>%
  mutate(`Proportion Endorsed` = round(Count / Total, 3)) %>%
  select(Variable, `Proportion Endorsed`, Count)

gt(dframe) %>%
  tab_header(
    title = md("**LCA Indicator Endorsement**"),
    subtitle = md("&nbsp;")
  ) %>%
  tab_options(
    column_labels.font.weight = "bold",
    row_group.font.weight = "bold"
  )
```

Save as image

## LCA Indicator Endorsement

| Variable       | Proportion Endorsed | Count |
|----------------|---------------------|-------|
| counselors_fte | 0.453               | 919   |
| law_fte        | 0.124               | 251   |
| psych_fte      | 0.467               | 947   |
| report_dis     | 0.042               | 85    |
| report_race    | 0.102               | 206   |
| report_sex     | 0.168               | 340   |

```
gtsave(prop_table, here("figures", "prop_table.png"))
```

## 4.4 3-Class Probability Plot

Continuing the example from the bookdown, use the `plot_lca` function provided in the folder to plot the item probability plot. This function requires one argument: - `model_name`: The name of the Mplus `readModels` object (e.g., `output_enum$c3_bully.out`)

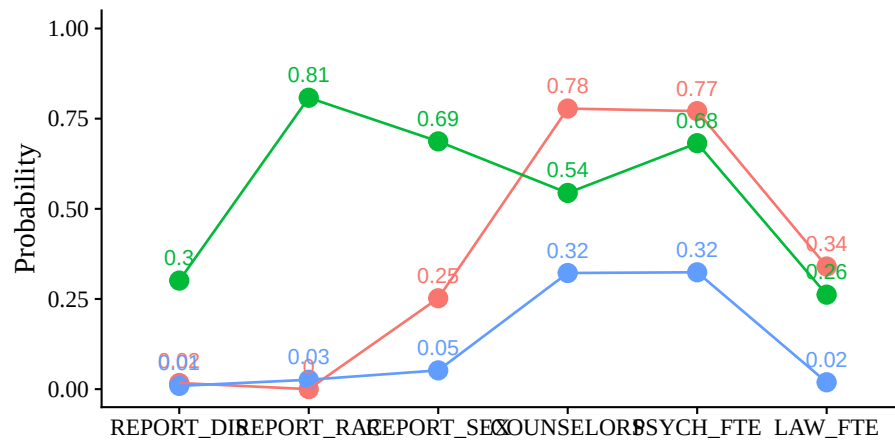
```
source(here("functions", "plot_lca.R"))

# Extract outputs from the enumeration folder
output_enum <- readModels(here("enum"), filefilter = "bully", quiet = TRUE)

plot_lca(model_name = output_enum$c3_bully.out)
```

**3-Class Probability Plot**

● Class 1 (24.94%) 
 ● Class 2 (10.63%) 
 ● Class 3 (64.43%)




---

Save figure:

```
ggsave(here("figures", "C3_bully_LCA_Plot.png"), dpi="retina", height=5, width=7, units="in")
```

---

## Part III

# Polytomous LCA





## Chapter 5

# Enumeration

Polytomous LCA deals with variables that have more than two categories, such as survey questions with responses like **never**, **sometimes**, and **always**. The workflow of a polytomous LCA model is similar to that of an LCA model with binary indicators. However, polytomous LCA captures more complex response patterns, which can make interpretation more complex. The following code demonstrates an example, along with a visualization of the model.

---

### 5.1 Example: Elections

“Two sets of six questions with four responses each, asking respondents’ opinions of how well various traits describe presidential candidates Al Gore and George W. Bush. Also potential covariates vote choice, age, education, gender, and party ID. Source: The National Election Studies (2000).” (poLCA, 2016) See [documentation here](#)

Two sets of six questions with four responses each, asking respondents’ opinions of how well various traits describe presidential candidates Al Gore and George W. Bush. In the election data set, respondents to the 2000 American National Election Study public opinion poll were asked to evaluate how well a series of traits—moral, caring, knowledgeable, good leader, dishonest, and intelligent—described presidential candidates Al Gore and George W. Bush. Each question had four possible choices: (1) extremely well; (2) quite well; (3) not too well; and (4) not well at all.

---

Load packages

```
library(poLCA)
library(tidyverse)
library(janitor)
library(gt)
library(MplusAutomation)
library(here)
library(glue)
```

## 5.2 Prepare Data

```
data(election)

# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)

df_election <- election %>%
  clean_names() %>%
  select(moralb:intelb) %>%
  mutate(across(everything(),
    ~ as.factor(as.numeric(gsub("\\D", "", .))),
    .names = "{.col}1"))

# Quick summary
summary(df_election)
```

|    |                       |                       |
|----|-----------------------|-----------------------|
| #> | moralb                | caresb                |
| #> | 1 Extremely well :340 | 1 Extremely well :155 |
| #> | 2 Quite well :841     | 2 Quite well :625     |
| #> | 3 Not too well :330   | 3 Not too well :562   |
| #> | 4 Not well at all: 98 | 4 Not well at all:342 |
| #> | NA's :176             | NA's :101             |
| #> | knowb                 | leadb                 |
| #> | 1 Extremely well :274 | 1 Extremely well :266 |
| #> | 2 Quite well :933     | 2 Quite well :842     |
| #> | 3 Not too well :379   | 3 Not too well :407   |
| #> | 4 Not well at all:133 | 4 Not well at all:166 |
| #> | NA's : 66             | NA's :104             |
| #> | dishonb               | intelb moralb1        |

```
#> 1 Extremely well : 70 1 Extremely well :329 1 :340
#> 2 Quite well :288 2 Quite well :967 2 :841
#> 3 Not too well :653 3 Not too well :306 3 :330
#> 4 Not well at all:574 4 Not well at all:110 4 : 98
#> NA's :200 NA's : 73 NA's:176
#> caresb1 knowb1 leadb1 dishonb1 intelb1
#> 1 :155 1 :274 1 :266 1 : 70 1 :329
#> 2 :625 2 :933 2 :842 2 :288 2 :967
#> 3 :562 3 :379 3 :407 3 :653 3 :306
#> 4 :342 4 :133 4 :166 4 :574 4 :110
#> NA's:101 NA's: 66 NA's:104 NA's:200 NA's: 73
```

## 5.3 Descriptive Statistics

```
ds <- df_election %>%
  pivot_longer(moralb1:intelb1, names_to = "variable") %>%
  count(variable, value) %>% # Count occurrences of each value for each variable
  group_by(variable) %>%
  mutate(prop = n / sum(n)) %>%
  arrange(desc(variable))

# Create the table
prop_table <- ds %>%
  gt() %>%
  tab_header(title = md("**Descriptive Summary**")) %>%
  cols_label(
    variable = "Variable",
    n = md("*N*"),
    prop = md("Proportion")
  ) %>%
  fmt_number(c("n", "prop"), decimals = 2) %>% # Format both n and prop columns
  cols_align(
    align = "center",
    columns = c(prop, n)
  )

# View the table
prop_table
```

```
# Save as a Word doc
#gtsave(prop_table, here("figures", "prop_table.docx"))
```

## 5.4 Enumeration

This code uses the `mplusObject` function in the `MplusAutomation` package.

```
lca_enumeration <- lapply(1:6, function(k) {
  lca_enum <- mplusObject(

    TITLE = glue("{k}-Class"),

    VARIABLE = glue(
      "categorical = moralb1-intelb1;
      usevar = moralb1-intelb1;
      classes = c({k}); "),

    ANALYSIS =
      "estimator = mlr;
      type = mixture;
      starts = 500 100;
      processors = 10;",

    OUTPUT = "sampstat residual tech11 tech14 svalues;",

    usevariables = colnames(df_election),
    rdata = df_election)

  lca_enum_fit <- mplusModeler(lca_enum,
                              dataout=glue(here("poLCA", "election.dat")),
                              modelout=glue(here("poLCA", "c{k}_election.inp")) ,
                              check=TRUE, run = TRUE, hashfilename = FALSE)
})
```

### 5.4.1 Table of Fit

```
source(here("functions","enum_table.R"))

output_election <- readModels(here("poLCA"), filefilter = "election", quiet = TRUE)

# To see rows:
#seeRows(output_election)

# Arguments for `enum_table`
# 1. readModels objects
# 2-5. Rows of successfully estimated models
fit_table <- enum_table(output_election, 1:6)
fit_table
```

---

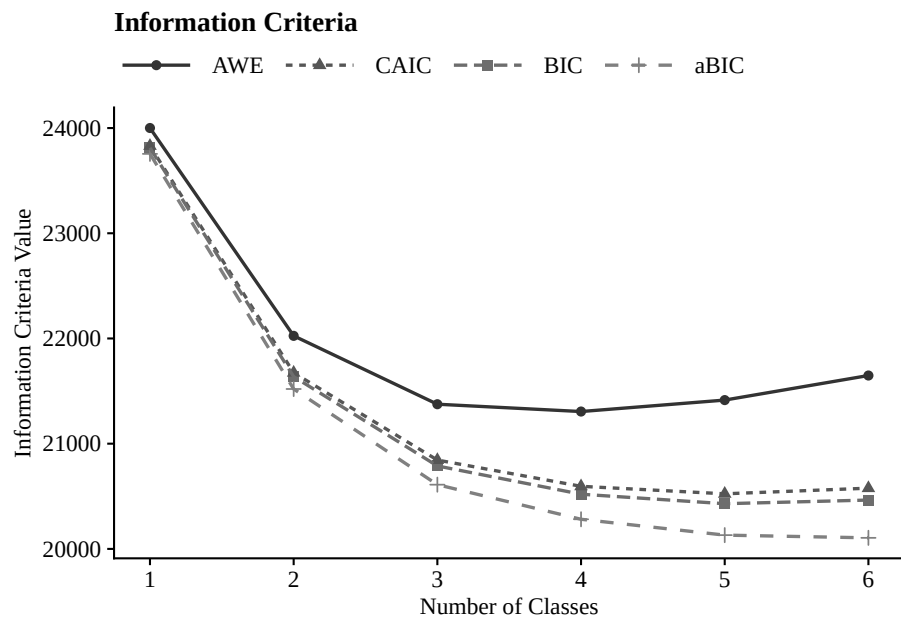
Save table:

```
gtsave(fit_table, here("figures", "fit_table.png"))
```

---

### 5.4.2 Information Criteria Plot

```
ic_plot(output_election)
```



---

Save figure:

```
ggsave(here("figures", "info_criteria.png"), dpi="retina", bg = "white", height=5, width=10)
```

---

## Descriptive Summary

| value    | $N$    | Proportion |
|----------|--------|------------|
| moralb1  |        |            |
| 1        | 340.00 | 0.19       |
| 2        | 841.00 | 0.47       |
| 3        | 330.00 | 0.18       |
| 4        | 98.00  | 0.05       |
| NA       | 176.00 | 0.10       |
| leadb1   |        |            |
| 1        | 266.00 | 0.15       |
| 2        | 842.00 | 0.47       |
| 3        | 407.00 | 0.23       |
| 4        | 166.00 | 0.09       |
| NA       | 104.00 | 0.06       |
| knowb1   |        |            |
| 1        | 274.00 | 0.15       |
| 2        | 933.00 | 0.52       |
| 3        | 379.00 | 0.21       |
| 4        | 133.00 | 0.07       |
| NA       | 66.00  | 0.04       |
| intelb1  |        |            |
| 1        | 329.00 | 0.18       |
| 2        | 967.00 | 0.54       |
| 3        | 306.00 | 0.17       |
| 4        | 110.00 | 0.06       |
| NA       | 73.00  | 0.04       |
| dishonb1 |        |            |
| 1        | 70.00  | 0.04       |
| 2        | 288.00 | 0.16       |
| 3        | 653.00 | 0.37       |
| 4        | 574.00 | 0.32       |
| NA       | 200.00 | 0.11       |
| caresb1  |        |            |
| 1        | 155.00 | 0.09       |
| 2        | 625.00 | 0.35       |
| 3        | 562.00 | 0.31       |
| 4        | 342.00 | 0.19       |
| NA       | 101.00 | 0.06       |

Model Fit Summary Table<sup>1</sup>

| Classes | Par | <i>LL</i>  | BIC              | aBIC             | CAIC             | AWE              | BLRT  |
|---------|-----|------------|------------------|------------------|------------------|------------------|-------|
| 1-Class | 18  | -11,838.63 | 23,811.84        | 23,754.66        | 23,829.84        | 24,000.42        | –     |
| 2-Class | 37  | -10,680.33 | 21,637.30        | 21,519.75        | 21,674.30        | 22,024.92        | <.001 |
| 3-Class | 56  | -10,184.89 | 20,788.46        | 20,610.55        | 20,844.46        | 21,375.14        | <.001 |
| 4-Class | 75  | -9,979.58  | 20,519.89        | 20,281.62        | 20,594.89        | <b>21,305.63</b> | <.001 |
| 5-Class | 94  | -9,863.42  | <b>20,429.62</b> | 20,130.99        | <b>20,523.62</b> | 21,414.41        | <.001 |
| 6-Class | 113 | -9,809.63  | 20,464.09        | <b>20,105.10</b> | 20,577.09        | 21,647.93        | <.001 |

<sup>1</sup>Note. Par = Parameters; *LL* = model log likelihood; BIC = Bayesian information criterion; aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion; AWE = approximate weight of evidence criterion; BLRT = bootstrapped likelihood ratio test p-value; cmPk = approximate correct model probability.



## Chapter 6

# Item Probability Plots

Polytomous LCA deals with variables that have more than two categories, such as survey questions with responses like **never**, **sometimes**, and **always**. The workflow of a polytomous LCA model is similar to that of an LCA model with binary indicators. However, polytomous LCA captures more complex response patterns, which can make interpretation more complex. The following code demonstrates an example, along with a visualization of the model.

---

### 6.1 Example: Elections

“Two sets of six questions with four responses each, asking respondents’ opinions of how well various traits describe presidential candidates Al Gore and George W. Bush. Also potential covariates vote choice, age, education, gender, and party ID. Source: The National Election Studies (2000).” (poLCA, 2016) See [documentation here](#)

Two sets of six questions with four responses each, asking respondents’ opinions of how well various traits describe presidential candidates Al Gore and George W. Bush. In the election data set, respondents to the 2000 American National Election Study public opinion poll were asked to evaluate how well a series of traits—moral, caring, knowledgeable, good leader, dishonest, and intelligent—described presidential candidates Al Gore and George W. Bush. Each question had four possible choices: (1) extremely well; (2) quite well; (3) not too well; and (4) not well at all.

Load packages

```
library(poLCA)
library(tidyverse)
library(janitor)
library(gt)
library(MplusAutomation)
library(here)
library(glue)
```

## 6.2 Prepare Data

```
data(election)

# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)

df_election <- election %>%
  clean_names() %>%
  select(moralb:intelb) %>%
  mutate(across(everything(),
    ~ as.factor(as.numeric(gsub("\\D", "", .))),
    .names = "{.col}1"))

# Quick summary
summary(df_election)
#>           moralb           caresb
#> 1 Extremely well :340  1 Extremely well :155
#> 2 Quite well     :841  2 Quite well     :625
#> 3 Not too well   :330  3 Not too well   :562
#> 4 Not well at all: 98  4 Not well at all:342
#> NA's            :176  NA's            :101
#>           knowb           leadb
#> 1 Extremely well :274  1 Extremely well :266
#> 2 Quite well     :933  2 Quite well     :842
#> 3 Not too well   :379  3 Not too well   :407
#> 4 Not well at all:133  4 Not well at all:166
#> NA's            : 66  NA's            :104
#>           dishonb           intelb           moralb1
```

```
#> 1 Extremely well : 70 1 Extremely well :329 1 :340
#> 2 Quite well :288 2 Quite well :967 2 :841
#> 3 Not too well :653 3 Not too well :306 3 :330
#> 4 Not well at all:574 4 Not well at all:110 4 : 98
#> NA's :200 NA's : 73 NA's:176
#> caresb1 knowb1 leadb1 dishonb1 intelb1
#> 1 :155 1 :274 1 :266 1 : 70 1 :329
#> 2 :625 2 :933 2 :842 2 :288 2 :967
#> 3 :562 3 :379 3 :407 3 :653 3 :306
#> 4 :342 4 :133 4 :166 4 :574 4 :110
#> NA's:101 NA's: 66 NA's:104 NA's:200 NA's: 73
```

## 6.3 Descriptive Statistics

```
ds <- df_election %>%
  pivot_longer(moralb1:intelb1, names_to = "variable") %>%
  count(variable, value) %>% # Count occurrences of each value for each variable
  group_by(variable) %>%
  mutate(prop = n / sum(n)) %>%
  arrange(desc(variable))

# Create the table
prop_table <- ds %>%
  gt() %>%
  tab_header(title = md("**Descriptive Summary**")) %>%
  cols_label(
    variable = "Variable",
    n = md("*N*"),
    prop = md("Proportion")
  ) %>%
  fmt_number(c("n", "prop"), decimals = 2) %>% # Format both n and prop columns
  cols_align(
    align = "center",
    columns = c(prop, n)
  )

# View the table
prop_table
```

```
# Save as a Word doc
#gtsave(prop_table, here("figures", "prop_table.docx"))
```

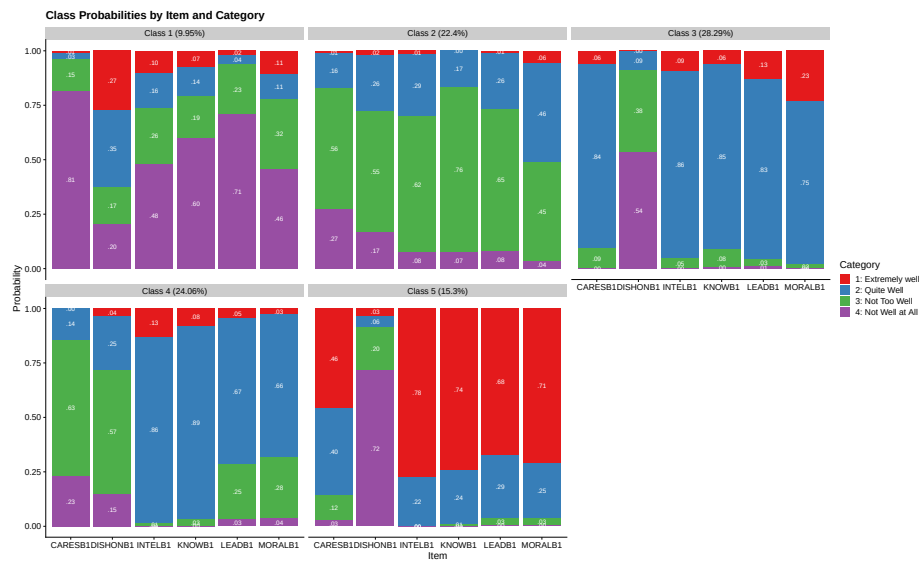
## 6.4 Item Probability Plot

The functions `poLCA_stacked` and `poLCA_grouped` create visualizations of class probabilities for LCA with polytomous indicators. Each function takes the following arguments:

- **model\_name**: The LCA model read into R using the `readModels` function from the `MplusAutomation` package.
- **category\_labels**: A character vector of category labels for the response options (e.g., survey answers).

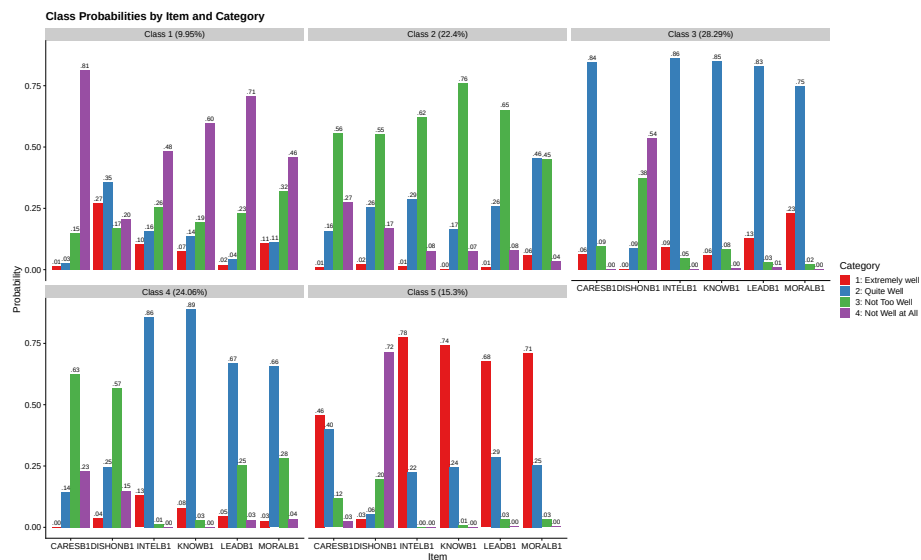
Note: Double check that the labels are in the correct order!

```
source(here("functions", "poLCA_plot.R"))  
  
# Read in models  
output_election <- readModels(here("poLCA"), filefilter = "election", quiet = TRUE)  
  
poLCA_stacked(output_election$c5_election.out, category_labels = c("1" = "1: Extremely  
                             "2" = "2: Quite Well  
                             "3" = "3: Not Too W  
                             "4" = "4: Not Well
```



Alternative plot

```
polCA_grouped(output_election$c5_election.out, category_labels = c("1" = "1: Extremely well",
                                                                    "2" = "2: Quite Well",
                                                                    "3" = "3: Not Too Well",
                                                                    "4" = "4: Not Well at All"))
```



## 6.5 APA-formatted Plot

```
# Model
model <- output_election$c5_election.out

# Title
title <- "2000 Descriptions of Presidential Candidate George W. Bush; Item Probabiliti

# Item names
item_labels <- c("CARES1" = "Caring",
                 "DISHON1" = "Dishonest",
                 "INTEL1" = "Intelligent",
                 "KNOW1" = "Knowledgeable",
                 "LEAD1" = "Good Leader",
                 "MORAL1" = "Moral")

# Item Category
category_labels <- c("1" = "1: Extremely well",
                    "2" = "2: Quite Well",
                    "3" = "3: Not Too Well",
                    "4" = "4: Not Well at All")

# Class labels
class_labels <- c("1" = "Poor Decsription (9.95%)",
                  "2" = "Mostly Poor Description (22.40%)",
                  "3" = "In-Between (24.06%)",
                  "4" = "Mostly Well-Described But Not Intelligent (28.29%)",
                  "5" = "Well-Described But Not Intelligent (15.30%)")

#### END EDIT ####
```

### 6.5.1 Extract data needed for plotting

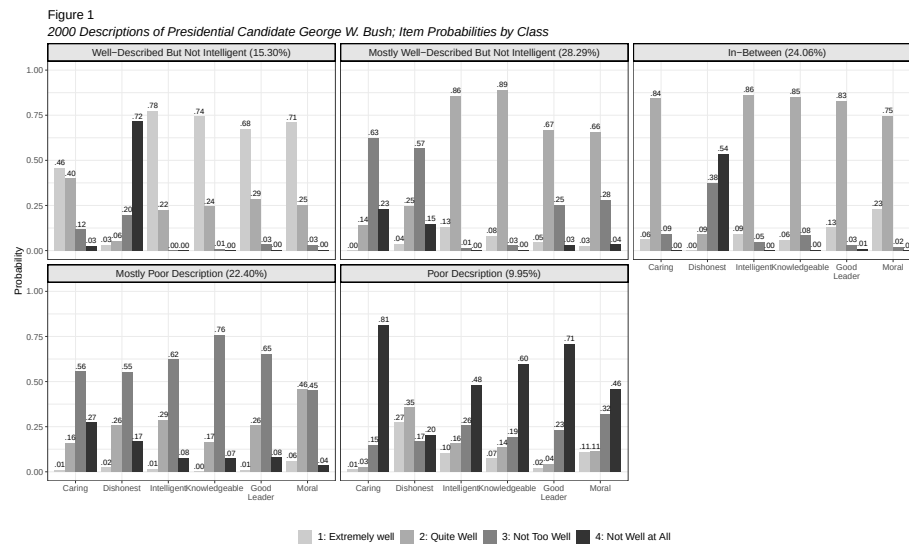
```
# Extract data needed for plotting
plot_data <- data.frame(model$parameters$probability.scale) %>%
  dplyr::select(est, LatentClass, param, category) %>%
  mutate(
    items = factor(param, labels = item_labels),
    class = factor(LatentClass, labels = class_labels),
    cat = factor(category, labels = category_labels)
```

```
) %>%
mutate(class = factor(class, levels = rev(levels(factor(class)))))
```

---

### 6.5.2 Final grouped bar plot

```
## Plot data
plot_data %>%
  ggplot(aes(
    x = items,
    y = est,
    fill = cat,
    group = cat
  )) +
  geom_bar(stat = "identity", position = "dodge") +
  geom_text(aes(label = sub("^0\\.\"", ".", sprintf("%.2f", est))),
    position = position_dodge(width = 0.9),
    vjust = -0.5, size = 3) +
  facet_wrap(~ class) +
  ylim(0, 1) +
  scale_x_discrete(
    "",
    labels = function(x)
      str_wrap(x, width = 10)
  ) +
  labs(title = "Figure 1",
    subtitle = title,
    y = "Probability") +
  theme_bw(12) +
  scale_fill_grey(start = 0.8, end = 0.2) + # Gives different shades
  theme(
    text = element_text(family = "sans", size = 12),
    legend.text = element_text(family = "sans", size = 12, color = "black"),
    legend.title = element_blank(),
    legend.position = "bottom",
    legend.justification = "center",
    axis.text.x = element_text(vjust = 1),
    plot.subtitle = element_text(face = "italic", size = 15),
    plot.title = element_text(size = 15),
    strip.background = element_rect(fill = "grey90", color = "black", size = 1),
    strip.text = element_text(size = 12)
  )
```



Save figure:

```
ggsave(here("figures", "APA_plot1.png"), dpi="retina", bg = "white", height=9, width=10)
```

### 6.5.3 Alternative

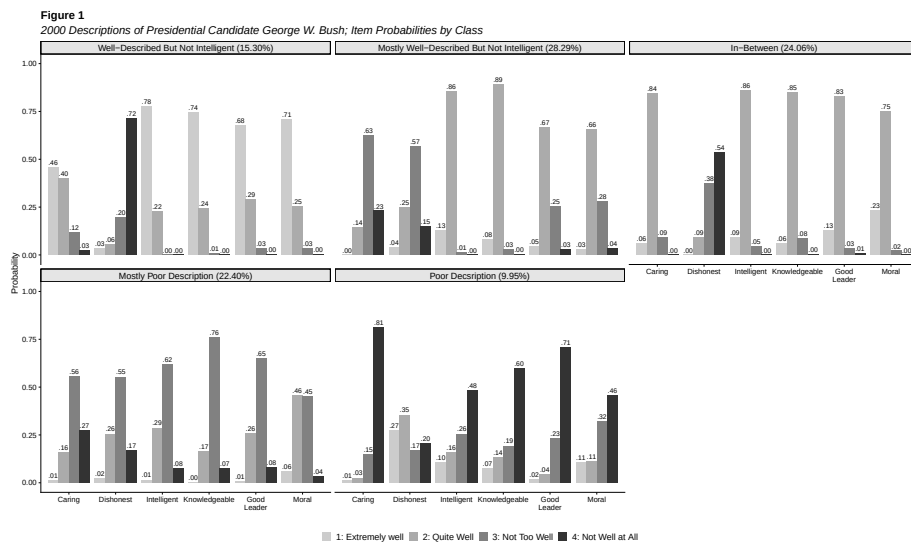
```
## Plot data
plot_data %>%
  ggplot(aes(
    x = items,
    y = est,
    fill = cat,
    group = cat
  )) +
  geom_bar(stat = "identity", position = "dodge") +
  geom_text(aes(label = sub("^0\\.?", ".", sprintf("%.2f", est))),
    position = position_dodge(width = 0.9),
    vjust = -0.5, size = 3) +
  facet_wrap(~ class) +
  ylim(0, 1) +
```



```

scale_x_discrete(
  "",
  labels = function(x)
    str_wrap(x, width = 10)
) +
labs(title = "Figure 1",
     subtitle = title,
     y = "Probability") +
theme_cowplot(12) +
scale_fill_grey(start = 0.8, end = 0.2) + # Gives different shades
theme(
  text = element_text(family = "sans", size = 12),
  legend.text = element_text(family = "sans", size = 12, color = "black"),
  legend.title = element_blank(),
  legend.position = "bottom",
  legend.justification = "center",
  axis.text.x = element_text(vjust = 1),
  plot.subtitle = element_text(face = "italic", size = 15),
  plot.title = element_text(size = 15),
  strip.background = element_rect(fill = "grey90", color = "black", size = 1),
  strip.text = element_text(size = 12)
)

```



Save figure:

```
ggsave(here("figures", "APA_plot2.png"), dpi="retina", bg = "white", height=10, width=
```

**Descriptive Summary**

| value    | <i>N</i> | Proportion |
|----------|----------|------------|
| moralb1  |          |            |
| 1        | 340.00   | 0.19       |
| 2        | 841.00   | 0.47       |
| 3        | 330.00   | 0.18       |
| 4        | 98.00    | 0.05       |
| NA       | 176.00   | 0.10       |
| leadb1   |          |            |
| 1        | 266.00   | 0.15       |
| 2        | 842.00   | 0.47       |
| 3        | 407.00   | 0.23       |
| 4        | 166.00   | 0.09       |
| NA       | 104.00   | 0.06       |
| knowb1   |          |            |
| 1        | 274.00   | 0.15       |
| 2        | 933.00   | 0.52       |
| 3        | 379.00   | 0.21       |
| 4        | 133.00   | 0.07       |
| NA       | 66.00    | 0.04       |
| intelb1  |          |            |
| 1        | 329.00   | 0.18       |
| 2        | 967.00   | 0.54       |
| 3        | 306.00   | 0.17       |
| 4        | 110.00   | 0.06       |
| NA       | 73.00    | 0.04       |
| dishonb1 |          |            |
| 1        | 70.00    | 0.04       |
| 2        | 288.00   | 0.16       |
| 3        | 653.00   | 0.37       |
| 4        | 574.00   | 0.32       |
| NA       | 200.00   | 0.11       |
| caresb1  |          |            |
| 1        | 155.00   | 0.09       |
| 2        | 625.00   | 0.35       |
| 3        | 562.00   | 0.31       |
| 4        | 342.00   | 0.19       |
| NA       | 101.00   | 0.06       |



## Part IV

# Latent Profile Analysis



## Chapter 7

# Enumeration

---

*Example: PISA Student Data*

1. The first example closely follows the vignette used to demonstrate the tidyLPA package (Rosenberg, 2019).
  - This model utilizes the PISA data collected in the U.S. in 2015. To learn more about this data see [here](#).
  - To access the 2015 US PISA data & documentation in R use the following code:

Variables:

**broad\_interest** composite measure of students' self reported broad interest

**enjoyment** composite measure of students' self reported enjoyment

**instrumental\_mot** composite measure of students' self reported instrumental motivation

**self\_efficacy** composite measure of students' self reported self efficacy

```
#devtools::install_github("jrosen48/pisaUSA15")  
#library(pisaUSA15)
```

## 7.1 Latent Profile Models

Latent Profile Analysis (LPA) is a statistical modeling approach for estimating distinct profiles of variables and uses continuous - rather than categorical - indicators. Unlike latent class analysis, which uses categorical indicators, LPA models the means, variances, and covariances of continuous indicators within each latent profile.

---

## 7.2 Terminology for specifying variance-covariance matrix

The code used to estimate LPA models in this walkthrough is from the `tidyLPA` package. `TidyLPA`(source) is an R package designed to estimate latent profile models using a tidy framework. It can interface with Mplus via the `MplusAutomation` package, enabling the estimation of latent profile models with different variance-covariance structures.

- `model 1` Profile-invariant / Diagonal: Equal variances, and covariances fixed to 0
- `model 2` Profile-varying / Diagonal: Free variances and covariances fixed to 0
- `model 3` Profile-invariant / Non-Diagonal: Equal variances and equal covariances
  - Note: an alternative to Model 3 is freely estimating the covariances
- `model 4` Free variances, and equal covariances
- `model 5` Equal variances, and free covariances
- `model 6` Profile Varying / Non-Diagonal: Free variances and free covariances

### 7.2.1 Model 1

*Profile-invariant/diagonal:*

- *Equal Variances:* Variances are fixed to equality across the profiles (i.e., variances are constrained to be equal for each profile).
- *Covariances fixed to zero* (i.e., the off-diagonal cells of the matrix are zero).



The most parsimonious model and the most restricted.

$$\begin{pmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}$$

### 7.2.2 Model 2

*Profile-varying/diagonal:*

- *Free variances:* Variances parameters are freely estimated across the profiles (i.e., variances vary by profile).
- *Covariances fixed to zero* (i.e., the off-diagonal cells of the matrix are zero).

This model is more flexible and less parsimonious than model 1.

$$\begin{pmatrix} \sigma_{1p}^2 & 0 & 0 \\ 0 & \sigma_{2p}^2 & 0 \\ 0 & 0 & \sigma_{3p}^2 \end{pmatrix}$$

### 7.2.3 Model 3

*Profile-invariant/ non-diagonal or unrestricted:*

- *Equal variances:* Variances are fixed to equality across profile. (i.e., variances are constrained to be same for each profile).
- *Equal Covariances:* The covariances are now estimated and constrained to be equal.
  - An alternative to Model 3 is freely estimating the covariances (Model 5 here).

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 \end{pmatrix}$$

### 7.2.4 Model 4

Varying means, varying variances, and equal covariances:

- *Free variances*: Variances parameters are freely estimated across profiles (i.e., variances vary by profile).
- *Equal Covariances*: Covariances are constrained to be equal.

This model is also considered to be an extension of Model 3.

$$\begin{pmatrix} \sigma_{1p}^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{2p}^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{3p}^2 \end{pmatrix}$$

### 7.2.5 Model 5

Varying means, equal variances, and varying covariances:

- *Equal variances*: Variances are fixed to equality across the profiles. (i.e., variances are constrained to be same for each profile).
- *Free Covariances*: Covariances are now freely estimated across the profiles.

This model is also considered to be an extension of Model 3.

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12p} & \sigma_{13p} \\ \sigma_{12p} & \sigma_2^2 & \sigma_{23p} \\ \sigma_{13p} & \sigma_{23p} & \sigma_3^2 \end{pmatrix}$$

### 7.2.6 Model 6

*Profile-varying / Non-diagonal*:

- *Free variances*: Variances parameters are freely estimated across profiles (i.e., variances vary by profile).
- *Free Covariances*: Covariances are now freely estimated across the profiles.

This is the most complex and unrestricted model. It is also the least parsimonious

*Note*: The unrestricted model is also sometimes known as Model 4.

$$\begin{pmatrix} \sigma_{1p}^2 & \sigma_{12p} & \sigma_{13p} \\ \sigma_{12p} & \sigma_{2p}^2 & \sigma_{23p} \\ \sigma_{13p} & \sigma_{23p} & \sigma_{3p}^2 \end{pmatrix}$$

---

## 7.3 Load packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

---

## 7.4 Prepare Data

```
pisa <- pisaUSA15[1:500,] %>%
  dplyr::select(broad_interest, enjoyment, instrumental_mot, self_efficacy)
```

---

## 7.5 Descriptive Statistics

Quick Summary

```
summary(pisa)
#> broad_interest      enjoyment      instrumental_mot
#> Min.      :1.000    Min.      :1.00    Min.      :1.000
#> 1st Qu.:2.200    1st Qu.:2.40    1st Qu.:1.750
#> Median :2.800    Median :3.00    Median :2.000
#> Mean   :2.666    Mean   :2.82    Mean   :2.129
#> 3rd Qu.:3.200    3rd Qu.:3.00    3rd Qu.:2.500
#> Max.   :5.000    Max.   :4.00    Max.   :4.000
#> NA's   :23      NA's   :14      NA's   :21
#> self_efficacy
#> Min.      :1.000
#> 1st Qu.:1.750
#> Median :2.000
#> Mean   :2.125
#> 3rd Qu.:2.500
#> Max.   :4.000
#> NA's   :23
```

Mean Table

```
ds <- pisa %>%
  pivot_longer(broad_interest:self_efficacy, names_to = "variable") %>%
  group_by(variable) %>%
  summarise(mean = mean(value, na.rm = TRUE),
            sd = sd(value, na.rm = TRUE))

ds %>%
  gt () %>%
  tab_header(title = md("**Descriptive Summary**")) %>%
  cols_label(
    variable = "Variable",
    mean = md("M"),
    sd = md("SD")
  ) %>%
  fmt_number(c(2:3),
             decimals = 2) %>%
  cols_align(
    align = "center",
    columns = mean
  )
```

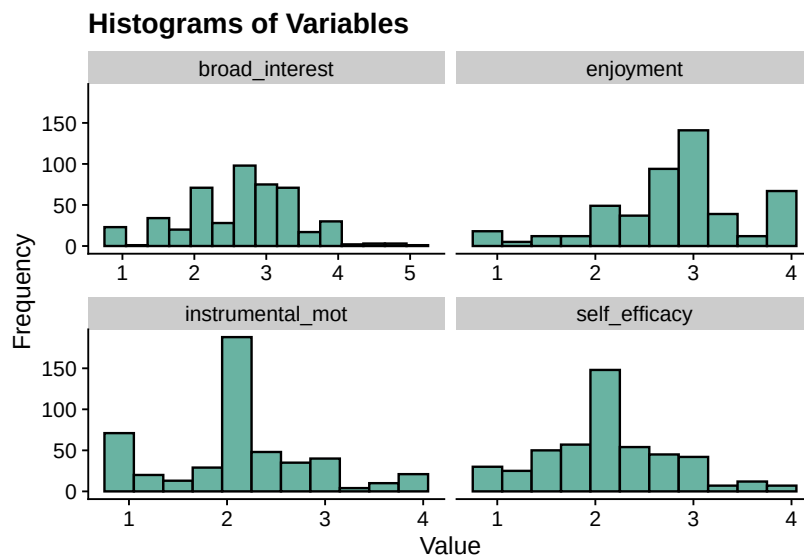
Histograms

### Descriptive Summary

| Variable         | M    | SD   |
|------------------|------|------|
| broad_interest   | 2.67 | 0.77 |
| enjoyment        | 2.82 | 0.72 |
| instrumental_mot | 2.13 | 0.75 |
| self_efficacy    | 2.12 | 0.64 |

```
data_long <- pisa %>%
  pivot_longer(broad_interest:self_efficacy, names_to = "variable")

ggplot(data_long, aes(x = value)) +
  geom_histogram(binwidth = .3, fill = "#69b3a2", color = "black") +
  facet_wrap(~ variable, scales = "free_x") +
  labs(title = "Histograms of Variables", x = "Value", y = "Frequency") +
  theme_cowplot()
```



## 7.6 Enumeration

### 7.6.1 tidyLPA

---

Enumerate using `estimate_profiles()`:

- Estimate models with profiles  $K = 1 : 5$
- Model has 4 continuous indicators
- Default variance-covariance specifications (model 1)
- Change `variances` and `covariances` to indicate the model you want to specify, in this example, we are estimating all six models.

```
# Run LPA models
lpa_fit <- pisa %>%
  estimate_profiles(1:5,
    package = "MplusAutomation",
    ANALYSIS = "starts = 500 100;",
    OUTPUT = "sampstat residual tech11 tech14",
    variances = c("equal", "varying", "equal", "varying", "equal", "varying"),
    covariances = c("zero", "zero", "equal", "equal", "varying", "varying"),
    keepfiles = TRUE)

# Compare fit statistics
get_fit(lpa_fit)

# Move files to folder
files <- list.files(here(), pattern = "~model")
move_files(files, here("lpa", "tidyLPA"), overwrite = TRUE)
```

---

### 7.6.2 Mplus

---

Alternative method to `estimate_profiles()`: Run enumeration using `mplusObject` method

You can change the model specification for LPA using the syntax provided in lecture.

## 7.6.2.1 Model 1

When estimating LPA in Mplus, the default variance/covariance specification is the most restricted model (Model 1). So we don't have to specify anything here.

```
lpa_k14 <- lapply(1:5, function(k) {
  lpa_enum <- mplusObject(

    TITLE = glue("Profile {k}"),

    VARIABLE = glue(
      "usevar = broad_interest-self_efficacy;
      classes = c({k}); "),

    ANALYSIS =
      "estimator = mlr;
      type = mixture;
      starts = 500 100;",

    OUTPUT = "sampstat svalues residual tech11 tech14;",

    usevariables = colnames(pisa),
    rdata = pisa)

  lpa_enum_fit <- mplusModeler(lpa_enum,
    dataout=glue(here("lpa", "enum_lpa", "lpa_pisa")),
    modelout=glue(here("lpa", "enum_lpa", "c{k}_lpa_m1.inp")),
    check=TRUE, run = TRUE, hashfilename = FALSE)
})
```

## 7.6.2.2 Model 2

Here, an addition loop adds the variance/covariance specifications for each class-specific statement. For the profile-varying/diagonal specification, you must specify the variances to be freely estimated:

```
broad_interest-self_efficacy;
```

```
lpa_m2_k14 <- lapply(1:5, function(k){

  # This MODEL section changes the model specification
  MODEL <- paste(sapply(1:k, function(i) {
    glue("
      %c#{i}%
```

```

    broad_interest-self_efficacy;      ! variances are freely estimated
  ")
}), collapse = "\n")

lpa_enum_m2 <- mplusObject(
  TITLE = glue("Profile {k} - Model 2"),

  VARIABLE = glue(
    "usevar = broad_interest-self_efficacy;
    classes = c({k});"),

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;",

  MODEL = MODEL,

  OUTPUT = "sampstat svalues residual tech11 tech14;",

  usevariables = colnames(pisa),
  rdata = pisa)

lpa_m2_fit <- mplusModeler(lpa_enum_m2,
  dataout = here("lpa", "enum_lpa", "lpa_pisa"),
  modelout = glue(here("lpa", "enum_lpa", "c{k}_lpa_m2.inp"),
  check = TRUE, run = TRUE, hashfilename = FALSE)
})

```

For reference, here is the Mplus syntax for different specifications:

**Fixed covariance to zero (DEFAULT):**

```
broad_interest WITH enjoyment@0;
```

**Free covariance:**

```
broad_interest WITH enjoyment;
```

**Equal covariances:**

```
%c#1%
```

```
broad_interest WITH enjoyment (1);
```

```
%c#2%
```

```
broad_interest WITH enjoyment (1);
```



**Equal variance (DEFAULT):**`%c#1%``broad_interest (1);``%c#2%``broad_interest (1);`**Free variance:**`mth_scor-bio_scor;`

You can also open the tidyLPA .inp files to see the specifications.

---

## 7.7 Table of Fit

Evaluate each model specification separately using fit indices.

After examining the outputs (and increasing the random starts where necessary), there are a few models excluded from analysis:

Model 2:

- Profile 4
- Profile 5

Model 3:

- Profile 5

Model 4:

- Profile 4
- Profile 5

Model 5:

- Profile 3
- Profile 4
- Profile 5

Model 6:

- Profile 4
- Profile 5

```
source(here("functions","enum_table_lpa.R"))

# Read in model
output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

# Preview with numbered rows
enum_fit(output_enum)
#> # A tibble: 29 x 12
#>   row Title      Parameters    LL    BIC  aBIC  CAIC  AWE
#>   <int> <chr>          <dbl> <dbl> <dbl> <dbl> <dbl>
#> 1     1 Model 1 ~      8 -2089. 4227. 4201. 4235. 4300.
#> 2     2 Model 1 ~     13 -1997. 4074. 4032. 4087. 4193.
#> 3     3 Model 1 ~     18 -1953. 4017. 3960. 4035. 4183.
#> 4     4 Model 1 ~     23 -1889. 3921. 3848. 3944. 4133.
#> 5     5 Model 1 ~     28 -1871. 3915. 3826. 3943. 4172.
#> 6     6 Model 2 ~      8 -2089. 4227. 4201. 4235. 4300.
#> 7     7 Model 2 ~     17 -1989. 4083. 4029. 4100. 4239.
#> 8     8 Model 2 ~     26 -1878. 3917. 3834. 3943. 4156.
#> 9     9 Model 2 ~     35 -1851. 3919. 3808. 3954. 4241.
#> 10    10 Model 2 ~     44 -1825. 3923. 3783. 3967. 4327.
#> # i 19 more rows
#> # i 4 more variables: BLRT_PValue <dbl>,
#> #   T11_VLMR_PValue <dbl>, BF <dbl>, cmPk <dbl>

select_models <- LatexSummaryTable(output_enum,
  keepCols=c("Title", "Parameters", "LL", "BIC", "aBIC",
    "BLRT_PValue", "T11_VLMR_PValue","Observations")) %>%
  slice( # Remove the models that we don't want to consider!!!! Because we looked at e
    # Model 2
    -9, -10,
    # Model 3
    -15,
    # Model 4
    -19, -20,
    # Model 5
    -23, -24, -25,
    # Model 6
    -29, -30
  )
```

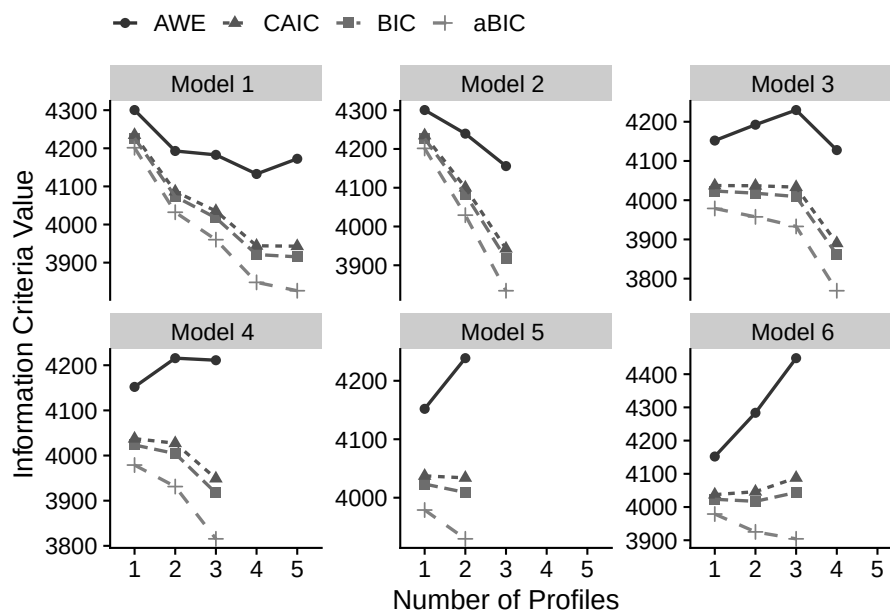
```
# Check to make sure that the rows of the models we don't want are removed
#View(select_models)

enum_table(select_models, 1:5, 6:8, 9:12, 13:15, 16:17, 18:20)
```

## 7.8 Information Criteria Plot

Look for “elbow” to help with profile selection

```
source(here("functions", "ic_plot_lpa.R"))
ic_plot(select_models)
```



Based on fit indices, I am choosing the following candidate models:

1. Model 1: 2 Profile
2. Model 2: 3 Profile
3. Model 3: 4 Profile
4. Model 4: 3 Profile
5. Model 5: 2 Profile

## 6. Model 6: 2 Profile

## 7.9 Compare models

### 7.9.1 Correct Model Probability (cmpK) recalculation

Take the candidate models and recalculate the approximate correct model probabilities (Masyn, 2013)

```
# CmpK recalculation:
enum_fit1 <- enum_fit(output_enum)

stage2_cmpk <- enum_fit1 %>%
  slice(2, 8, 14, 18, 22, 27) %>%
  mutate(SIC = -.5 * BIC,
         expSIC = exp(SIC - max(SIC)),
         cmPk = expSIC / sum(expSIC),
         BF = exp(SIC - lead(SIC))) %>%
  select(Title, Parameters, BIC:AWE, cmPk, BF)

# Format Fit Table
stage2_cmpk %>%
  gt() %>%
  tab_options(column_labels.font.weight = "bold") %>%
  fmt_number(
    7,
    decimals = 2,
    drop_trailing_zeros = TRUE,
    suffixing = TRUE
  ) %>%
  fmt_number(c(3:6),
             decimals = 2) %>%
  fmt_number(8, decimals = 2,
             drop_trailing_zeros=TRUE,
             suffixing = TRUE) %>%
  fmt(8, fns = function(x)
    ifelse(x>100, ">100",
           scales::number(x, accuracy = .1))) %>%
  tab_style(
    style = list(
      cell_text(weight = "bold")
```

```

    ),
    locations = list(cells_body(
      columns = BIC,
      row = BIC == min(BIC[1:nrow(stage2_cmpk)])
    ),
    cells_body(
      columns = aBIC,
      row = aBIC == min(aBIC[1:nrow(stage2_cmpk)])
    ),
    cells_body(
      columns = CAIC,
      row = CAIC == min(CAIC[1:nrow(stage2_cmpk)])
    ),
    cells_body(
      columns = AWE,
      row = AWE == min(AWE[1:nrow(stage2_cmpk)])
    ),
    cells_body(
      columns = cmPk,
      row = cmPk == max(cmPk[1:nrow(stage2_cmpk)])
    ),
    cells_body(
      columns = BF,
      row = BF > 10)
  )
)

```

### 7.9.2 Compare loglikelihood

You can also compare models using nested model testing directly with `MplusAutomation`. Note that you can only compare across models but the profiles must stay the same.

```

# MplusAutomation Method using `compareModels`

compareModels(output_enum[["model_2_class_3.out"]],
  output_enum[["model_4_class_3.out"]], diffTest = TRUE)
#>
#> =====
#>
#> Mplus model comparison
#> -----
#>
#> -----

```

```

#> Model 1: C:\Users\dna\arch\Box\lca-bookdown\lpa\tidyLPA\model_2_class_3.out
#> Model 2: C:\Users\dna\arch\Box\lca-bookdown\lpa\tidyLPA\model_4_class_3.out
#> -----
#>
#> Model Summary Comparison
#> -----
#>
#>
#>
#>      m1                      m2
#> Title      model 2 with 3 classes model 4 with 3 classes
#> Observations 488                      488
#> Estimator    MLR                      MLR
#> Parameters    26                      32
#> LL           -1877.965                 -1859.466
#> AIC          3807.929                 3782.932
#> BIC          3916.877                 3917.022
#>
#> MLR Chi-Square Difference Test for Nested Models Based on Loglikelihood
#> -----
#>
#> Difference Test Scaling Correction: 1.177167
#> Chi-square difference: 31.4297
#> Diff degrees of freedom: 6
#> P-value: 0
#>
#> Note: The chi-square difference test assumes that these models are nested.
#> It is up to you to verify this assumption.
#>
#> MLR Chi-Square Difference test for nested models
#> -----
#>
#> Difference Test Scaling Correction:
#> Chi-square difference:
#> Diff degrees of freedom:
#> P-value:
#>
#> Note: The chi-square difference test assumes that these models are nested.
#> It is up to you to verify this assumption.
#>
#> =====
#>
#> Model parameter comparison
#> -----
#> Parameters present in both models
#> =====
#>

```

```

#> Approximately equal in both models (param. est. diff <= 1e-04)
#> -----
#> paramHeader      param LatentClass m1_est m2_est . m1_se
#>      Means ENJOYMENT          1  2.968  2.968 | 0.012
#> m2_se . m1_est_se m2_est_se . m1_pval m2_pval
#> 0.021 | 238.242 143.432 | 0 0
#>
#>
#> Parameter estimates that differ between models (param. est. diff > 1e-04)
#> -----
#> paramHeader      param                               LatentClass
#> BROAD_IN.WITH ENJOYMENT                               1
#> BROAD_IN.WITH ENJOYMENT                               2
#> BROAD_IN.WITH ENJOYMENT                               3
#> BROAD_IN.WITH INSTRUMENT                              1
#> BROAD_IN.WITH INSTRUMENT                              2
#> BROAD_IN.WITH INSTRUMENT                              3
#> BROAD_IN.WITH SELF_EFFIC                              1
#> BROAD_IN.WITH SELF_EFFIC                              2
#> BROAD_IN.WITH SELF_EFFIC                              3
#> ENJOYMEN.WITH INSTRUMENT                              1
#> ENJOYMEN.WITH INSTRUMENT                              2
#> ENJOYMEN.WITH INSTRUMENT                              3
#> ENJOYMEN.WITH SELF_EFFIC                              1
#> ENJOYMEN.WITH SELF_EFFIC                              2
#> ENJOYMEN.WITH SELF_EFFIC                              3
#> INSTRUME.WITH SELF_EFFIC                              1
#> INSTRUME.WITH SELF_EFFIC                              2
#> INSTRUME.WITH SELF_EFFIC                              3
#>      Means BROAD_INTE                                1
#>      Means BROAD_INTE                                2
#>      Means BROAD_INTE                                3
#>      Means      C1#1 Categorical.Latent.Variables
#>      Means      C1#2 Categorical.Latent.Variables
#>      Means ENJOYMENT                                  2
#>      Means ENJOYMENT                                  3
#>      Means INSTRUMENT                                  1
#>      Means INSTRUMENT                                  2
#>      Means INSTRUMENT                                  3
#>      Means SELF_EFFIC                                  1
#>      Means SELF_EFFIC                                  2
#>      Means SELF_EFFIC                                  3
#>      Variances BROAD_INTE                              1
#>      Variances BROAD_INTE                              2
#>      Variances BROAD_INTE                              3

```

```

#>      Variances  ENJOYMENT      1
#>      Variances  ENJOYMENT      2
#>      Variances  ENJOYMENT      3
#>      Variances INSTRUMENT      1
#>      Variances INSTRUMENT      2
#>      Variances INSTRUMENT      3
#>      Variances SELF_EFFIC      1
#>      Variances SELF_EFFIC      2
#>      Variances SELF_EFFIC      3
#> m1_est m2_est . m1_se m2_se . m1_est_se m2_est_se .
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |
#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> 2.912 2.926 | 0.049 0.060 | 59.366 49.094 |
#> 3.205 3.264 | 0.086 0.126 | 37.165 25.960 |
#> 2.257 2.159 | 0.082 0.114 | 27.478 18.918 |
#> -0.285 0.143 | 0.188 0.341 | -1.517 0.418 |
#> -0.891 -1.085 | 0.183 0.271 | -4.877 -4.013 |
#> 3.808 3.948 | 0.048 0.016 | 80.133 251.321 |
#> 2.305 2.271 | 0.072 0.149 | 32.038 15.214 |
#> 2.042 1.991 | 0.082 0.051 | 24.923 39.148 |
#> 1.735 1.801 | 0.092 0.110 | 18.769 16.401 |
#> 2.357 2.400 | 0.068 0.093 | 34.505 25.743 |
#> 2.072 2.086 | 0.057 0.044 | 36.269 47.523 |
#> 1.724 1.758 | 0.067 0.076 | 25.718 23.035 |
#> 2.330 2.294 | 0.049 0.074 | 47.308 30.997 |
#> 0.262 0.279 | 0.056 0.053 | 4.671 5.292 |
#> 0.405 0.384 | 0.104 0.222 | 3.887 1.729 |
#> 0.594 0.578 | 0.083 0.099 | 7.189 5.844 |
#> 0.010 0.051 | 0.003 0.019 | 3.283 2.630 |

```



```

#> 0.060 0.011 | 0.016 0.004 | 3.701 3.089 |
#> 0.397 0.448 | 0.044 0.070 | 9.011 6.373 |
#> 0.358 0.312 | 0.119 0.061 | 3.008 5.143 |
#> 0.636 0.797 | 0.134 0.178 | 4.738 4.479 |
#> 0.560 0.654 | 0.069 0.069 | 8.137 9.507 |
#> 0.298 0.328 | 0.038 0.032 | 7.771 10.250 |
#> 0.309 0.377 | 0.048 0.067 | 6.484 5.626 |
#> 0.435 0.449 | 0.045 0.054 | 9.689 8.370 |
#> m1_pval m2_pval
#> 999.000 0.194
#> 999.000 0.194
#> 999.000 0.194
#> 999.000 0.596
#> 999.000 0.596
#> 999.000 0.596
#> 999.000 0.197
#> 999.000 0.197
#> 999.000 0.197
#> 999.000 0.024
#> 999.000 0.024
#> 999.000 0.024
#> 999.000 0.000
#> 999.000 0.000
#> 999.000 0.000
#> 999.000 0.005
#> 999.000 0.005
#> 999.000 0.005
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.129 0.676
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.084
#> 0.000 0.000
#> 0.001 0.009
#> 0.000 0.002

```

```

#> 0.000 0.000
#> 0.003 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#> 0.000 0.000
#>
#>
#> P-values that differ between models (p-value diff > 1e-04)
#> -----
#> paramHeader      param      LatentClass
#> BROAD_IN.WITH ENJOYMENT      1
#> BROAD_IN.WITH ENJOYMENT      2
#> BROAD_IN.WITH ENJOYMENT      3
#> BROAD_IN.WITH INSTRUMENT      1
#> BROAD_IN.WITH INSTRUMENT      2
#> BROAD_IN.WITH INSTRUMENT      3
#> BROAD_IN.WITH SELF_EFFIC      1
#> BROAD_IN.WITH SELF_EFFIC      2
#> BROAD_IN.WITH SELF_EFFIC      3
#> ENJOYMEN.WITH INSTRUMENT      1
#> ENJOYMEN.WITH INSTRUMENT      2
#> ENJOYMEN.WITH INSTRUMENT      3
#> ENJOYMEN.WITH SELF_EFFIC      1
#> ENJOYMEN.WITH SELF_EFFIC      2
#> ENJOYMEN.WITH SELF_EFFIC      3
#> INSTRUME.WITH SELF_EFFIC      1
#> INSTRUME.WITH SELF_EFFIC      2
#> INSTRUME.WITH SELF_EFFIC      3
#> Means      C1#1 Categorical.Latent.Variables
#> Variances BROAD_INTE      2
#> Variances ENJOYMENT      1
#> Variances ENJOYMENT      2
#> Variances INSTRUMENT      1
#> m1_est m2_est . m1_se m2_se . m1_est_se m2_est_se .
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 0.010 | 0.000 0.008 | 999.000 1.300 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.013 | 0.000 0.024 | 999.000 -0.530 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |
#> 0.000 -0.029 | 0.000 0.023 | 999.000 -1.290 |

```

```

#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.034 | 0.000 0.015 | 999.000 -2.252 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 -0.029 | 0.000 0.008 | 999.000 -3.702 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> 0.000 0.056 | 0.000 0.020 | 999.000 2.777 |
#> -0.285 0.143 | 0.188 0.341 | -1.517 0.418 |
#> 0.405 0.384 | 0.104 0.222 | 3.887 1.729 |
#> 0.010 0.051 | 0.003 0.019 | 3.283 2.630 |
#> 0.060 0.011 | 0.016 0.004 | 3.701 3.089 |
#> 0.358 0.312 | 0.119 0.061 | 3.008 5.143 |
#> m1_pval m2_pval
#> 999.000 0.194
#> 999.000 0.194
#> 999.000 0.194
#> 999.000 0.596
#> 999.000 0.596
#> 999.000 0.596
#> 999.000 0.197
#> 999.000 0.197
#> 999.000 0.197
#> 999.000 0.024
#> 999.000 0.024
#> 999.000 0.024
#> 999.000 0.000
#> 999.000 0.000
#> 999.000 0.000
#> 999.000 0.005
#> 999.000 0.005
#> 999.000 0.005
#> 0.129 0.676
#> 0.000 0.084
#> 0.001 0.009
#> 0.000 0.002
#> 0.003 0.000
#>
#>
#> Parameters unique to model 1: 0
#> -----
#>
#> None
#>

```

```
#>
#>   Parameters unique to model 2: 0
#>   -----
#>
#>   None
#>
#>
#>   =====
```

Here, Model 1 (restricted, fewer parameters) is nested in Model 2. The chi-square difference test, assuming nested models, shows a significant improvement in fit for Model 2 over Model 1, despite the added parameters.

---

**Model Fit Summary Table<sup>1</sup>**

| Classes                | Par | <i>LL</i> | BIC             | aBIC            | CAIC            | AWE             | BLRT  |
|------------------------|-----|-----------|-----------------|-----------------|-----------------|-----------------|-------|
| Model 1                |     |           |                 |                 |                 |                 |       |
| Model 1 With 1 Classes | 8   | -2,088.66 | 4,226.84        | 4,201.45        | 4,234.84        | 4,300.36        | –     |
| Model 1 With 2 Classes | 13  | -1,996.51 | 4,073.50        | 4,032.24        | 4,086.50        | 4,192.97        | <.001 |
| Model 1 With 3 Classes | 18  | -1,952.98 | 4,017.38        | 3,960.25        | 4,035.38        | 4,182.81        | <.001 |
| Model 1 With 4 Classes | 23  | -1,889.43 | 3,921.23        | 3,848.23        | 3,944.23        | <b>4,132.61</b> | <.001 |
| Model 1 With 5 Classes | 28  | -1,870.91 | <b>3,915.16</b> | <b>3,826.28</b> | <b>3,943.15</b> | 4,172.48        | <.001 |
| Model 2                |     |           |                 |                 |                 |                 |       |
| Model 2 With 1 Classes | 8   | -2,088.66 | 4,226.84        | 4,201.45        | 4,234.84        | 4,300.36        | –     |
| Model 2 With 2 Classes | 17  | -1,988.99 | 4,083.22        | 4,029.26        | 4,100.22        | 4,239.45        | <.001 |
| Model 2 With 3 Classes | 26  | -1,877.96 | <b>3,916.88</b> | <b>3,834.35</b> | <b>3,942.88</b> | <b>4,155.83</b> | <.001 |
| Model 3                |     |           |                 |                 |                 |                 |       |
| Model 3 With 1 Classes | 14  | -1,968.35 | 4,023.36        | 3,978.93        | <b>4,037.36</b> | <b>4,152.02</b> | –     |
| Model 3 With 2 Classes | 19  | -1,950.11 | 4,017.84        | 3,957.53        | 4,036.84        | 4,192.45        | <.001 |
| Model 3 With 3 Classes | 24  | -1,930.36 | 4,009.29        | 3,933.12        | 4,033.29        | 4,229.86        | <.001 |
| Model 3 With 4 Classes | 29  | -1,840.85 | <b>3,861.23</b> | <b>3,769.18</b> | <b>3,890.23</b> | <b>4,127.75</b> | <.001 |
| Model 4                |     |           |                 |                 |                 |                 |       |
| Model 4 With 1 Classes | 14  | -1,968.35 | 4,023.36        | 3,978.93        | <b>4,037.36</b> | <b>4,152.02</b> | –     |
| Model 4 With 2 Classes | 23  | -1,930.96 | 4,004.30        | 3,931.30        | 4,027.30        | 4,215.67        | <.001 |
| Model 4 With 3 Classes | 32  | -1,859.47 | <b>3,917.02</b> | <b>3,815.45</b> | <b>3,949.02</b> | 4,211.11        | <.001 |
| Model 5                |     |           |                 |                 |                 |                 |       |
| Model 5 With 1 Classes | 14  | -1,968.35 | 4,023.36        | 3,978.93        | <b>4,037.36</b> | <b>4,152.02</b> | –     |
| Model 5 With 2 Classes | 25  | -1,927.10 | <b>4,008.95</b> | <b>3,929.60</b> | <b>4,033.95</b> | 4,238.71        | <.001 |
| Model 6                |     |           |                 |                 |                 |                 |       |
| Model 6 With 1 Classes | 14  | -1,968.35 | 4,023.36        | 3,978.93        | <b>4,037.36</b> | <b>4,152.02</b> | –     |
| Model 6 With 2 Classes | 29  | -1,918.84 | <b>4,017.19</b> | 3,925.15        | 4,046.19        | 4,283.71        | <.001 |
| Model 6 With 3 Classes | 44  | -1,885.73 | 4,043.84        | <b>3,904.18</b> | 4,087.83        | 4,448.21        | <.001 |

<sup>1</sup>Note. Par = Parameters; *LL* = model log likelihood; BIC = Bayesian information criterion; aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion; AWE = approximate weight of evidence criterion; BLRT = bootstrapped likelihood ratio test p-value; cmPk = approximate correct model probability.

| Title                  | Parameters | BIC             | aBIC            | CAIC            | AWE             | cr |
|------------------------|------------|-----------------|-----------------|-----------------|-----------------|----|
| Model 1 With 2 Classes | 13         | 4,073.50        | 4,032.24        | 4,086.50        | 4,192.97        |    |
| Model 2 With 3 Classes | 26         | 3,916.88        | 3,834.35        | 3,942.88        | 4,155.83        |    |
| Model 3 With 4 Classes | 29         | <b>3,861.23</b> | <b>3,769.18</b> | <b>3,890.23</b> | <b>4,127.75</b> |    |
| Model 4 With 3 Classes | 32         | 3,917.02        | 3,815.45        | 3,949.02        | 4,211.11        |    |
| Model 5 With 2 Classes | 25         | 4,008.95        | 3,929.60        | 4,033.95        | 4,238.71        |    |
| Model 6 With 2 Classes | 29         | 4,017.19        | 3,925.15        | 4,046.19        | 4,283.71        |    |

## Chapter 8

# Item Mean Plots

---

*Example: PISA Student Data*

1. The first example closely follows the vignette used to demonstrate the tidyLPA package (Rosenberg, 2019).
  - This model utilizes the PISA data collected in the U.S. in 2015. To learn more about this data see [here](#).
  - To access the 2015 US PISA data & documentation in R use the following code:

Variables:

**broad\_interest** composite measure of students' self reported broad interest

**enjoyment** composite measure of students' self reported enjoyment

**instrumental\_mot** composite measure of students' self reported instrumental motivation

**self\_efficacy** composite measure of students' self reported self efficacy

```
#devtools::install_github("jrosen48/pisaUSA15")  
#library(pisaUSA15)
```

## 8.1 Load packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

---

## 8.2 Prepare Data

```
pisa <- pisaUSA15[1:500,] %>%
  dplyr::select(broad_interest, enjoyment, instrumental_mot, self_efficacy)
```

---

## 8.3 Descriptive Statistics

Quick Summary

```
summary(pisa)
#> broad_interest      enjoyment      instrumental_mot
#> Min.      :1.000    Min.      :1.00    Min.      :1.000
#> 1st Qu.:2.200    1st Qu.:2.40    1st Qu.:1.750
#> Median :2.800    Median :3.00    Median :2.000
#> Mean    :2.666    Mean    :2.82    Mean    :2.129
#> 3rd Qu.:3.200    3rd Qu.:3.00    3rd Qu.:2.500
#> Max.    :5.000    Max.    :4.00    Max.    :4.000
```



## Descriptive Summary

| Variable         | M    | SD   |
|------------------|------|------|
| broad_interest   | 2.67 | 0.77 |
| enjoyment        | 2.82 | 0.72 |
| instrumental_mot | 2.13 | 0.75 |
| self_efficacy    | 2.12 | 0.64 |

```
#> NA's :23      NA's :14      NA's :21
#> self_efficacy
#> Min. :1.000
#> 1st Qu.:1.750
#> Median :2.000
#> Mean :2.125
#> 3rd Qu.:2.500
#> Max. :4.000
#> NA's :23
```

Mean Table

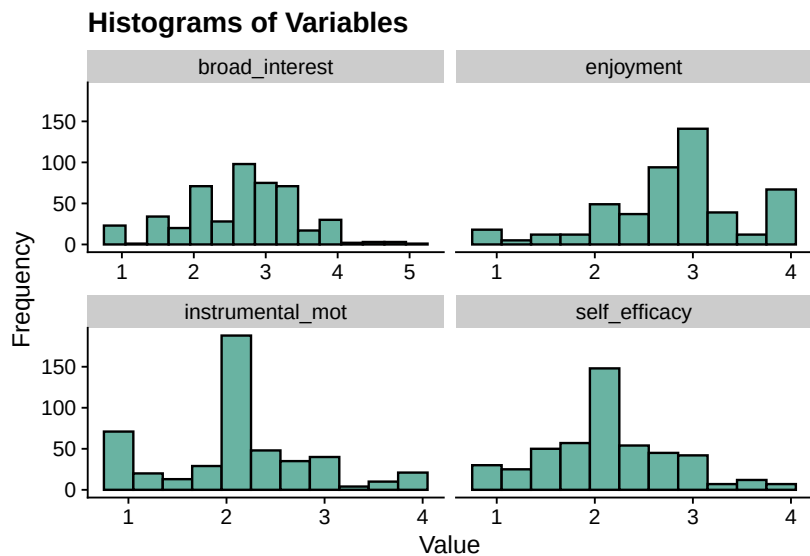
```
ds <- pisa %>%
  pivot_longer(broad_interest:self_efficacy, names_to = "variable") %>%
  group_by(variable) %>%
  summarise(mean = mean(value, na.rm = TRUE),
            sd = sd(value, na.rm = TRUE))

ds %>%
  gt () %>%
  tab_header(title = md("**Descriptive Summary**")) %>%
  cols_label(
    variable = "Variable",
    mean = md("M"),
    sd = md("SD")
  ) %>%
  fmt_number(c(2:3),
             decimals = 2) %>%
  cols_align(
    align = "center",
    columns = mean
  )
```

Histograms

```
data_long <- pisa %>%
  pivot_longer(broad_interest:self_efficacy, names_to = "variable")

ggplot(data_long, aes(x = value)) +
  geom_histogram(binwidth = .3, fill = "#69b3a2", color = "black") +
  facet_wrap(~ variable, scales = "free_x") +
  labs(title = "Histograms of Variables", x = "Value", y = "Frequency") +
  theme_cowplot()
```



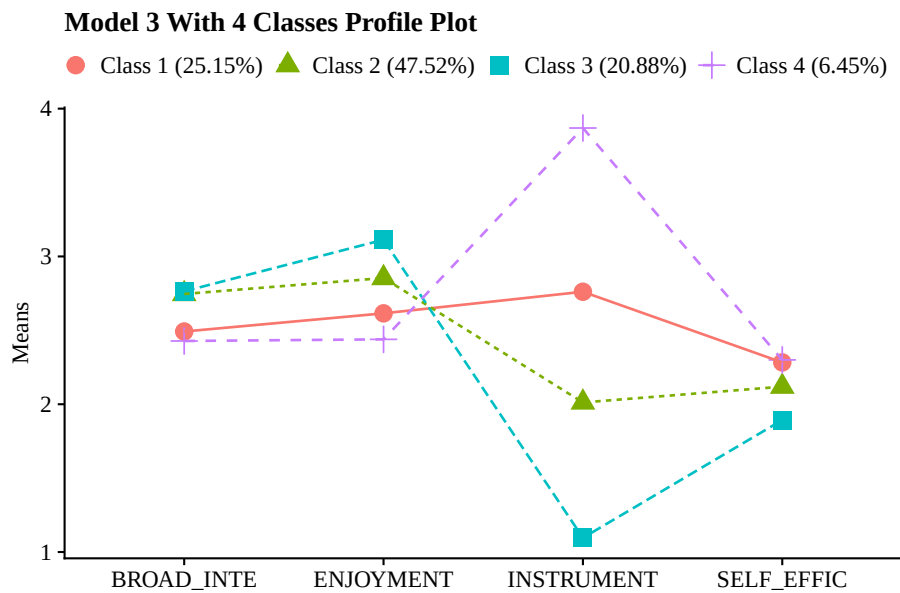
## 8.4 Visualization

### 8.4.1 Latent Profile Plot

```
source(here("functions", "plot_lpa.R"))

# Read in models
output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

plot_lpa(model_name = output_enum$model_3_class_4.out)
```

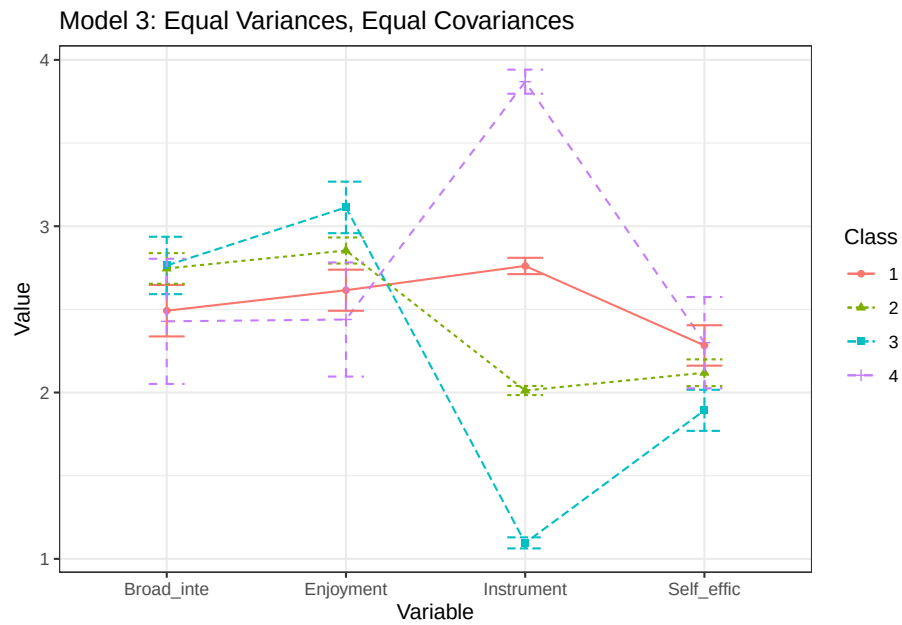


Save figure

```
ggsave(here("figures", "model3_profile4.png"), dpi = "retina", bg = "white", height=5, width=8, u
```

### 8.4.2 Plots Means and Variances

```
plotMixtures(output_enum$model_3_class_4.out, ci = 0.95, bw = FALSE) +  
  labs(title = "Model 3: Equal Variances, Equal Covariances")
```



### 8.4.3 Plot comparison

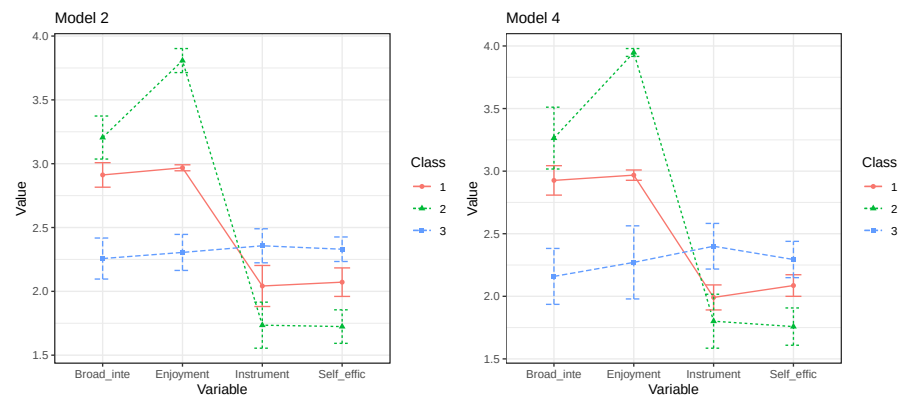
We can also plot the comparisons and look at the error bars.

NOTE: The `plotMixtures()` function is used for plotting LPA models only (i.e., means & variances)

```
a <- plotMixtures(output_enum$model_2_class_3.out,
  ci = 0.95, bw = FALSE)

b <- plotMixtures(output_enum$model_4_class_3.out,
  ci = 0.95, bw = FALSE)

a + labs(title = "Model 2") +
  theme(plot.title = element_text(size = 12)) +
b + labs(title = "Model 4") +
  theme(plot.title = element_text(size = 12))
```





## Part V

# Understanding the Data





## Chapter 9

# Observed Response Patterns

---

### 9.1 Load Packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

## 9.2 Prepare Data

```
df_bully <- read_csv(here("data", "crdc_lca_data.csv")) %>%
  clean_names() %>%
  dplyr::select(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_ft
```

---

Continuing the LCA example (??) in this bookdown, save response frequencies for the 3-class model with `response` is `____.dat` under `SAVEDATA`.

```
patterns <- mplusObject(

  TITLE = "C3 LCA - Save response patterns",

  VARIABLE =
    "categorical = report_dis-law_fte;
     usevar = report_dis-law_fte;
     classes = c(3);",

  ANALYSIS =
    "estimator = mlr;
     type = mixture;
     starts = 0;
     processors = 10;
     optseed = 802779;",

  SAVEDATA =
    "File=savedata.dat;
     Save=cprob;

     ! Code to save response frequency data

     response is resp_patterns.dat;",

  OUTPUT = "residual patterns tech11 tech14",

  usevariables = colnames(df_bully),
  rdata = df_bully)

patterns_fit <- mplusModeler(patterns,
  dataout=here("mplus", "bully.dat"),
  modelout=here("mplus", "patterns.inp") ,
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

---

Read in observed response pattern data and relabel the columns

```
# Read in response frequency data that we just created:
patterns <- read_table(here("mplus", "resp_patterns.dat"),
  col_names=FALSE, na = "*")

# Extract the column names
names <- names(readModels(here("mplus", "patterns.out"))[['savedata']])

# Add the names back to the dataset
colnames(patterns) <- c("Frequency", names)

# Rename the variable
patterns <- patterns %>%
  rename(C = dplyr::any_of(c("C", "MLCC")))
```

Create a table with the top 5 unconditional response pattern, then top of conditional response pattern for each modal class assignment

```
# Order responses by highest frequency
order_highest <- patterns %>%
  rename(C = dplyr::any_of(c("C", "MLCC"))) %>%
  arrange(desc(Frequency))

# Loop `patterns` data to list top 5 conditional response patterns for each class
loop_cond <- lapply(1:max(patterns$C), function(k) {
  order_cond <- patterns %>%
    filter(C == k) %>%
    arrange(desc(Frequency)) %>%
    head(5)
})

# Convert loop into data frame
table_data <- as.data.frame(bind_rows(loop_cond))

# Combine unconditional and conditional responses patterns
response_patterns <- rbind(order_highest[1:5,], table_data)
```

Finally, use {gt} to make a nicely formatted table

```
resp_table <- response_patterns %>%
  gt() %>%
```

```

tab_header(
  title = "Observed Response Patterns",
  subtitle = html("Response patterns, estimated frequencies, estimated posterior cla
tab_source_note(
  source_note = md("Data Source: **Civil Rights Data Collection (CRDC)**")) %>%
cols_label(
  Frequency = html("<i>f</i><sub>r</sub>"),
  REPORT_D = "Harrassment: Disability",
  REPORT_R = "Harrassment: Race",
  REPORT_S = "Harrassment: Sex",
  COUNSELO = "Staff: Counselor",
  PSYCH_FT = "Staff: Psychologist",
  LAW_FTE = "Staff: Law Enforcement",
  CPROB1 = html("P<sub><i>k</i></sub>=1"),
  CPROB2 = html("P<sub><i>k</i></sub>=2"),
  CPROB3 = html("P<sub><i>k</i></sub>=3"),
  C = md("**k**")) %>%
tab_row_group(
  label = "Unconditional response patterns",
  rows = 1:5) %>%
tab_row_group(
  label = md("**k* = 1 Conditional response patterns"),
  rows = 6:10) %>% #EDIT THESE VALUES BASED ON THE LAST COLUMN
tab_row_group(
  label = md("**k* = 2 Conditional response patterns"),
  rows = 11:15) %>% #EDIT THESE VALUES BASED ON THE LAST COLUMN
tab_row_group(
  label = md("**k* = 3 Conditional response patterns"),
  rows = 16:20) %>% #EDIT THESE VALUES BASED ON THE LAST COLUMN
row_group_order(
  groups = c("Unconditional response patterns",
             md("**k* = 1 Conditional response patterns"),
             md("**k* = 2 Conditional response patterns"),
             md("**k* = 3 Conditional response patterns"))) %>%
tab_footnote(
  footnote = html(
    "<i>Note.</i> <i>f</i><sub>r</sub> = response pattern frequency; P<sub><i>k</i></sub>=
  )
) %>%
cols_align(align = "center") %>%
opt_align_table_header(align = "left") %>%
gt::tab_options(table.font.names = "Times New Roman")

resp_table

```

---

Save table:

```
gtsave(resp_table, here("figures", "resp_table.png"))
```

## Observed Response Patterns

Response patterns, estimated frequencies, estimated posterior class probabilities and modal assignments

| $f_r$                                 | Harrassment: Disability | Harrassment: Race | Harrassment: Sex |
|---------------------------------------|-------------------------|-------------------|------------------|
| Unconditional response patterns       |                         |                   |                  |
| 525                                   | 0                       | 0                 | 0                |
| 299                                   | 0                       | 0                 | 0                |
| 293                                   | 0                       | 0                 | 0                |
| 251                                   | 0                       | 0                 | 0                |
| 75                                    | 0                       | 0                 | 0                |
| $k = 1$ Conditional response patterns |                         |                   |                  |
| 251                                   | 0                       | 0                 | 0                |
| 75                                    | 0                       | 0                 | 0                |
| 72                                    | 0                       | 0                 | 1                |
| 38                                    | 0                       | 0                 | 1                |
| 34                                    | 0                       | 0                 | 0                |
| $k = 2$ Conditional response patterns |                         |                   |                  |
| 24                                    | 0                       | 1                 | 0                |
| 20                                    | 0                       | 1                 | 1                |
| 19                                    | 0                       | 1                 | 1                |
| 18                                    | 0                       | 1                 | 1                |
| 12                                    | 0                       | 1                 | 1                |
| $k = 3$ Conditional response patterns |                         |                   |                  |
| 525                                   | 0                       | 0                 | 0                |
| 299                                   | 0                       | 0                 | 0                |
| 293                                   | 0                       | 0                 | 0                |
| 36                                    | 0                       | 0                 | 1                |
| 27                                    | 0                       | 0                 | 0                |

Note.  $f_r$  = response pattern frequency;  $P_k$  = posterior class probabilities  
 Data Source: **Civil Rights Data Collection (CRDC)**

## Chapter 10

# Classification Diagnostics

---

### 10.1 Load Packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

---

### 10.2 LCA

Prepare Data

```
df_bully <- read_csv(here("data", "crdc_lca_data.csv")) %>%
  clean_names() %>%
  dplyr::select(report_dis, report_race, report_sex, counselors_fte, psych_fte, law_ft
```

---

Continuing the LCA example (??) in this bookdown, use Mplus to calculate k-class confidence intervals (Note: Change the syntax to make your chosen  $k$ -class model):

```
classification <- mplusObject(

  TITLE = "C3 LCA - Calculated k-Class 95% CI",

  VARIABLE =
    "categorical = report_dis-law_fte;
    usevar = report_dis-law_fte;
    classes = c(3);",

  ANALYSIS =
    "estimator = ml;
    type = mixture;
    starts = 0;
    processors = 10;
    optseed = 802779;
    bootstrap = 1000;",

  MODEL =
    "
    !CHANGE THIS SECTION TO YOUR CHOSEN k-CLASS MODEL

    %OVERALL%
    [C#1](c1);

    [C#2](C2);

    Model Constraint:
    New(p1 p2 p3);

    p1 = exp(c1)/(1+exp(c1)+exp(c2));
    p2 = exp(c2)/(1+exp(c1)+exp(c2));
    p3 = 1/(1+exp(c1)+exp(c2));",
```



```

OUTPUT = "cinterval(bcbootstrap)",

usevariables = colnames(df_bully),
rdata = df_bully)

classification_fit <- mplusModeler(classification,
  dataout=here("mplus", "bully.dat"),
  modelout=here("mplus", "class.inp") ,
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

*Note:* Ensure that the classes did not shift during this step (i.g., Class 1 in the enumeration run is now Class 4). Evaluate output and compare the class counts and proportions for the latent classes. Using the OPTSEED function ensures replication of the best loglikelihood value run.

---

Read in the 3-class model:

```

# Read in the 3-class model and extract information needed
output_enum <- readModels(here("mplus", "class.out"))

# 95% k-Class and k-class 95% Confidence Intervals
k_ci <- output_enum$parameters$ci.unstandardized %>%
  filter(paramHeader == "New.Additional.Parameters") %>%
  unite(CI, c(low2.5, up2.5), sep=" ", remove = TRUE) %>%
  mutate(CI = paste0("[", CI, "]")) %>%
  rename(kclass=est) %>%
  dplyr::select(kclass, CI)

# AvePPk = Average Latent Class Probabilities for Most Likely Latent Class Membership (Row) by L
avePPk <- tibble(avePPk = diag(output_enum$class_counts$avgProbs.mostLikely))

# mcaPk = modal class assignment proportion
mcaPk <- round(output_enum$class_counts$mostLikely, 3) %>%
  mutate(model = paste0("Class ", class)) %>%
  add_column(avePPk, k_ci) %>%
  rename(mcaPk = proportion) %>%
  dplyr::select(model, kclass, CI, mcaPk, avePPk)

# OCCk = odds of correct classification
OCCk <- mcaPk %>%
  mutate(OCCk = round((avePPk/(1-avePPk))/(kclass/(1-kclass)), 3))

```

## Model Classification Diagnostics for the 3-Class Solution

| $k$ -Class | $k$ -Class Proportions | 95% CI         | McaP <sub><math>k</math></sub> | AvePP <sub><math>k</math></sub> | OCC <sub><math>k</math></sub> |
|------------|------------------------|----------------|--------------------------------|---------------------------------|-------------------------------|
| Class 1    | 0.249                  | [0.166, 0.329] | 0.282                          | 0.675                           | 6.264                         |
| Class 2    | 0.106                  | [0.083, 0.136] | 0.095                          | 0.904                           | 79.420                        |
| Class 3    | 0.644                  | [0.561, 0.731] | 0.623                          | 0.893                           | 4.614                         |

Note. McaP <sub>$k$</sub>  = Modal class assignment proportion; AvePP <sub>$k$</sub>  = Average posterior class probabilities; OCC <sub>$k$</sub>  = Odds of correct classification;

```
# Put everything together
class_data <- data.frame(OCCk)
```

Now, use {gt} to make a nicely formatted table

```
class_table <- class_data %>%
  gt() %>%
  tab_header(
    title = "Model Classification Diagnostics for the 3-Class Solution") %>%
  cols_label(
    model = md("*k*-Class"),
    kclass = md("*k*-Class Proportions"),
    CI = "95% CI",
    mcaPk = html("McaP<sub>k</sub>"),
    avePPk = md("AvePP<sub>k</sub>"),
    OCCk = md("OCC<sub>k</sub>")
  ) %>%
  tab_footnote(
    footnote = html(
      "<i>Note.</i> McaP<sub>k</sub> = Modal class assignment proportion; AvePP<sub>k</sub>"
    )
  ) %>%
  cols_align(aligned = "center") %>%
  opt_align_table_header(aligned = "left") %>%
  gt::tab_options(table.font.names = "Times New Roman")

class_table
```

## 10.3 LPA

Prepare Data

```
pisa <- pisaUSA15[1:500,] %>%
  dplyr::select(broad_interest, enjoyment, instrumental_mot, self_efficacy)
```

---

Continuing the LPA example (??) in this bookdown, use Mplus to calculate k-class confidence intervals (Note: Change the syntax to make your chosen  $k$ -class model):

```
classification <- mplusObject(

  TITLE = "LPA - Calculated k-Class 95% CI",

  VARIABLE =
    "usevar = broad_interest-self_efficacy;
    classes = c1(4);",

  ANALYSIS =
    "estimator = ml;
    type = mixture;
    starts = 0;
    processors = 10;
    optseed = 468036; ! This seed is taken from chosen model output
    bootstrap = 1000;",

  MODEL =
    "
    ! This is copied and pasted from the chosen model input
    %c1#1%
    broad_interest (vbroad_interest);
    enjoyment (venjoyment);
    instrumental_mot (vinstrumental_mot);
    self_efficacy (vself_efficacy);

    broad_interest WITH enjoyment (broad_interestWenjoyment);
    broad_interest WITH instrumental_mot (broad_interestWinstrumental_mot);
    broad_interest WITH self_efficacy (broad_interestWself_efficacy);
    enjoyment WITH instrumental_mot (enjoymentWinstrumental_mot);
    enjoyment WITH self_efficacy (enjoymentWself_efficacy);
    instrumental_mot WITH self_efficacy (instrumental_motWself_efficacy);
```

```

%c1#2%
broad_interest (vbroad_interest);
enjoyment (venjoyment);
instrumental_mot (vinstrumental_mot);
self_efficacy (vself_efficacy);

broad_interest WITH enjoyment (broad_interestWenjoyment);
broad_interest WITH instrumental_mot (broad_interestWinstrumental_mot);
broad_interest WITH self_efficacy (broad_interestWself_efficacy);
enjoyment WITH instrumental_mot (enjoymentWinstrumental_mot);
enjoyment WITH self_efficacy (enjoymentWself_efficacy);
instrumental_mot WITH self_efficacy (instrumental_motWself_efficacy);

%c1#3%
broad_interest (vbroad_interest);
enjoyment (venjoyment);
instrumental_mot (vinstrumental_mot);
self_efficacy (vself_efficacy);

broad_interest WITH enjoyment (broad_interestWenjoyment);
broad_interest WITH instrumental_mot (broad_interestWinstrumental_mot);
broad_interest WITH self_efficacy (broad_interestWself_efficacy);
enjoyment WITH instrumental_mot (enjoymentWinstrumental_mot);
enjoyment WITH self_efficacy (enjoymentWself_efficacy);
instrumental_mot WITH self_efficacy (instrumental_motWself_efficacy);

%c1#4%
broad_interest (vbroad_interest);
enjoyment (venjoyment);
instrumental_mot (vinstrumental_mot);
self_efficacy (vself_efficacy);

broad_interest WITH enjoyment (broad_interestWenjoyment);
broad_interest WITH instrumental_mot (broad_interestWinstrumental_mot);
broad_interest WITH self_efficacy (broad_interestWself_efficacy);
enjoyment WITH instrumental_mot (enjoymentWinstrumental_mot);
enjoyment WITH self_efficacy (enjoymentWself_efficacy);
instrumental_mot WITH self_efficacy (instrumental_motWself_efficacy);

!CHANGE THIS SECTION TO YOUR CHOSEN k-CLASS MODEL

%OVERALL%
[C1#1] (c1);
[C1#2] (c2);

```

```

[C1#3](c3);

Model Constraint:
New(p1 p2 p3 p4);

p1 = exp(c1)/(1+exp(c1)+exp(c2)+exp(c3));
p2 = exp(c2)/(1+exp(c1)+exp(c2)+exp(c3));
p3 = exp(c3)/(1+exp(c1)+exp(c2)+exp(c3));
p4 = 1/(1+exp(c1)+exp(c2)+exp(c3));",

OUTPUT = "cinterval(bcbootstrap)",

usevariables = colnames(pisa),
rdata = pisa)

classification_fit <- mplusModeler(classification,
                                   dataout=here("lpa", "mplus", "class.dat"),
                                   modelout=here("lpa", "mplus", "class.inp") ,
                                   check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

Create table using `diagnostics_table` function

```

source(here("functions", "diagnostics_table.R"))

class_output <- readModels(here("mplus", "class.out"))

diagnostics_table(class_output)

```

## Model Classification Diagnostics

| <i>k</i> -Class | <i>k</i> -Class Proportions | 95% CI         | <i>mcaPk</i> | <i>AvePPk</i> | <i>OCCk</i> |
|-----------------|-----------------------------|----------------|--------------|---------------|-------------|
| Class 1         | 0.249                       | [0.166, 0.329] | 0.282        | 0.675         | 6.264       |
| Class 2         | 0.106                       | [0.083, 0.136] | 0.095        | 0.904         | 79.420      |
| Class 3         | 0.644                       | [0.561, 0.731] | 0.623        | 0.893         | 4.614       |

*Note.*  $McaP_{k\cdot}$  = Modal class assignment proportion;  $AvePP_{k\cdot}$  = Average posterior class probabilities;  $OCC_{k\cdot}$  = Odds of correct classification;

# Chapter 11

## Class Homogeneity

---

### 11.1 Load packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

---

### 11.2 LCA

Continuing the LCA example (??) in this bookdown, use R to evaluate the LCA class homogeneity.

We want a high degree of homogeneity (small variance) within each class (i.e., people in each class are not identical but are more similar to each other than to people in different classes).

Class-specific response probabilities above .70 or below .30 indicate high homogeneity in that particular class for that specific item. For a binary variable,  $X$ ,  $Var(X) = p(1 - p)$ . The largest variance possible for a binary variable is when  $p = .5$ , the smallest variance (0) is when  $p = 0$  or  $p = 1$ .

The easiest way is to visualize the item probability plot:

```
source(here("functions", "plot_lca.R"))

# Read in model
output_enum <- readModels(here("enum"), filefilter = "bully", quiet = TRUE)

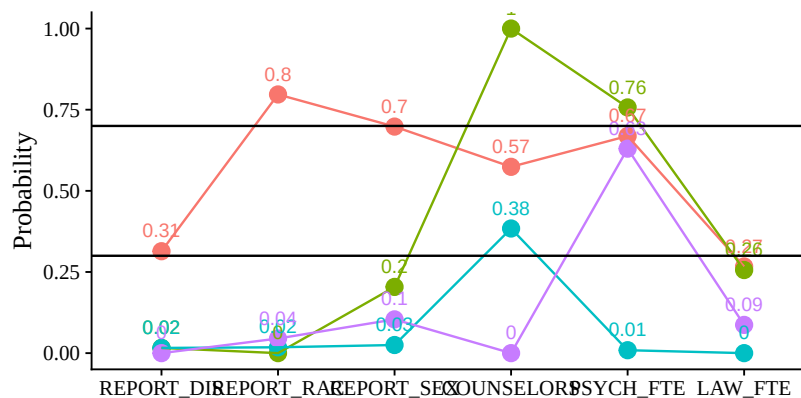
# Change this to look at your chosen LCA model
chosen_model <- output_enum$c4_bully.out

# Add lines
plot_lca(chosen_model) + geom_hline(yintercept = c(0.7, 0.3))
```

#### 4-Class Probability Plot

● Class 1 (10.47%) ● Class 2 (27.77%)

● Class 3 (31.55%) ● Class 4 (30.21%)



Or create a table:



```

# Read in model
output_enum <- readModels(here("enum"), filefilter = "bully", quiet = TRUE)

# Change this to look at your chosen LCA model
chosen_model <- output_enum$c4_bully.out

# Extract table of probabilities
probabilities <- data.frame(chosen_model$parameters$probability.scale) %>%
  mutate(LatentClass = sub("^", "Class ", LatentClass)) %>%
  filter(category == 2) %>%
  dplyr::select(est, LatentClass, param) %>%
  pivot_wider(names_from = LatentClass, values_from = est) %>%
  relocate(param, .after = last_col()) %>%
  rename(Items = param)

# Bold above .7 and below .3
bold_condition <- function(x) {
  ifelse(x > 0.7 | x < 0.3, paste0("***", sprintf("%.3f", x), "***"),
        sprintf("%.3f", x))
}

# Apply the bold formatting condition to the table
homogeneity <- probabilities %>%
  mutate(across(where(is.numeric), ~ sapply(., bold_condition))) %>%
  relocate(Items, .before = 1)

# Display the formatted table and percentage
homogeneity %>%
  gt() %>%
  cols_label(
    Items = "Item"
  ) %>%
  fmt_number(
    columns = everything(),
    decimals = 3
  ) %>%
  tab_header(
    title = "Class Homogeneity Table"
  ) %>%
  fmt_markdown(columns = everything()) %>%
  tab_footnote(
    footnote = "Cells in bold indicate probabilities > 0.7 or < 0.3"
  )

```

Class Homogeneity Table

| Item       | Class 1 | Class 2 | Class 3 | Class 4 |
|------------|---------|---------|---------|---------|
| REPORT_DIS | 0.314   | 0.016   | 0.016   | 0.000   |
| REPORT_RAC | 0.797   | 0       | 0.018   | 0.045   |
| REPORT_SEX | 0.698   | 0.204   | 0.025   | 0.104   |
| COUNSELORS | 0.574   | 1       | 0.384   | 0.000   |
| PSYCH_FTE  | 0.669   | 0.757   | 0.009   | 0.630   |
| LAW_FTE    | 0.267   | 0.256   | 0       | 0.087   |

Cells in bold indicate probabilities  $> 0.7$  or  $< 0.3$

### 11.3 LPA

Continuing the LPA example (??) in this bookdown, use R to evaluate the LPA profile homogeneity.

Profile homogeneity is the model-estimated within-profile variances for each indicator  $m$  across the  $k$ -profiles and comparing them to the total overall sample variance:

$$\frac{\theta_{mk}}{\theta_m}$$

Where  $\theta_{mk}$  is the item variance for item  $m$ , class  $k$  and  $\theta_m$  is the item variance for item  $m$  not conditioned on class.

Individuals belonging to the same class are more similar to other members of that class than they are compared to members of other classes. Individuals belonging to the same class are closer to the class mean than they are to the overall population mean. Within-class variance for each indicator is smaller than overall population variance, therefore:

- Values  $> .90$  corresponds to a high degree of homogeneity
- Values  $< .60$  corresponds to a low degree of homogeneity

To calculate the values in a table:

```
# Read in model
output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

# Change this to look at your chosen LPA model
chosen_model <- output_enum$model_3_class_4.out

# Extract overall and profile-specific variances
overall_variances <- data.frame(chosen_model$sampstat$univariate.sample.statistics) %>%
  dplyr::select(Variance) %>%
  rownames_to_column(var = "item") %>%
  clean_names() %>%
  mutate(item = str_sub(item, 1, 8))

profile_variances <- data.frame(chosen_model$parameters$unstandardized) %>%
  filter(paramHeader == "Variances") %>%
  dplyr::select(param, est, LatentClass) %>%
  rename(item = param,
         variance = est,
         k = LatentClass) %>%
  mutate(k = paste0("Profile ", k)) %>%
  mutate(item = str_sub(item, 1, 8))

# Bold above 0.90 and italicize below 0.60
format_condition <- function(x) {
  ifelse(x > 0.9,
        paste0("**", sprintf("%.3f", x), "**"), # bold
        ifelse(x < 0.6,
              paste0("*", sprintf("%.3f", x), "*"), # italic
              sprintf("%.3f", x))) # normal
}

# Evaluate the ratio
homogeneity <- profile_variances %>%
  left_join(overall_variances, by = "item") %>%
  mutate(homogeneity_ratio = round((variance.x / variance.y), 2)) %>%
  mutate(across(where(is.numeric), ~ as.character(sapply(., format_condition))))

# Create a gt table
homogeneity %>%
  dplyr::select(homogeneity_ratio, k, item) %>%
  pivot_wider(
    names_from = k,
    values_from = homogeneity_ratio
```

Profile Homogeneity Table

| Item     | Profile 1    | Profile 2    | Profile 3    | Profile 4    |
|----------|--------------|--------------|--------------|--------------|
| BROAD_IN | <b>0.970</b> | <b>0.970</b> | <b>0.970</b> | <b>0.970</b> |
| ENJOYMEN | <b>0.930</b> | <b>0.930</b> | <b>0.930</b> | <b>0.930</b> |
| INSTRUME | <i>0.060</i> | <i>0.060</i> | <i>0.060</i> | <i>0.060</i> |
| SELF_EFF | <b>0.950</b> | <b>0.950</b> | <b>0.950</b> | <b>0.950</b> |

Cells in bold indicate proportions  $> 0.90$  and cells in italics indicate proportions  $< 0.60$ . Note: In this example, the class variances are set to be equal.

```

) %>%
gt() %>%
cols_label(
  item = "Item"
) %>%
fmt_number(
  columns = everything(),
  decimals = 3
) %>%
tab_header(
  title = "Profile Homogeneity Table"
) %>%
tab_footnote(
  footnote = "Cells in bold indicate proportions  $> 0.90$  and cells in italics indicate
  Note: In this example, the class variances are set to be equal." ) %>%
fmt_markdown(columns = everything())

```

You can also visualize the plot:

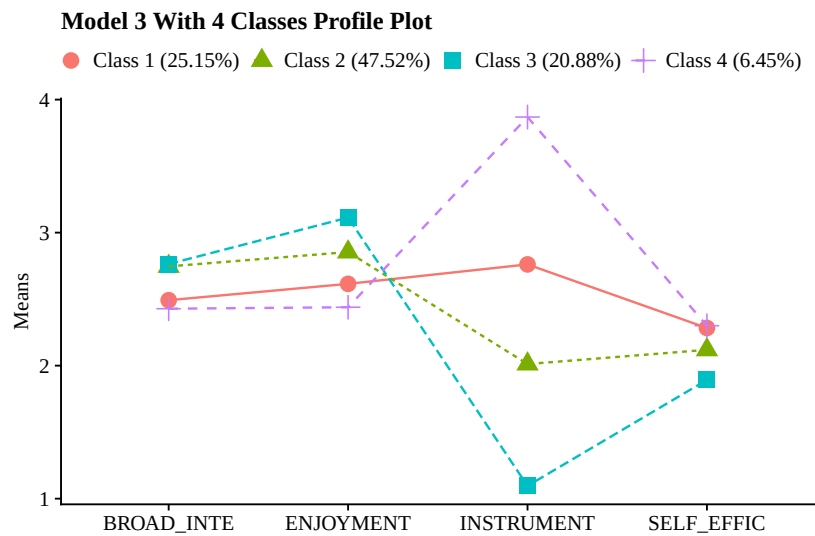
```

source(here("functions", "plot_lpa.R"))

output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

plot_lpa(model_name = output_enum$model_3_class_4.out)

```





## Chapter 12

# Class Separation

---

### 12.1 Load Packages

```
library(naniar)
library(tidyverse)
library(haven)
library(glue)
library(MplusAutomation)
library(here)
library(janitor)
library(gt)
library(tidyLPA)
library(pisaUSA15)
library(cowplot)
library(filesstrings)
library(patchwork)
library(RcppAlgos)
```

---

### 12.2 LCA

Continuing the LCA example (??) in this bookdown, use R to evaluate the LCA class separation.

$$\widehat{OR}_{1v2} = \frac{\frac{P(u_1=1|c_1)}{P(u_1=0|c_1)}}{\frac{P(u_1=1|c_2)}{P(u_1=0|c_2)}}$$

Odds ratios (ORs) of item endorsements between two classes  $>5$  (corresponding to class-specific endorsement probabilities of  $>.7$  and  $<.3$ ) or  $<.2$  (corresponding to class-specific endorsement probabilities of  $<.3$  and  $>.7$ ), indicate high separation between those two classes on that specific item. No single binary (0/1) indicator can separate more than two subsets of latent classes, no matter what the total number of classes in the model. We want a high degree of separation (large standardized mean differences) between classes.

```
# Read in model
output_enum <- readModels(here("enum"), filefilter = "bully", quiet = TRUE)

# Change this to look at your chosen LCA model
chosen_model <- output_enum$c4_bully.out

# Extract table of probabilities
c_prob <- data.frame(chosen_model$parameters$probability.scale) %>%
  mutate(LatentClass = sub("^", "Class ", LatentClass)) %>%
  filter(category == 2) %>%
  dplyr::select(est, LatentClass, param) %>%
  rename(item = param, prob = est, k = LatentClass) %>%
  mutate(# Adjust probabilities by setting a minimum of 0.01 and a maximum of 0.99
    prob = pmin(pmax(prob, 0.01), 0.99),
    prob = as.numeric(prob))

# Relative class sizes
class_sizes <- data.frame(chosen_model$class_counts$modelEstimated) %>%
  rename(size = proportion, k = class) %>%
  mutate(k = paste0("Class ", k)) %>%
  dplyr::select(-count)

# Create the combinations of classes
class_combinations <- data.frame(comboGrid(
  k1 = unique(class_sizes$k),
  k2 = unique(class_sizes$k)
)) %>%
  filter(k1 != k2) %>%
  arrange(k1, k2)
```



```

# Calculate pairwise comparisons for probabilities (e.g.,  $P(u1=1/c1)$  /  $P(u1=0/c1)$ ) for each combination
combined_results <- class_combinations %>%
  rowwise() %>%
  do({
    pair <- .
    k1 <- pair$k1
    k2 <- pair$k2

    # Filter for the probabilities ( $P(u1=1/c1)$  and  $P(u1=0/c1)$ )
    data_k1_prob <- c_prob %>% filter(k == k1)
    data_k2_prob <- c_prob %>% filter(k == k2)

    # Combine the probability data for the two profiles
    prob_data <- data_k1_prob %>%
      inner_join(data_k2_prob,
        by = "item",
        suffix = c("_k1", "_k2")) %>%
      mutate(
        prob_ratio = (prob_k1 / (1 - prob_k1)) / (prob_k2 / (1 - prob_k2)),
        # Division formula
        comparison = paste(k1, "vs", k2)
      ) %>%
      mutate(prob_ratio = round(ifelse(
        is.infinite(prob_ratio) | is.nan(prob_ratio),
        NA,
        prob_ratio
      ), 3)) %>%
      dplyr::select(item, prob_ratio, comparison)

    prob_data
  }) %>%
  bind_rows()

# Define a function to apply the Markdown bold formatting
bold_condition <- function(x) {
  # If prob_ratio is > 5 or < 0.2, add **
  ifelse(x > 5 | x < 0.2, paste0("**", sprintf("%.3f", x), "**"),
    sprintf("%.3f", x))
}

# Assuming combined_results is already created as mentioned
formatted_results <- combined_results %>%
  dplyr::select(prob_ratio, comparison, item) %>%
  pivot_wider(names_from = comparison, values_from = prob_ratio) %>%

```

Class Separation Table

| Item       | Class 1 vs Class 2 | Class 1 vs Class 3 | Class 1 vs Class 4 | Class 2 vs Class 3 |
|------------|--------------------|--------------------|--------------------|--------------------|
| REPORT_DIS | <b>28.150</b>      | <b>28.150</b>      | <b>45.315</b>      |                    |
| REPORT_RAC | <b>388.685</b>     | <b>214.191</b>     | <b>83.321</b>      |                    |
| REPORT_SEX | <b>9.018</b>       | <b>90.139</b>      | <b>19.912</b>      |                    |
| COUNSELORS | <b>0.014</b>       | 2.161              | <b>133.394</b>     |                    |
| PSYCH_FTE  | 0.649              | <b>200.094</b>     | 1.187              |                    |
| LAW_FTE    | 1.059              | <b>36.061</b>      | 3.823              |                    |

Cells in bold indicate probabilities  $> 5$  or  $< 0.2$ .

```
mutate(across(where(is.numeric), ~ supply(., bold_condition)))

# Create a gt table
formatted_results %>%
  gt() %>%
  cols_label(item = "Item") %>%
  fmt_missing(columns = everything(), missing_text = "-") %>%
  tab_header(title = "Class Separation Table") %>%
  fmt_markdown(columns = everything()) %>%
  tab_footnote(footnote = "Cells in bold indicate probabilities > 5 or < 0.2.")
```

## 12.3 LPA

Continuing the LPA example (??) in this bookdown, use R to evaluate the LPA profile separation.

You can evaluate the degree of profile separation by assessing the actual distance between the profile-specific means. To quantify profile separation between Profile j and Profile k with respect to a particular item m, compute a standard mean difference:

$$\hat{d}_{mjk} = \frac{\hat{\alpha}_{mj} - \hat{\alpha}_{mk}}{\sigma_{mjk}}$$

Pooled variance:

$$\hat{\sigma}_{mjk} = \sqrt{\frac{(\hat{\pi}_j)(n)(\hat{\theta}_{mj}) + (\hat{\pi}_k)(n)(\hat{\theta}_{mk})}{(\hat{\pi}_j + \hat{\pi}_k)n}}$$

Well-separated classes have a small degree of overlap of the class-specific indicator distributions; that is, standardized mean difference is large.

- A mean difference  $< .85$  corresponds to low separation—more than 50% overlap
- A mean difference  $> 2$  corresponds to high separation—less than 20% overlap

```
# Read in model
output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

# Change this to look at your chosen LPA model
chosen_model <- output_enum$model_3_class_4.out

# Profile-specific means
profile_means <- data.frame(chosen_model$parameters$unstandardized) %>%
  filter(paramHeader == "Means") %>%
  filter(!str_detect(param, "#")) %>%
  dplyr::select(param, est, LatentClass) %>%
  rename(item = param,
         means = est,
         k = LatentClass) %>%
  mutate(k = paste0("Profile ", k)) %>%
  mutate(item = str_sub(item, 1, 8))

# Relative profile sizes
profile_sizes <- data.frame(chosen_model$class_counts$modelEstimated) %>%
  rename(size = proportion,
         k = class) %>%
  mutate(k = paste0("Profile ", k)) %>%
  dplyr::select(-count)

# Sample size
n <- chosen_model$summaries$Observations

# Profile-specific variance
profile_variances <- data.frame(chosen_model$parameters$unstandardized) %>%
  filter(paramHeader == "Variances") %>%
  dplyr::select(param, est, LatentClass) %>%
  rename(item = param,
```

```

    variance = est,
    k = LatentClass) %>%
mutate(k = paste0("Profile ", k)) %>%
mutate(item = str_sub(item, 1, 8))

# Combine profile variances with profile sizes
variance_with_sizes <- profile_variances %>%
  left_join(profile_sizes, by = "k")

# Create the combinations
profile_combinations <- data.frame(comboGrid(k1 = unique(profile_sizes$k), k2 = unique(
  filter(k1 != k2) %>%
  arrange(k1, k2)

# Calculate pooled variance for each item across the profiles
combined_results <- profile_combinations %>%
  rowwise() %>%
  do({
    pair <- .
    k1 <- pair$k1
    k2 <- pair$k2

    # Filter for the two profiles (variance)
    data_k1_var <- variance_with_sizes %>% filter(k == k1)
    data_k2_var <- variance_with_sizes %>% filter(k == k2)

    # Filter for the two profiles (means)
    data_k1_mean <- profile_means %>% filter(k == k1)
    data_k2_mean <- profile_means %>% filter(k == k2)

    # Combine variance data for the two profiles
    variance_data <- data_k1_var %>%
      inner_join(data_k2_var, by = "item", suffix = c("_k1", "_k2")) %>%
      mutate(
        pooled_variance = sqrt(
          ((size_k1 * n * variance_k1) + (size_k2 * n * variance_k2)) / ((size_k1 + si
        ),
        comparison = paste(k1, "vs", k2)
      ) %>%
      dplyr::select(item, pooled_variance, comparison)

    # Combine mean data for the two profiles
    mean_data <- data_k1_mean %>%
      inner_join(data_k2_mean, by = "item", suffix = c("_k1", "_k2")) %>%
      mutate(

```

```

    mean_diff = means_k1 - means_k2,
    comparison = paste(k1, "vs", k2)
  ) %>%
  dplyr::select(item, mean_diff, comparison)

  # Combine both variance and mean differences data
  combined_data <- variance_data %>%
    left_join(mean_data, by = c("item", "comparison")) %>%
    mutate(
      mean_diff_by_pooled_variance = mean_diff / pooled_variance
    )

  combined_data
}) %>%
bind_rows() %>%
mutate(mean_diff_by_pooled_variance = round(mean_diff_by_pooled_variance, 3)) %>%
dplyr::select(mean_diff_by_pooled_variance, comparison, item) %>%
pivot_wider(
  names_from = comparison,
  values_from = mean_diff_by_pooled_variance
)

# Create a gt table
gt_table <- combined_results %>%
  gt() %>%
  cols_label(
    item = "Item"
  ) %>%
  tab_header(
    title = "Profile Separation Table"
  ) %>%
  tab_footnote(
    footnote = "Green cells indicate >2; Red cells indicate <0.85."
  )

# Formatting thresholds
high_threshold <- 2
low_threshold <- 0.85

# Apply conditional colors for each numeric cell
num_cols <- gt_table$`_data` %>% dplyr::select(-item) %>% names() # numeric columns
for(col in num_cols){

```

Profile Separation Table

| Item     | Profile 1 vs Profile 2 | Profile 1 vs Profile 3 | Profile 1 vs Profile 4 | Profile 1 vs Profile 5 |
|----------|------------------------|------------------------|------------------------|------------------------|
| BROAD_IN | -0.334                 | -0.357                 | 0.084                  | 0.084                  |
| ENJOYMEN | -0.346                 | -0.722                 | 0.255                  | 0.255                  |
| INSTRUME | 4.004                  | 8.900                  | -5.923                 | -5.923                 |
| SELF_EFF | 0.261                  | 0.621                  | -0.027                 | -0.027                 |

Green cells indicate >2; Red cells indicate <0.85.

```

for(i in 1:nrow(combined_results)){
  val <- combined_results[[col]][i] # numeric value
  color <- if(val > 2) "#66BB6A" else if(val < 0.85) "#E57373" else NA
  if(!is.na(color)){
    gt_table <- gt_table %>%
      tab_style(
        style = cell_fill(color = color),
        locations = cells_body(columns = col, rows = i)
      )
  }
}
}

# Display table
gt_table

```

You can also visualize the plot:

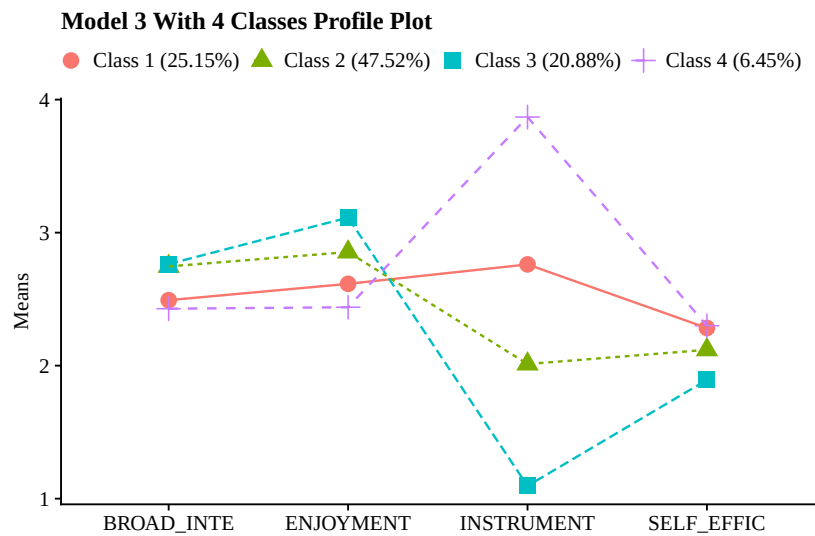
```

source(here("functions", "plot_lpa.R"))

output_enum <- readModels(here("lpa", "tidyLPA"), quiet = TRUE)

plot_lpa(model_name = output_enum$model_3_class_4.out)

```







Part VI

Including Auxiliary  
Variables



## Chapter 13

# Manual 3-Step Covariate Only

---

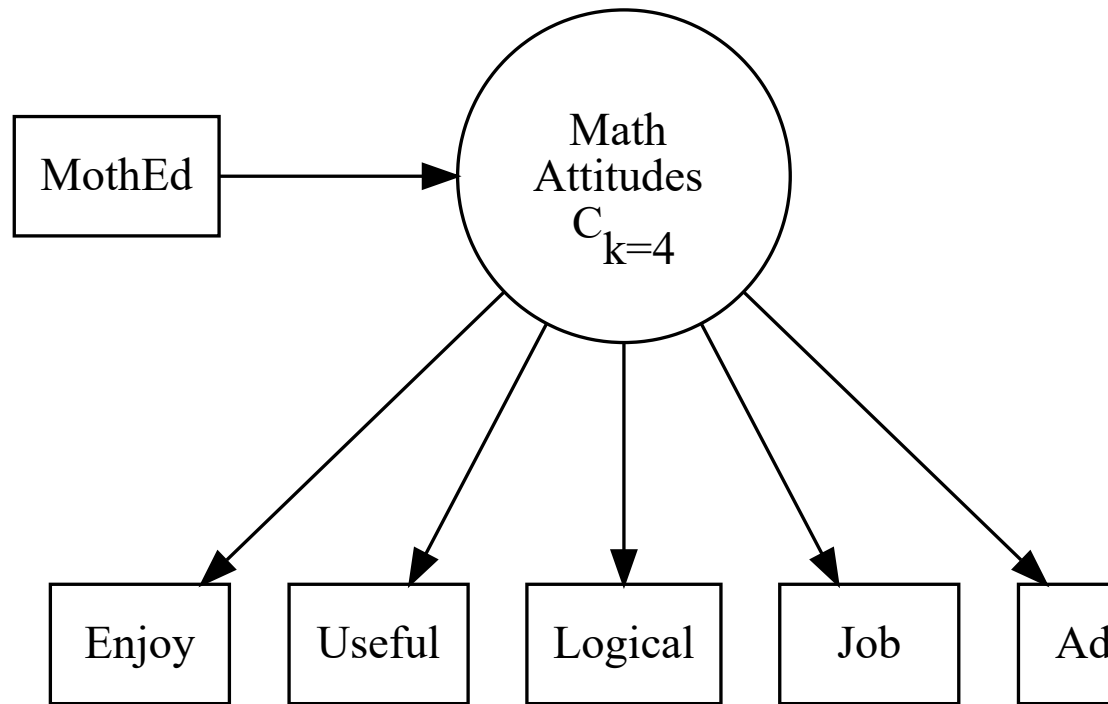
### Data source:

1. The first examples utilizes the public-use dataset, *The Longitudinal Survey of American Youth* (**LSAY**): See documentation [here](#)
  2. The second example utilizes a dataset on undergraduate *Cheating* available from the **poLCA** package (Dayton, 1998): See documentation [here](#)
- 

### 13.1 Load packages

```
library(MplusAutomation)
library(tidyverse) #collection of R packages designed for data science
library(here) #helps with filepaths
library(janitor) #clean_names
library(gt) # create tables
library(cowplot) # a ggplot theme
library(DiagrammeR) # create path diagrams
library(glue) # allows us to paste expressions into R code
library(data.table) # used for `melt()` function
library(poLCA)
library(reshape2)
```

Our example is Mother's Education as a predictor of latent class membership of Math Attitudes



### LCA Indicators & Auxiliary Variables: Math Attitudes Example

| Name                | Variable Description                                                                     |
|---------------------|------------------------------------------------------------------------------------------|
| enjoy               | I enjoy math.                                                                            |
| useful              | Math is useful in everyday problems.                                                     |
| logical             | Math helps a person think logically.                                                     |
| job                 | It is important to know math to get a good job.                                          |
| adult               | I will use math in many ways as an adult.                                                |
| Auxiliary Variables |                                                                                          |
| mothed              | Level of education: (1) less than high school, (2) high school diploma, (3) some college |

Application: Longitudinal Study of American Youth, **Science Attitudes**

The data can be found in the `data` folder and is called `lsay_subset.csv`.

```
lsay_data <- read_csv(here("three_step", "data", "lsay_subset.csv")) %>%
  clean_names() %>% # make variable names lowercase
  mutate(female = recode(gender, `1` = 0, `2` = 1)) # relabel values from 1,2 to 0,1
```

## 13.2 Descriptive Statistics

```
dframe <- lsay_data %>%
  pivot_longer(
    c(enjoy, useful, logical, job, adult),
    names_to = "Variable"
  ) %>%
  group_by(Variable) %>%
  summarise(
```

### LCA Indicator Endorsement

| Variable | Proportion Endorsed | Count |
|----------|---------------------|-------|
| adult    | 0.596               | 1858  |
| enjoy    | 0.573               | 1784  |
| job      | 0.625               | 1947  |
| logical  | 0.541               | 1686  |
| useful   | 0.589               | 1835  |

```

    Count = sum(value == 1, na.rm = TRUE),
    Total = n(),
    .groups = "drop"
  ) %>%
  mutate(`Proportion Endorsed` = round(Count / Total, 3)) %>%
  dplyr::select(Variable, `Proportion Endorsed`, Count)

gt(dframe) %>%
  tab_header(
    title = md("**LCA Indicator Endorsement**"),
    subtitle = md("&nbsp;")
  ) %>%
  tab_options(
    column_labels.font.weight = "bold",
    row_group.font.weight = "bold"
  )

```

Mother's Education

```

summary(factor(lsay_data$mothed))
#>   1    2    3    4    5 NA's
#> 452 1732 304 368 158 102

```

## 13.3 Manual ML Three-step

### 13.3.1 Step 1 - Class Enumeration w/ Auxiliary Specification

---

This step is done after class enumeration (or after you have selected the best latent class model). In this example, the four class model was the best. Now, we re-estimate the four-class model using `optseed` for efficiency. The difference here is the `SAVEDATA` command, where I can save the posterior probabilities and the modal class assignment that will be used in steps two and three.

```
step1 <- mplusObject(
  TITLE = "Step 1 - Three-Step using LSAL",
  VARIABLE =
    "categorical = enjoy useful logical job adult;
    usevar = enjoy useful logical job adult;

    classes = c(4);

    auxiliary = mothed ! covariate ",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;
    optseed = 568405;",

  SAVEDATA =
    "File=savedata_cov.dat;
    Save=cprob;",

  OUTPUT = "residual tech11 tech14",

  PLOT =
    "type = plot3;
    series = enjoy-adult(*)",

  usevariables = colnames(lsay_data),
  rdata = lsay_data)

step1_fit <- mplusModeler(step1,
  dataout=here("three_step", "manual_3step", "Step1.dat"),
  modelout=here("three_step", "manual_3step", "one_cov.inp")
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

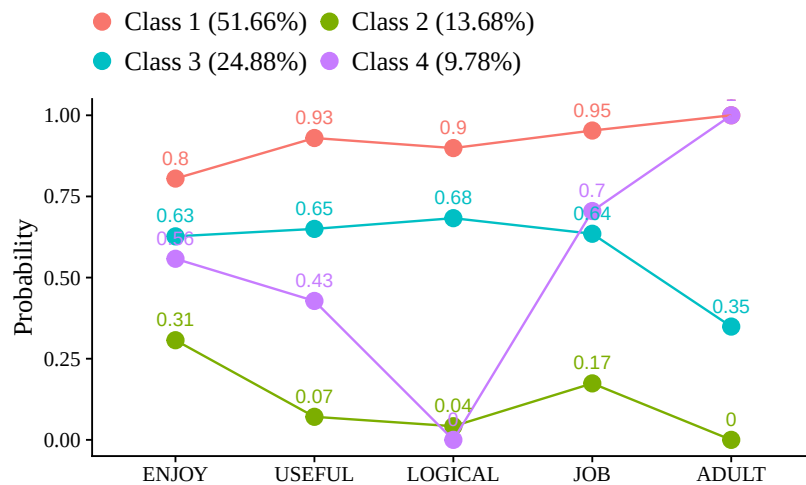


---

```
source(here("functions", "plot_lca.R"))
output_lsay <- readModels(here("three_step", "manual_3step", "one_cov.out"))

plot_lca(model_name = output_lsay)
```

### Step 1 – Three-Step using LSAL Probability Plot




---

### 13.3.2 Step 2 - Determine Measurement Error

---

Extract logits for the classification probabilities for the most likely latent class

```
logit_cprobs <- as.data.frame(output_lsay[["class_counts"]]  
                             [["logitProbs.mostLikely"]])
```

Extract saved dataset which is part of the mplusObject “step1\_fit”

```
savedata <- as.data.frame(output_lsay[["savedata"]])
```

Rename the column in savedata named “C” and change to “N”

```
colnames(savedata)[colnames(savedata) %in% c("C", "MLCC")] <- "N"
```

---

### 13.3.3 Step 3 - LCA Auxiliary Variable Model with 1 Covariate

---

```
step3 <- mplusObject(
  TITLE = "Step3 - 3step LSAY",

  VARIABLE =
    "nominal=N;
    usevar = n;

    classes = c(4);

    usevar = mothed;" ,

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;",

  DEFINE =
    "center mothed (grandmean);",

  MODEL =
    glue(
      " %OVERALL%

      C on mothed;          ! covariate as predictor of C

      %C#1%
      [n#1@{logit_cprobs[1,1]}}; ! MUST EDIT if you do not have a 4-class model.
      [n#2@{logit_cprobs[1,2]}};
      [n#3@{logit_cprobs[1,3]}};

      %C#2%
      [n#1@{logit_cprobs[2,1]}};
      [n#2@{logit_cprobs[2,2]}};
      [n#3@{logit_cprobs[2,3]}};
```

```

%C#3%
[n#1@{logit_cprobs[3,1]}};
[n#2@{logit_cprobs[3,2]}};
[n#3@{logit_cprobs[3,3]}};

%C#4%
[n#1@{logit_cprobs[4,1]}};
[n#2@{logit_cprobs[4,2]}};
[n#3@{logit_cprobs[4,3]}};"),

usevariables = colnames(savedata),
rdata = savedata)

step3_fit <- mplusModeler(step3,
  dataout=here("three_step", "manual_3step", "Step3.dat"),
  modelout=here("three_step", "manual_3step", "three_cov.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

### 13.3.3.1 Covariates Relations

---

As of 1/1/2026, this code is outdated. See here: <https://github.com/michaelhallquist/MplusAutomation/issues/228>

```

modelParams <- readModels(here("three_step", "manual_3step", "three_cov.out"))

# Extract information as data frame
cov <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(str_detect(paramHeader, "^C#\\d+\\.ON$")) %>%
# mutate(param = str_replace(param, "FEMALE", "Gender")) %>% # Change this to your own covariate
mutate(param = str_replace(param, "MOTHEd", "Mother's Education")) %>%
mutate(est = format(round(est, 3), nsmall = 3),
  se = round(se, 2),
  pval = round(pval, 3)) %>%
mutate(latent_class = str_replace(paramHeader, "^C#(\\d+)\\.ON$", "Class \\1")) %>%
dplyr::select(param, est, se, pval, latent_class) %>%
mutate(se = paste0("(", format(round(se,2), nsmall =2), ")")) %>%

```

```

unite(logit, est, se, sep = " ") %>%
dplyr::select(param, logit, pval, latent_class) %>%
mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
                    ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001),
                    scales::number(pval, accuracy = .001))))

or <- as.data.frame(modelParams[["parameters"]][["odds"]])%>%
  filter(str_detect(paramHeader, "^C#\\d+\\.ON$")) %>%
  # mutate(param = str_replace(param, "FEMALE", "Gender")) %>% # Change this to your own
  mutate(param = str_replace(param, "MOTHED", "Mother's Education")) %>%
  mutate(est = format(round(est, 3), nsmall = 3)) %>%
  mutate(latent_class = str_replace(paramHeader, "C#(\\d+)\\.ON$", "Class \\1")) %>%
  mutate(CI = paste0("[", format(round(lower_2.5ci, 3), nsmall = 3), ", ", format(round(
dplyr::select(param, est, CI, latent_class) %>%
rename(or = est)

combined <- or %>%
  full_join(cov) %>%
  dplyr::select(param, latent_class, logit, pval, or, CI)

# Create table

combined %>%
  gt(groupname_col = "latent_class", rowname_col = "param") %>%
  tab_header(
    title = "Covariate Results: Mother's Education on Class") %>%
  cols_label(
    logit = md("Logit (*se*)"),
    or = md("Odds Ratio"),
    CI = md("95% CI"),
    pval = md("*p*-value")) %>%
  sub_missing(1:3,
    missing_text = "") %>%
  sub_values(values = c("999.000"), replacement = "-") %>%
  cols_align(align = "center") %>%
  opt_align_table_header(align = "left") %>%
  gt::tab_options(table.font.names = "serif") %>%
  tab_footnote(
    footnote = "Reference Class: 4",
    locations = cells_title(groups = "title")
  )

```

---

## 13.4 Automated Three-Step

---

*Application: Undergraduate Cheating behavior*

“Dichotomous self-report responses by 319 undergraduates to four questions about cheating behavior” (poLCA, 2016).

---

Prepare data

```
data(cheating)

cheating <- cheating %>% clean_names()

df_cheap <- cheating %>%
  dplyr::select(1:4) %>%
  mutate_all(funs(.-1)) %>%
  mutate(gpa = cheating$gpa)

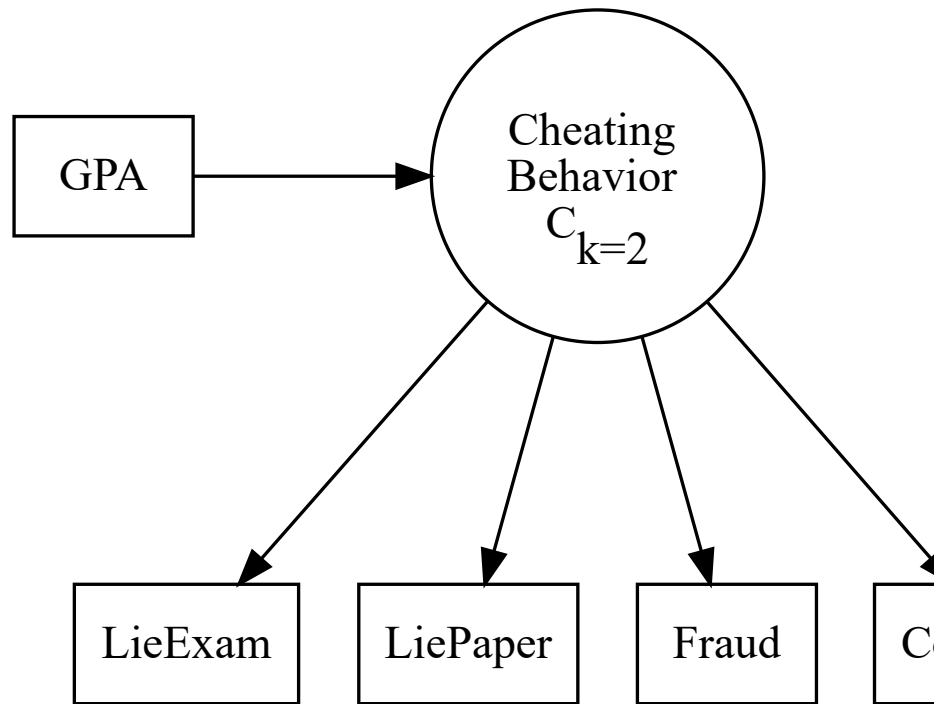
# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)
```

---

### 13.4.1 R3STEP

**R3STEP** incorporates latent class predictors with mixture models. However, it is recommended to use the manual three-step.

---



#### 13.4.1.1 Run the R3STEP model with gpa as the latent class predictor

---

```

m_stepr <- mplusObject(
  TITLE = "R3STEP - GPA as Predictor",
  VARIABLE =
    "categorical = lieexam-copyexam;
    usevar = lieexam-copyexam;
    auxiliary = gpa (R3STEP);
    classes = c(2);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors = 10;",

  OUTPUT = "sampstat patterns tech11 tech14;",

  PLOT =
    "type = plot3;
    series = lieexam-copyexam(*)",

  usevariables = colnames(df_cheat),
  rdata = df_cheat)

m_stepr_fit <- mplusModeler(m_stepr,
  dataout=here("three_step", "auto_3step", "r3step.dat"),
  modelout=here("three_step", "auto_3step", "c2_r3step.inp") ,
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

#### 13.4.1.2 Regression slopes and odds ratios

TESTS OF CATEGORICAL LATENT VARIABLE MULTINOMIAL LOGISTIC REGRESSIONS USING THE 3-STEP PROCEDURE

WARNING: LISTWISE DELETION IS APPLIED TO THE AUXILIARY VARIABLES IN THE ANALYSIS. TO AVOID LISTWISE DELETION, DATA IMPUTATION CAN BE USED

FOR THE AUXILIARY VARIABLES FOLLOWED BY ANALYSIS WITH TYPE=IMPUTATION.

NUMBER OF DELETED OBSERVATIONS: 4

NUMBER OF OBSERVATIONS USED: 315

|                                          |     |  | Estimate | S.E.  | Est./S.E. | Two-Tailed<br>P-Value |
|------------------------------------------|-----|--|----------|-------|-----------|-----------------------|
| C#1                                      | ON  |  |          |       |           |                       |
|                                          | GPA |  | -0.698   | 0.255 | -2.739    | 0.006                 |
| Intercepts                               |     |  |          |       |           |                       |
|                                          | C#1 |  | -0.241   | 0.460 | -0.523    | 0.601                 |
| Parameterization using Reference Class 1 |     |  |          |       |           |                       |
| C#2                                      | ON  |  |          |       |           |                       |
|                                          | GPA |  | 0.698    | 0.255 | 2.739     | 0.006                 |
| Intercepts                               |     |  |          |       |           |                       |
|                                          | C#2 |  | 0.241    | 0.460 | 0.523     | 0.601                 |

ODDS RATIOS FOR TESTS OF CATEGORICAL LATENT VARIABLE MULTINOMIAL LOGISTIC REGRESSIONS  
USING THE 3-STEP PROCEDURE

|                                          |     |  | Estimate | S.E.  | 95% C.I.   |            |
|------------------------------------------|-----|--|----------|-------|------------|------------|
|                                          |     |  |          |       | Lower 2.5% | Upper 2.5% |
| C#1                                      | ON  |  |          |       |            |            |
|                                          | GPA |  | 0.498    | 0.127 | 0.302      | 0.820      |
| Parameterization using Reference Class 1 |     |  |          |       |            |            |
| C#2                                      | ON  |  |          |       |            |            |
|                                          | GPA |  | 2.009    | 0.512 | 1.220      | 3.310      |



## Chapter 14

# Manual 3-Step Distal Only

---

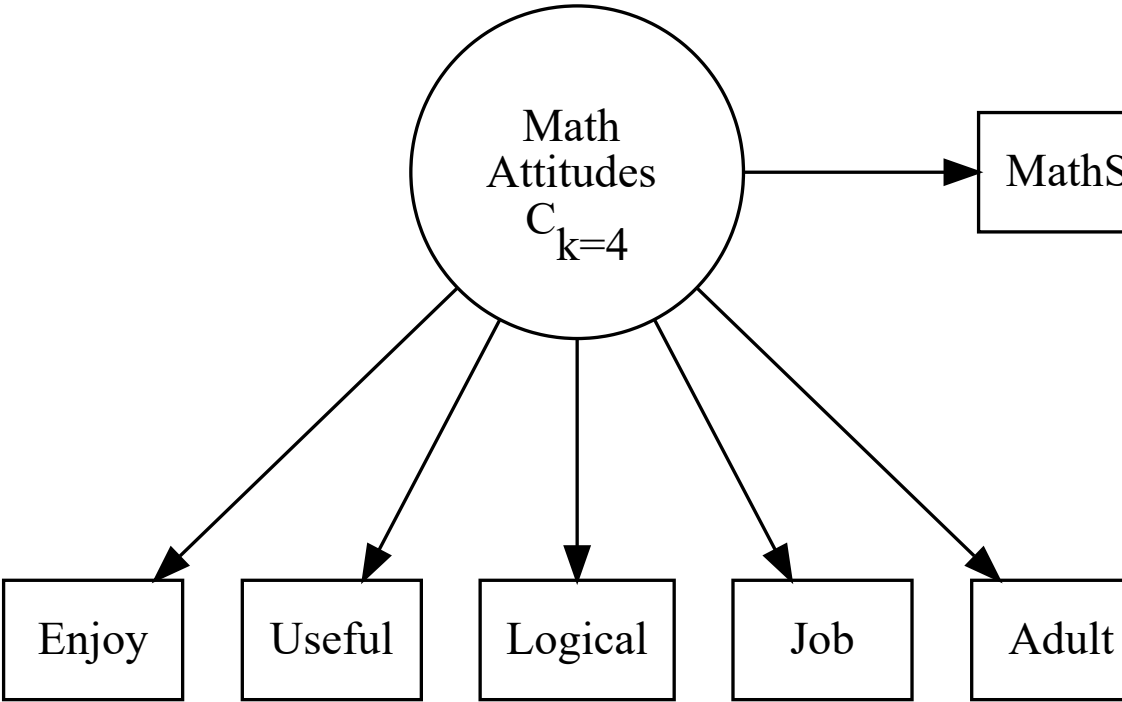
### Data source:

1. The first examples utilizes the public-use dataset, *The Longitudinal Survey of American Youth* (LSAY): See documentation [here](#)
  2. The second example utilizes a dataset on undergraduate *Cheating* available from the poLCA package (Dayton, 1998): See documentation [here](#)
- 

## 14.1 Load packages

```
library(MplusAutomation)
library(tidyverse) #collection of R packages designed for data science
library(here) #helps with filepaths
library(janitor) #clean_names
library(gt) # create tables
library(cowplot) # a ggplot theme
library(DiagrammeR) # create path diagrams
library(glue) # allows us to paste expressions into R code
library(data.table) # used for `melt()` function
library(poLCA)
library(reshape2)
```

Our example is Math IRT Score as a distal outcome of Math Attitude classes



### LCA Indicators & Auxiliary Variables: Math Attitudes Example

| Name                | Variable Description                            |
|---------------------|-------------------------------------------------|
| enjoy               | I enjoy math.                                   |
| useful              | Math is useful in everyday problems.            |
| logical             | Math helps a person think logically.            |
| job                 | It is important to know math to get a good job. |
| adult               | I will use math in many ways as an adult.       |
| Auxiliary Variables |                                                 |
| math_irt            | Standardized IRT math test score - 12th grade.  |

Application: Longitudinal Study of American Youth, **Math Attitudes**

The data can be found in the `data` folder and is called `lsay_subset.csv`.

```
lsay_data <- read_csv(here("three_step", "data", "lsay_subset.csv")) %>%
  clean_names() %>% # make variable names lowercase
  mutate(female = recode(gender, `1` = 0, `2` = 1)) # relabel values from 1,2 to 0,1
```

## 14.2 Descriptive Statistics

```
dframe <- lsay_data %>%
  pivot_longer(
    c(enjoy, useful, logical, job, adult),
    names_to = "Variable"
  ) %>%
  group_by(Variable) %>%
  summarise(
```

## LCA Indicator Endorsement

| Variable | Proportion Endorsed | Count |
|----------|---------------------|-------|
| adult    | 0.596               | 1858  |
| enjoy    | 0.573               | 1784  |
| job      | 0.625               | 1947  |
| logical  | 0.541               | 1686  |
| useful   | 0.589               | 1835  |

```

Count = sum(value == 1, na.rm = TRUE),
Total = n(),
.groups = "drop"
) %>%
mutate(`Proportion Endorsed` = round(Count / Total, 3)) %>%
dplyr::select(Variable, `Proportion Endorsed`, Count)

gt(dframe) %>%
  tab_header(
    title = md("**LCA Indicator Endorsement**"),
    subtitle = md("&nbsp;")
  ) %>%
  tab_options(
    column_labels.font.weight = "bold",
    row_group.font.weight = "bold"
  )

```

Math IRT Score

```

summary(lsay_data$math_irt)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
#>  26.57  50.00   59.30   58.81  68.21   94.19    875

```

---

## 14.3 Manual ML Three-step

---

### 14.3.1 Step 1 - Class Enumeration w/ Auxiliary Specification

---

This step is done after class enumeration (or after you have selected the best latent class model). In this example, the four class model was the best. Now, we re-estimate the four-class model using `optseed` for efficiency. The difference here is the `SAVEDATA` command, where I can save the posterior probabilities and the modal class assignment that will be used in steps two and three.

```
step1 <- mplusObject(
  TITLE = "Step 1 - Three-Step using LSAL",
  VARIABLE =
    "categorical = enjoy useful logical job adult;
    usevar = enjoy useful logical job adult;

    classes = c(4);

    auxiliary = math_irt ! distal outcome ",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;
    optseed = 568405;",

  SAVEDATA =
    "File=savedata_dis.dat;
    Save=cprob;",

  OUTPUT = "residual tech11 tech14",

  PLOT =
    "type = plot3;
    series = enjoy-adult(*)",

  usevariables = colnames(lsay_data),
  rdata = lsay_data)

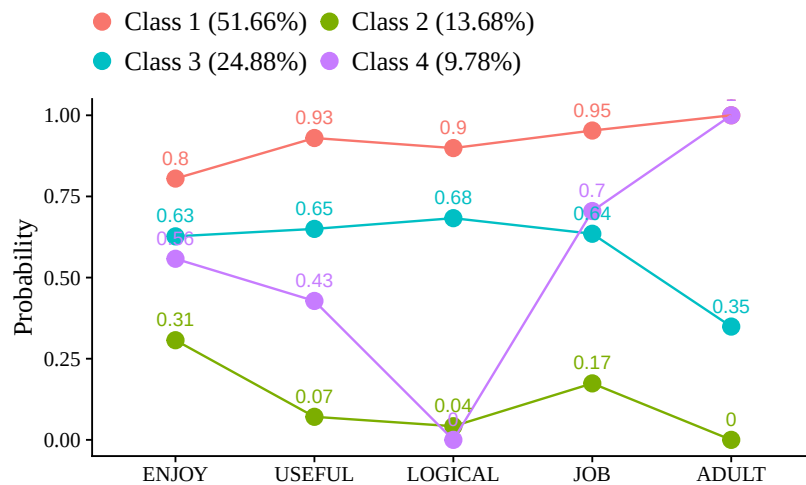
step1_fit <- mplusModeler(step1,
  dataout=here("three_step", "manual_3step", "Step1.dat"),
  modelout=here("three_step", "manual_3step", "one_dis.inp")
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

---

```
source(here("functions", "plot_lca.R"))
output_lsay <- readModels(here("three_step", "manual_3step", "one_dis.out"))

plot_lca(model_name = output_lsay)
```

### Step 1 – Three-Step using LSAL Probability Plot



### 14.3.2 Step 2 - Determine Measurement Error

---

Extract logits for the classification probabilities for the most likely latent class

```
logit_cprobs <- as.data.frame(output_lsay[["class_counts"]]  
                             [["logitProbs.mostLikely"]])
```

Extract saved dataset which is part of the mplusObject “step1\_fit”

```
savedata <- as.data.frame(output_lsay[["savedata"]])
```

Rename the column in savedata named “C” and change to “N”

```
colnames(savedata)[colnames(savedata)=="C"] <- "N"
```

---

### 14.3.3 Step 3 - LCA Auxiliary Variable Model with 1 Distal Outcome

---

```
step3 <- mplusObject(
  TITLE = "Step3 - 3step LSAY",

  VARIABLE =
    "nominal=N;
    usevar = n;

    classes = c(4);

    usevar = math_irt;" ,

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;",

  MODEL =
    glue(
      " %OVERALL%

      %C#1%
      [n#1@{logit_cprobs[1,1]}}; ! MUST EDIT if you do not have a 4-class model.
      [n#2@{logit_cprobs[1,2]}};
      [n#3@{logit_cprobs[1,3]}};

      [math_irt](m1);      ! conditional distal mean
      math_irt;           ! conditional distal variance (freely estimated)

      %C#2%
      [n#1@{logit_cprobs[2,1]}};
      [n#2@{logit_cprobs[2,2]}};
      [n#3@{logit_cprobs[2,3]}};
```



```

[math_irt](m2);
math_irt;

%C#3%
[n#1@{logit_cprobs[3,1]}];
[n#2@{logit_cprobs[3,2]}];
[n#3@{logit_cprobs[3,3]}];

[math_irt](m3);
math_irt;

%C#4%
[n#1@{logit_cprobs[4,1]}];
[n#2@{logit_cprobs[4,2]}];
[n#3@{logit_cprobs[4,3]}];

[math_irt](m4);
math_irt; "),

MODELCONSTRAINT =
  "New (diff12 diff13 diff23
    diff14 diff24 diff34);

    diff12 = m1-m2; ! test pairwise distal mean differences
    diff13 = m1-m3;
    diff23 = m2-m3;
    diff14 = m1-m4;
    diff24 = m2-m4;
    diff34 = m3-m4;",

MODELTEST = "      ! omnibus test of distal means
  m1=m2;
  m2=m3;
  m3=m4;",

usevariables = colnames(savedata),
rdata = savedata)

step3_fit <- mplusModeler(step3,
  dataout=here("three_step", "manual_3step", "Step3.dat"),
  modelout=here("three_step", "manual_3step", "three_dis.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

## Wald Test Distal Means (Math IRT Scores)

| Wald Test ( <i>df</i> ) | <i>p</i> -value |
|-------------------------|-----------------|
| 69.108 (3)              | <.001*          |

## 14.3.3.1 Wald Test Table

This is testing if there is a relation between the latent class variable and the distal outcome (mathirt)

```
modelParams <- readModels(here("three_step", "manual_3step", "three_dis.out"))

# Extract information as data frame
wald <- as.data.frame(modelParams[["summaries"]]) %>%
  dplyr::select(WaldChiSq_Value:WaldChiSq_PValue) %>%
  mutate(WaldChiSq_DF = paste0("(", WaldChiSq_DF, ")")) %>%
  unite(wald_test, WaldChiSq_Value, WaldChiSq_DF, sep = " ") %>%
  rename(pval = WaldChiSq_PValue) %>%
  mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
                      ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001),
                                                scales::number(pval, accuracy = .001))))

# Create table

wald_table <- wald %>%
  gt() %>%
  tab_header(
    title = "Wald Test Distal Means (Math IRT Scores)" %>%
    cols_label(
      wald_test = md("Wald Test (*df*)"),
      pval = md("*p*-value") %>%
    cols_align(aligned = "center") %>%
    opt_align_table_header(aligned = "left") %>%
    gt::tab_options(table.font.names = "serif")

wald_table
```

Save figure

```
gtsave(wald_table, here("figures", "wald_table.docx"))
```

## 14.3.3.2 Table of Pairwise Distal Outcome Differences

```

modelParams <- readModels(here("three_step", "manual_3step", "three_dis.out"))

# Extract information as data frame
diff <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(grepl("DIFF", param)) %>%
  dplyr::select(param:pval) %>%
  mutate(se = paste0("(", format(round(se,2), nsmall =2), ")")) %>%
  unite(estimate, est, se, sep = " ") %>%
  mutate(param = str_remove(param, "DIFF"),
         param = as.numeric(param)) %>%
  separate(param, into = paste0("Group", 1:2), sep = 1) %>%
  mutate(class = paste0("Class ", Group1, " vs ", Group2)) %>%
  dplyr::select(class, estimate, pval) %>%
  mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
                      ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001), "*"),
                             scales::number(pval, accuracy = .001))))

# Create table
diff %>%
  gt() %>%
  tab_header(
    title = "Distal Outcome Differences") %>%
  cols_label(
    class = "Class",
    estimate = md("Mean (*se*)"),
    pval = md("*p*-value") %>%
  sub_missing(1:3,
              missing_text = "") %>%
  cols_align(aligned = "center") %>%
  opt_align_table_header(aligned = "left") %>%
  gt::tab_options(table.font.names = "serif")

```

## 14.3.3.3 Plot Distal Outcome Means

```

modelParams <- readModels(here("three_step", "manual_3step", "three_dis.out"))

# Extract class size

```

## Distal Outcome Differences

| Class        | Mean ( <i>se</i> ) | <i>p</i> -value |
|--------------|--------------------|-----------------|
| Class 1 vs 2 | 6.731 (0.98)       | <.001*          |
| Class 1 vs 3 | 3.103 (0.96)       | 0.001*          |
| Class 2 vs 3 | -3.628 (1.31)      | 0.006*          |
| Class 1 vs 4 | 7.027 (1.29)       | <.001*          |
| Class 2 vs 4 | 0.296 (1.45)       | 0.838           |
| Class 3 vs 4 | 3.924 (1.49)       | 0.008*          |

```

c_size <- as.data.frame(modelParams[["class_counts"]][["modelEstimated"]][["proportion"]])
  rename("cs" = 1) %>%
  mutate(cs = round(cs*100, 2))

c_size_val <- paste0("C", 1:nrow(c_size), glue(" ({c_size[1:nrow(c_size),]}%)"))

# Extract information as data frame
estimates <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(paramHeader == "Means") %>%
  dplyr::select(param, est, se) %>%
  filter(param == "MATH_IRT") %>%
  mutate(across(c(est, se), as.numeric)) %>%
  mutate(LatentClass = c_size_val)

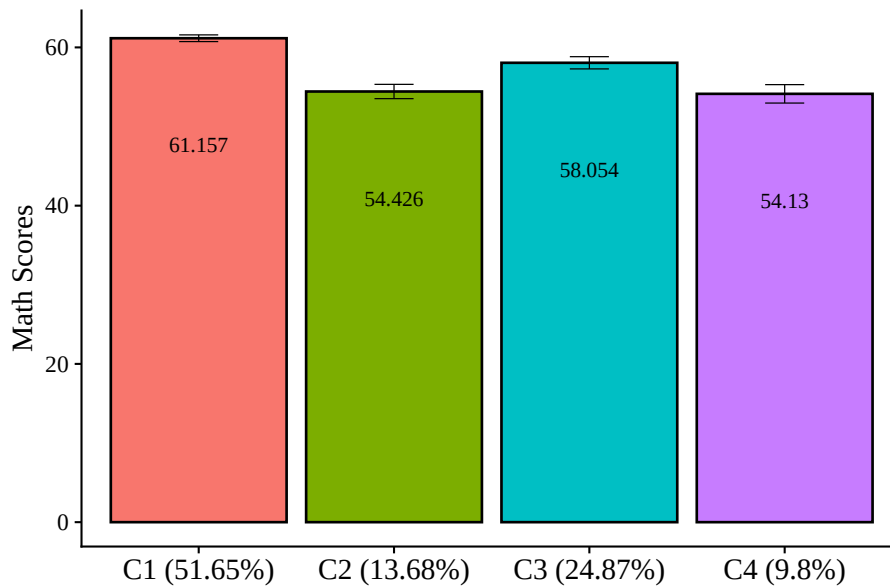
# Add labels (NOTE: You must change the labels to match the significance testing!!)
#value_labels <- paste0(estimates$est, c("a", " bc", " abd", " cd"))

estimates$LatentClass <- fct_inorder(estimates$LatentClass)

# Plot bar graphs
estimates %>%
  ggplot(aes(x=LatentClass, y = est, fill = LatentClass)) +
  geom_col(position = "dodge", color = "black") +
  geom_errorbar(aes(ymin=est-se, ymax=est+se),
    linewidth=.3, # Thinner lines
    width=.2,
    position=position_dodge(.9)) +
  geom_text(aes(label = est),
    family = "serif", size = 4,
    position=position_dodge(.9),
    vjust = 8) +

```

```
# scale_fill_grey(start = .4, end = .7) + # Remove for colorful bars
labs(y="Math Scores", x="") +
theme_cowplot() +
theme(text = element_text(family = "serif", size = 15),
      axis.text.x = element_text(size=15),
      legend.position="none")
```



```
# Save plot
ggsave(here("figures", "ManualDistal_Plot.jpeg"),
      dpi=300, width=10, height = 7, units="in")
```

---

## 14.4 Automated Three-Step

---

*Application: Undergraduate Cheating behavior*

“Dichotomous self-report responses by 319 undergraduates to four questions about cheating behavior” (poLCA, 2016).

Prepare data

```
data(cheating)

cheating <- cheating %>% clean_names()

df_cheat <- cheating %>%
  dplyr::select(1:4) %>%
  mutate_all(funs(.-1)) %>%
  mutate(gpa = cheating$gpa)

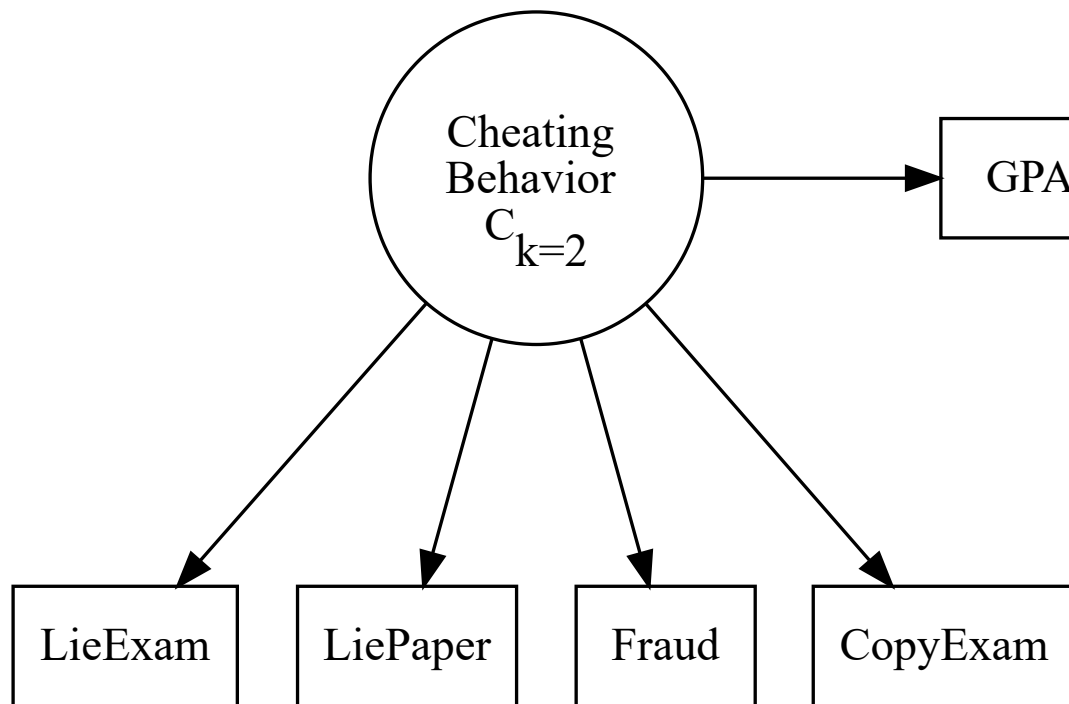
# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)
```

---

#### 14.4.1 DU3STEP

**DU3STEP** incorporates distal outcome variables (assumed to have unequal means and variances) with mixture models.

---



#### 14.4.1.1 Run the DU3step model with gpa as distal outcome

```
m_stepdu <- mplusObject(
  TITLE = "DU3STEP - GPA as Distal",
  VARIABLE =
    "categorical = lieexam-copyexam;
    usevar = lieexam-copyexam;
    auxiliary = gpa (du3step);
    classes = c(2);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors = 10;",

  OUTPUT = "sampstat patterns tech11 tech14;",

  PLOT =
    "type = plot3;
    series = lieexam-copyexam(*)",

  usevariables = colnames(df_cheat),
  rdata = df_cheat)

m_stepdu_fit <- mplusModeler(m_stepdu,
  dataout=here("three_step", "auto_3step", "du3step.dat"),
  modelout=here("three_step", "auto_3step", "c2_du3step.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

#### 14.4.1.2 Plot Distal Outcome mean differences

```
modelParams <- readModels(here("three_step", "auto_3step", "c2_du3step.out"))

# Extract class size
c_size <- as.data.frame(modelParams[["class_counts"]][["modelEstimated"]][["proportion"]])
  rename("cs" = 1) %>%
```



```

mutate(cs = round(cs*100, 2))

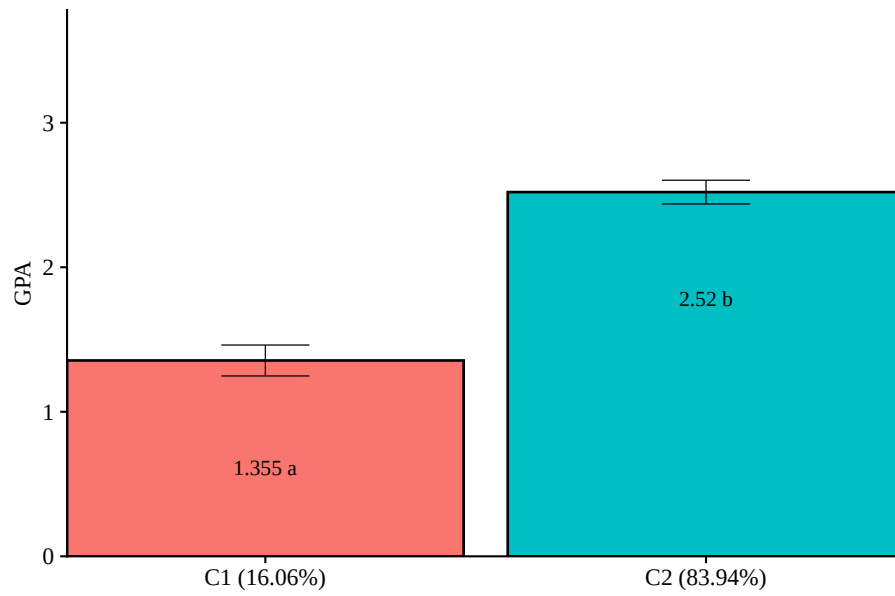
c_size_val <- paste0("C", 1:nrow(c_size), glue(" ({c_size[1:nrow(c_size),]}%)"))

# Extract information as data frame
estimates <- as.data.frame(modelParams[["lcCondMeans"]][["overall"]]) %>%
  reshape2::melt(id.vars = "var") %>%
  mutate(variable = as.character(variable),
         LatentClass = case_when(
           endsWith(variable, "1") ~ c_size_val[1],
           endsWith(variable, "2") ~ c_size_val[2])) %>% #Add to this based on the number of classes
  head(-3) %>%
  pivot_wider(names_from = variable, values_from = value) %>%
  unite("mean", contains("m"), na.rm = TRUE) %>%
  unite("se", contains("se"), na.rm = TRUE) %>%
  mutate(across(c(mean, se), as.numeric))

# Add labels (NOTE: You must change the labels to match the significance testing!!)
value_labels <- paste0(estimates$mean, c(" a", " b"))

# Plot bar graphs
estimates %>%
  ggplot(aes(fill = LatentClass, y = mean, x = LatentClass)) +
  geom_bar(position = "dodge", stat = "identity", color = "black") +
  geom_errorbar(aes(ymin=mean-se, ymax=mean+se),
               size=.3,
               width=.2,
               position=position_dodge(.9)) +
  geom_text(aes(y = mean, label = value_labels),
            family = "serif", size = 4,
            position=position_dodge(.9),
            vjust = 8) +
  #scale_fill_grey(start = .5, end = .7) +
  labs(y="GPA", x="") +
  theme_cowplot() +
  theme(text = element_text(family = "serif", size = 12),
        axis.text.x = element_text(size=12),
        legend.position="none") +
  coord_cartesian(expand = FALSE,
                  ylim=c(0,max(estimates$mean*1.5))) # Change ylim based on distal outcome range

```



```
# Save plot
ggsave(here("figures", "Du3STEP_plot.jpeg"),
       dpi=300, width=10, height = 7, units="in")
```

## Chapter 15

# Manual 3-Step Covariate and Distal

---

### Data source:

This example utilizes the public-use dataset, *The Longitudinal Survey of American Youth* (LSAY): See documentation [here](#)

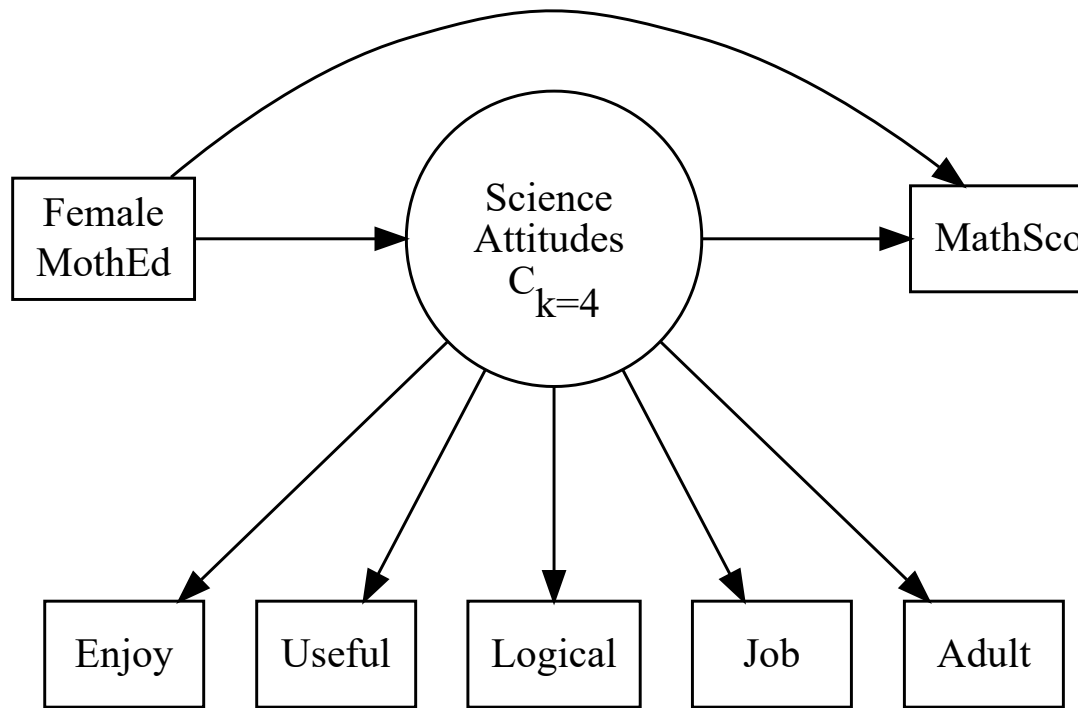
---

### 15.1 Load packages

```
library(MplusAutomation)
library(tidyverse) #collection of R packages designed for data science
library(here) #helps with filepaths
library(janitor) #clean_names
library(gt) # create tables
library(cowplot) # a ggplot theme
library(DiagrammeR) # create path diagrams
library(glue) # allows us to paste expressions into R code
library(data.table) # used for `melt()` function
library(poLCA)
library(reshape2)
```

---

This example uses Gender and Mother's Education as predictors of latent class membership and Math IRT scores as a distal outcome in a single model.



### LCA Indicators & Auxiliary Variables: Math Attitudes Example

| Name                | Variable Description                                                           |
|---------------------|--------------------------------------------------------------------------------|
| enjoy               | I enjoy math.                                                                  |
| useful              | Math is useful in everyday problems.                                           |
| logical             | Math helps a person think logically.                                           |
| job                 | It is important to know math to get a good job.                                |
| adult               | I will use math in many ways as an adult.                                      |
| Auxiliary Variables |                                                                                |
| female              | Self-reported student gender (0=Male, 1=Female).                               |
| math_irt            | Standardized IRT math test score - 12th grade.                                 |
| mothed              | Level of education: (1) less than high school, (2) high school diploma, (3) so |

Application: Longitudinal Study of American Youth, **Science Attitudes**

The data can be found in the `data` folder and is called `lsay_subset.csv`.

```
lsay_data <- read_csv(here("three_step", "data", "lsay_subset.csv")) %>%
  clean_names() %>% # make variable names lowercase
  mutate(female = recode(gender, `1` = 0, `2` = 1)) # relabel values from 1,2 to 0,1
```

## 15.2 Descriptive Statistics

```
dframe <- lsay_data %>%
  pivot_longer(
    c(enjoy, useful, logical, job, adult),
    names_to = "Variable"
```

## LCA Indicator Endorsement

| Variable | Proportion Endorsed | Count |
|----------|---------------------|-------|
| adult    | 0.596               | 1858  |
| enjoy    | 0.573               | 1784  |
| job      | 0.625               | 1947  |
| logical  | 0.541               | 1686  |
| useful   | 0.589               | 1835  |

```

) %>%
group_by(Variable) %>%
summarise(
  Count = sum(value == 1, na.rm = TRUE),
  Total = n(),
  .groups = "drop"
) %>%
mutate(`Proportion Endorsed` = round(Count / Total, 3)) %>%
dplyr::select(Variable, `Proportion Endorsed`, Count)

gt(dframe) %>%
  tab_header(
    title = md("**LCA Indicator Endorsement**"),
    subtitle = md("&nbsp;")
  ) %>%
  tab_options(
    column_labels.font.weight = "bold",
    row_group.font.weight = "bold"
  )

```

Gender

```

summary(factor(lsay_data$mothed))
#>   1    2    3    4    5 NA's
#> 452 1732 304 368 158 102

```

Mother's Education

```

summary(factor(lsay_data$female))
#>   0    1
#> 1490 1626

```

Math IRT Score

```
summary(lsay_data$math_irt)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
#>  26.57   50.00   59.30   58.81   68.21   94.19     875
```

## 15.3 Manual ML Three-step

### 15.3.1 Step 1 - Class Enumeration w/ Auxiliary Specification

This step is done after class enumeration (or after you have selected the best latent class model). In this example, the four class model was the best. Now, we re-estimate the four-class model using `optseed` for efficiency. The difference here is the `SAVEDATA` command, where I can save the posterior probabilities and the modal class assignment that will be used in steps two and three.

```
step1 <- mplusObject(
  TITLE = "Step 1 - Three-Step using LSAL",
  VARIABLE =
    "categorical = enjoy useful logical job adult;
     usevar = enjoy useful logical job adult;

     classes = c(4);

     auxiliary =    ! list all potential covariates and distals here
     female mothed    ! covariate
     math_irt;        ! distal math test score in 12th grade ",

  ANALYSIS =
    "estimator = mlr;
     type = mixture;
     starts = 0;
     optseed = 568405;",

  SAVEDATA =
```



```

"File=savedata.dat;
Save=cprob;";

OUTPUT = "residual tech11 tech14",

PLOT =
  "type = plot3;
  series = enjoy-adult(*)";

usevariables = colnames(lsay_data),
rdata = lsay_data)

step1_fit <- mplusModeler(step1,
                          dataout=here("three_step", "manual_3step", "Step1.dat"),
                          modelout=here("three_step", "manual_3step", "one.inp"),
                          check=TRUE, run = TRUE, hashfilename = FALSE)

```

```

source(here("functions", "plot_lca.R"))
output_lsay <- readModels(here("three_step", "manual_3step", "one.out"))

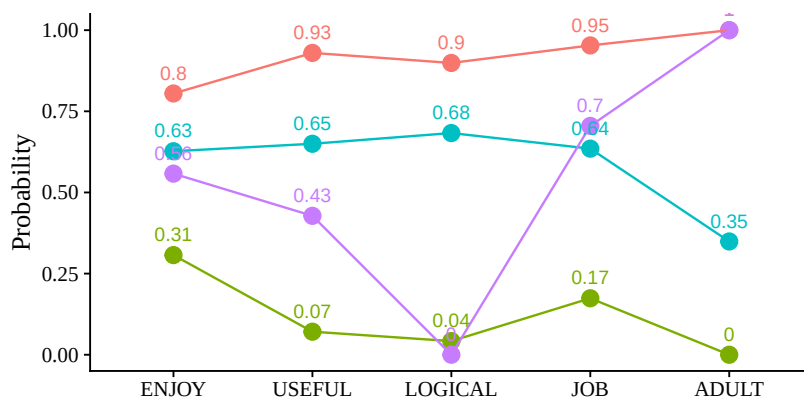
plot_lca(model_name = output_lsay)

```

### Step 1 – Three-Step using LSAL Probability Plot

● Class 1 (51.66%) ● Class 2 (13.68%)

● Class 3 (24.88%) ● Class 4 (9.78%)



### 15.3.2 Step 2 - Determine Measurement Error

---

Extract logits for the classification probabilities for the most likely latent class

```
logit_cprobs <- as.data.frame(output_lsay[["class_counts"]]  
                             [["logitProbs.mostLikely"]])
```

Extract saved dataset which is part of the mplusObject "step1\_fit"

```
savedata <- as.data.frame(output_lsay[["savedata"]])
```

Rename the column in savedata named "C" and change to "N"

```
colnames(savedata)[colnames(savedata)=="C"] <- "N"
```

---

### 15.3.3 Step 3 - LCA Auxiliary Variable Model with 2 covariates and 1 distal outcome

---

Model with 2 covariates (gender and mother's education) and 1 distal outcome (math IRT scores)

```
step3 <- mplusObject(  
  TITLE = "Step3 - 3step LSAY",  
  
  VARIABLE =  
    "nominal=N;  
    usevar = n;  
  
    classes = c(4);  
  
    usevar = female mothed math_irt;" ,  
  
  ANALYSIS =  
    "estimator = mlr;  
    type = mixture;  
    starts = 0;" ,
```

```

DEFINE =
  "center female mothed (grandmean);",

MODEL =
  glue(
" %OVERALL%

math_irt on female mothed; ! covariate as a related to the distal outcome
C on female (f1-f3);
c on mothed (e1-e3);      ! covariate as predictor of C

  %C#1%
[n#1@{logit_cprobs[1,1]}}; ! MUST EDIT if you do not have a 4-class model.
[n#2@{logit_cprobs[1,2]}};
[n#3@{logit_cprobs[1,3]}};

[math_irt](m1);      ! conditional distal mean
math_irt;            ! conditional distal variance (freely estimated)

  %C#2%
[n#1@{logit_cprobs[2,1]}};
[n#2@{logit_cprobs[2,2]}};
[n#3@{logit_cprobs[2,3]}};

[math_irt](m2);
math_irt;

  %C#3%
[n#1@{logit_cprobs[3,1]}};
[n#2@{logit_cprobs[3,2]}};
[n#3@{logit_cprobs[3,3]}};

[math_irt](m3);
math_irt;

  %C#4%
[n#1@{logit_cprobs[4,1]}};
[n#2@{logit_cprobs[4,2]}};
[n#3@{logit_cprobs[4,3]}};

[math_irt](m4);
math_irt; "),

MODELCONSTRAINT =
  "New (diff12 diff13 diff23

```

```

diff14 diff24 diff34

d_fem_12 d_fem_13
d_fem_23

d_ed_12 d_ed_13
d_ed_23
);

diff12 = m1-m2; ! test pairwise distal mean differences
diff13 = m1-m3;
diff23 = m2-m3;
diff14 = m1-m4;
diff24 = m2-m4;
diff34 = m3-m4;

d_fem_12 = f1-f2;
d_fem_13 = f1-f3;
d_fem_23 = f2-f3;

d_ed_12 = e1-e2;
d_ed_13 = e1-e3;
d_ed_23 = e2-e3;

",

MODELTEST = "      ! omnibus test of distal means
! m1=m2;
! m2=m3;
! m3=m4;

! f1=f2;      ! omnibus test of covariate logits (female)
! f1=f3;

e1=e2;      ! omnibus test of covariate logits (mothers ed)
e1=e3;

",

usevariables = colnames(savedata),
rdata = savedata)

step3_fit <- mplusModeler(step3,
  dataout=here("three_step", "manual_3step", "Step3.dat"),
  modelout=here("three_step", "manual_3step", "three.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

## Wald Test Distal Means (Math IRT Scores)

| Wald Test ( <i>df</i> ) | <i>p</i> -value |
|-------------------------|-----------------|
| 2.475 (2)               | 0.290           |

## 15.3.3.1 Wald Test Table

This is testing if there is a relation between the latent class variable and the distal outcome (mathirt)

```
modelParams <- readModels(here("three_step", "manual_3step", "three.out"))

# Extract information as data frame
wald <- as.data.frame(modelParams[["summaries"]]) %>%
  dplyr::select(WaldChiSq_Value:WaldChiSq_PValue) %>%
  mutate(WaldChiSq_DF = paste0("(", WaldChiSq_DF, ")")) %>%
  unite(wald_test, WaldChiSq_Value, WaldChiSq_DF, sep = " ") %>%
  rename(pval = WaldChiSq_PValue) %>%
  mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
    ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001), "*"),
    scales::number(pval, accuracy = .001))))

# Create table

wald_table <- wald %>%
  gt() %>%
  tab_header(
    title = "Wald Test Distal Means (Math IRT Scores)" %>%
    cols_label(
      wald_test = md("Wald Test (*df*"),
      pval = md("*p*-value")) %>%
    cols_align(aligned = "center") %>%
    opt_align_table_header(aligned = "left") %>%
    gt::tab_options(table.font.names = "serif")

wald_table
```

Save figure

```
gtsave(wald_table, here("figures", "wald_table.docx"))
```

### 15.3.3.2 Table of Pairwise Distal Outcome Differences

```
modelParams <- readModels(here("three_step", "manual_3step", "three.out"))

# Extract information as data frame
diff <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(grepl("DIFF", param)) %>%
  dplyr::select(param:pval) %>%
  mutate(se = paste0("(", format(round(se, 2), nsmall = 2), ")")) %>%
  unite(estimate, est, se, sep = " ") %>%
  mutate(param = str_remove(param, "DIFF"),
         param = as.numeric(param)) %>%
  separate(param, into = paste0("Group", 1:2), sep = 1) %>%
  mutate(class = paste0("Class ", Group1, " vs ", Group2)) %>%
  dplyr::select(class, estimate, pval) %>%
  mutate(pval = ifelse(pval < 0.001, paste0("<.001*"),
                      ifelse(pval < 0.05, paste0(scales::number(pval, accuracy = .001),
                                                scales::number(pval, accuracy = .001))))

# Create table
diff %>%
  gt() %>%
  tab_header(
    title = "Distal Outcome Differences" %>%
  cols_label(
    class = "Class",
    estimate = md("Mean (*se*)"),
    pval = md("*p*-value") %>%
  sub_missing(1:3,
    missing_text = "") %>%
  cols_align(align = "center") %>%
  opt_align_table_header(align = "left") %>%
  gt::tab_options(table.font.names = "serif")
```

## Distal Outcome Differences

| Class        | Mean ( <i>se</i> ) | <i>p</i> -value |
|--------------|--------------------|-----------------|
| Class 1 vs 2 | 6.471 (0.95)       | <.001*          |
| Class 1 vs 3 | 3.016 (0.96)       | 0.002*          |
| Class 2 vs 3 | -3.455 (1.31)      | 0.008*          |
| Class 1 vs 4 | 5.982 (1.24)       | <.001*          |
| Class 2 vs 4 | -0.489 (1.39)      | 0.725           |
| Class 3 vs 4 | 2.966 (1.46)       | 0.042*          |

## 15.3.3.3 Plot Distal Outcome Means

```

modelParams <- readModels(here("three_step", "manual_3step", "three.out"))

# Extract class size
c_size <- as.data.frame(modelParams[["class_counts"]][["modelEstimated"]][["proportion"]]) %>%
  rename("cs" = 1) %>%
  mutate(cs = round(cs*100, 2))

c_size_val <- paste0("C", 1:nrow(c_size), glue(" ({c_size[1:nrow(c_size),]}%)"))

# Extract information as data frame
estimates <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(paramHeader == "Intercepts") %>%
  dplyr::select(param, est, se) %>%
  filter(param == "MATH_IRT") %>%
  mutate(across(c(est, se), as.numeric)) %>%
  mutate(LatentClass = c_size_val)

# Add labels (NOTE: You must change the labels to match the significance testing!!)
#value_labels <- paste0(estimates$est, c("a", " bc", " abd", " cd"))

estimates$LatentClass <- fct_inorder(estimates$LatentClass)

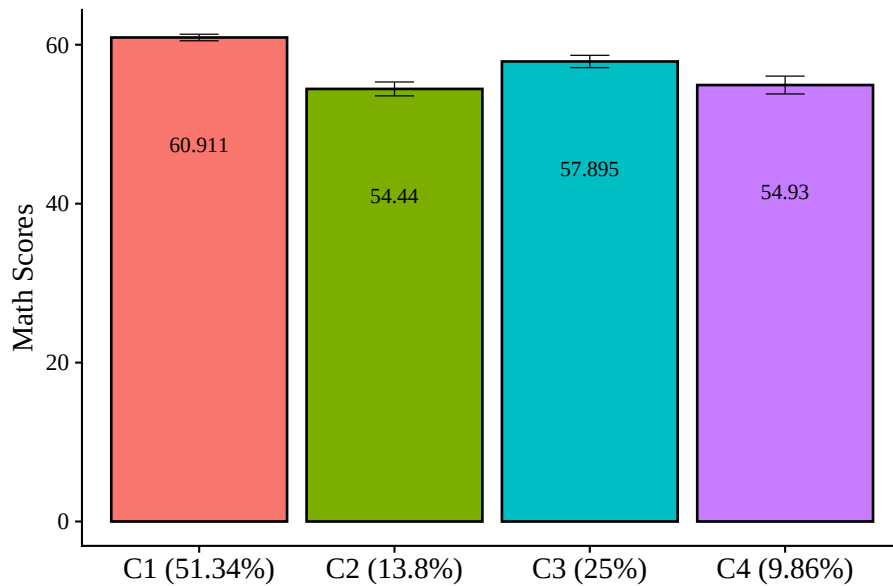
# Plot bar graphs
estimates %>%
  ggplot(aes(x=LatentClass, y = est, fill = LatentClass)) +
  geom_col(position = "dodge", stat = "identity", color = "black") +
  geom_errorbar(aes(ymin=est-se, ymax=est+se),
    size=.3,      # Thinner lines
    width=.2,

```

```

    position=position_dodge(.9)) +
  geom_text(aes(label = est),
    family = "serif", size = 4,
    position=position_dodge(.9),
    vjust = 8) +
  # scale_fill_grey(start = .4, end = .7) + # Remove for colorful bars
  labs(y="Math Scores", x="") +
  theme_cowplot() +
  theme(text = element_text(family = "serif", size = 15),
    axis.text.x = element_text(size=15),
    legend.position="none")

```



```

# Save plot
ggsave(here("figures", "ManualDistal_Plot.jpeg"),
  dpi=300, width=10, height = 7, units="in")

```

#### 15.3.3.4 Covariates Relations

As of 1/1/2026, this code is outdated. See here: <https://github.com/michaelhallquist/MplusAutomation/issues/228>



```

modelParams <- readModels(here("three_step", "manual_3step", "three.out"))

# Extract information as data frame
cov <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(str_detect(paramHeader, "^C#\\d+\\.ON$")) %>%
  mutate(param = str_replace(param, "FEMALE", "Gender")) %>% # Change this to your own covariates
  mutate(param = str_replace(param, "MOTHED", "Mother's Education")) %>%
  mutate(est = format(round(est, 3), nsmall = 3),
         se = round(se, 2),
         pval = round(pval, 3)) %>%
  mutate(latent_class = str_replace(paramHeader, "^C#(\\d+)\\.ON$", "Class \\1")) %>%
  dplyr::select(param, est, se, pval, latent_class) %>%
  mutate(se = paste0("(", format(round(se, 2), nsmall = 2), ")")) %>%
  unite(logit, est, se, sep = " ") %>%
  dplyr::select(param, logit, pval, latent_class) %>%
  mutate(pval = ifelse(pval < 0.001, paste0("<.001*"),
                      ifelse(pval < 0.05, paste0(scales::number(pval, accuracy = .001), "*"),
                             scales::number(pval, accuracy = .001))))

or <- as.data.frame(modelParams[["parameters"]][["odds"]]) %>%
  filter(str_detect(paramHeader, "^C#\\d+\\.ON$")) %>%
  mutate(param = str_replace(param, "FEMALE", "Gender")) %>% # Change this to your own covariates
  mutate(param = str_replace(param, "MOTHED", "Mother's Education")) %>%
  mutate(est = format(round(est, 3), nsmall = 3)) %>%
  mutate(latent_class = str_replace(paramHeader, "^C#(\\d+)\\.ON$", "Class \\1")) %>%
  mutate(CI = paste0("[", format(round(lower_2.5ci, 3), nsmall = 3), ", ", format(round(upper_2.5ci, 3), nsmall = 3), "]")) %>%
  dplyr::select(param, est, CI, latent_class) %>%
  rename(or = est)

combined <- or %>%
  full_join(cov) %>%
  dplyr::select(param, latent_class, logit, pval, or, CI)

# Create table
combined %>%
  gt(groupname_col = "latent_class", rowname_col = "param") %>%
  tab_header(
    title = "Predictors of Class Membership" %>%
  ) %>%
  cols_label(
    logit = md("Logit (*se*)"),
    or = md("Odds Ratio"),
    CI = md("95% CI"),
    pval = md("*p*-value") %>%
  )

```

```

sub_missing(1:3,
            missing_text = "") %>%
sub_values(values = c("999.000"), replacement = "-") %>%
cols_align(align = "center") %>%
opt_align_table_header(align = "left") %>%
gt::tab_options(table.font.names = "serif") %>%
tab_footnote(
  footnote = "Reference Class: 4",
  locations = cells_title(groups = "title")
)

```

### 15.3.3.5 Distal outcome regressed on the covariate

Is there a relation between the distal outcome (Math IRT Scores) and the covariate (Gender)?

```

modelParams <- readModels(here("three_step", "manual_3step", "three.out"))

# Extract information as data frame
donx <- as.data.frame(modelParams[["parameters"]][["unstandardized"]]) %>%
  filter(param %in% c("FEMALE", "MOTHEd")) %>%
  mutate(param = str_replace(param, "FEMALE", "Gender")) %>%
  mutate(param = str_replace(param, "MOTHEd", "Mother's Education")) %>%
  mutate(LatentClass = sub("^", "Class ", LatentClass)) %>%
  dplyr::select(!paramHeader) %>%
  mutate(se = paste0("(", format(round(se,2), nsmall =2), ")")) %>%
  unite(estimate, est, se, sep = " ") %>%
  dplyr::select(param, estimate, pval) %>%
  distinct(param, .keep_all=TRUE) %>%
  mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
                      ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001),
                      scales::number(pval, accuracy = .001))))

# Create table
donx %>%
  gt(groupname_col = "LatentClass", rowname_col = "param") %>%
  tab_header(
    title = "Gender Predicting Math Scores") %>%
  cols_label(
    estimate = md("Estimate (*se*"),

```

## Gender Predicting Math Scores

|                    | Estimate ( <i>se</i> ) | <i>p</i> -value |
|--------------------|------------------------|-----------------|
| Gender             | -1.527 (0.53)          | 0.004*          |
| Mother's Education | 3.589 (0.27)           | <.001*          |

```

pval = md("*p*-value")) %>%
sub_missing(1:3,
             missing_text = "") %>%
sub_values(values = c("999.000"), replacement = "-") %>%
cols_align(align = "center") %>%
opt_align_table_header(align = "left") %>%
gt::tab_options(table.font.names = "serif")

```



## Chapter 16

# Automatic 3-Step Covariate Only

---

### Data source:

This utilizes a dataset on undergraduate *Cheating* available from the poLCA package (Dayton, 1998): See documentation [here](#)

---

### 16.1 Load packages

```
library(MplusAutomation)
library(tidyverse) #collection of R packages designed for data science
library(here) #helps with filepaths
library(janitor) #clean_names
library(gt) # create tables
library(cowplot) # a ggplot theme
library(DiagrammeR) # create path diagrams
library(glue) # allows us to paste expressions into R code
library(data.table) # used for `melt()` function
library(poLCA)
library(reshape2)
```

---

## 16.2 Automated Three-Step

---

*Application: Undergraduate Cheating behavior*

“Dichotomous self-report responses by 319 undergraduates to four questions about cheating behavior” (poLCA, 2016).

---

Prepare data

```
data(cheating)

cheating <- cheating %>% clean_names()

df_cheat <- cheating %>%
  dplyr::select(1:4) %>%
  mutate_all(funs(.-1)) %>%
  mutate(gpa = cheating$gpa)

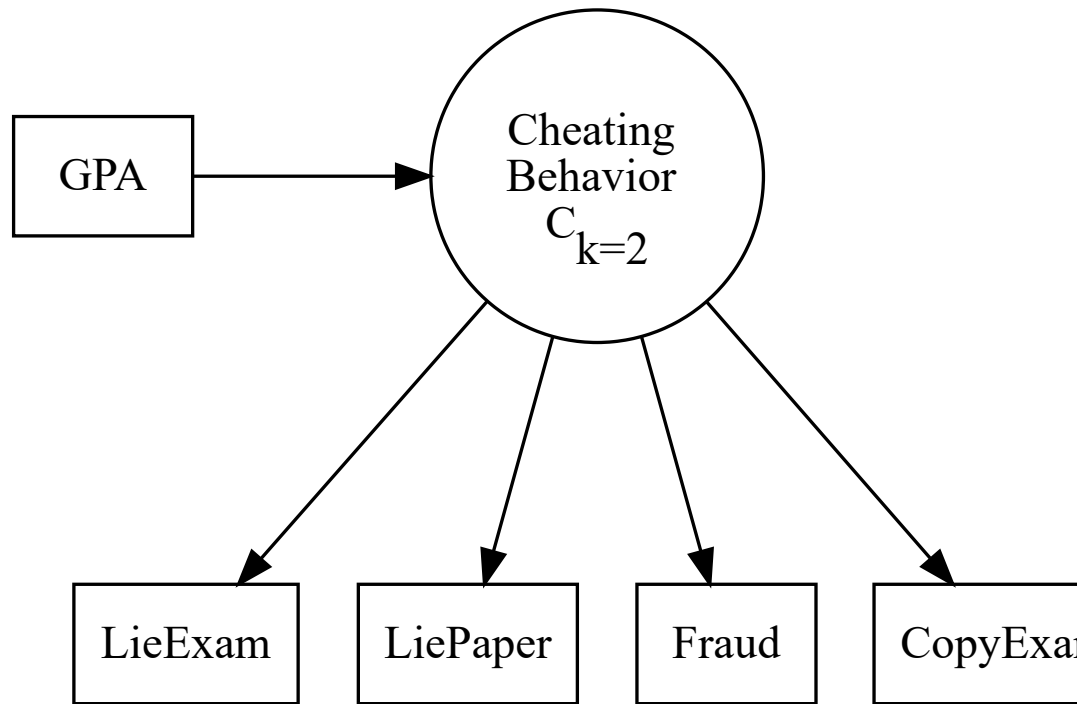
# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)
```

---

### 16.2.1 R3STEP

**R3STEP** incorporates latent class predictors with mixture models. However, it is recommended to use the manual three-step.

---



### 16.2.1.1 Run the R3STEP model with gpa as the latent class predictor

```
m_stepr <- mplusObject(
  TITLE = "R3STEP - GPA as Predictor",
  VARIABLE =
    "categorical = lieexam-copyexam;
    usevar = lieexam-copyexam;
    auxiliary = gpa (R3STEP);
    classes = c(2);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors = 10;",

  OUTPUT = "sampstat patterns tech11 tech14;",

  PLOT =
    "type = plot3;
    series = lieexam-copyexam(*)",

  usevariables = colnames(df_cheat),
  rdata = df_cheat)

m_stepr_fit <- mplusModeler(m_stepr,
  dataout=here("three_step", "auto_3step", "r3step.dat"),
  modelout=here("three_step", "auto_3step", "c2_r3step.inp")
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

### 16.2.1.2 Regression slopes and odds ratios

TESTS OF CATEGORICAL LATENT VARIABLE MULTINOMIAL LOGISTIC REGRESSIONS USING THE 3-STEP PROCEDURE

WARNING: LISTWISE DELETION IS APPLIED TO THE AUXILIARY VARIABLES IN THE ANALYSIS. TO AVOID LISTWISE DELETION, DATA IMPUTATION CAN BE USED



FOR THE AUXILIARY VARIABLES FOLLOWED BY ANALYSIS WITH TYPE=IMPUTATION.

NUMBER OF DELETED OBSERVATIONS: 4

NUMBER OF OBSERVATIONS USED: 315

|                                          |    | Estimate | S.E.  | Est./S.E. | Two-Tailed<br>P-Value |
|------------------------------------------|----|----------|-------|-----------|-----------------------|
| C#1                                      | ON |          |       |           |                       |
| GPA                                      |    | -0.698   | 0.255 | -2.739    | 0.006                 |
| Intercepts                               |    |          |       |           |                       |
| C#1                                      |    | -0.241   | 0.460 | -0.523    | 0.601                 |
| Parameterization using Reference Class 1 |    |          |       |           |                       |
| C#2                                      | ON |          |       |           |                       |
| GPA                                      |    | 0.698    | 0.255 | 2.739     | 0.006                 |
| Intercepts                               |    |          |       |           |                       |
| C#2                                      |    | 0.241    | 0.460 | 0.523     | 0.601                 |

ODDS RATIOS FOR TESTS OF CATEGORICAL LATENT VARIABLE MULTINOMIAL LOGISTIC REGRESSIONS  
USING THE 3-STEP PROCEDURE

|                                          |    | Estimate | S.E.  | 95% C.I.<br>Lower 2.5% Upper 2.5% |       |
|------------------------------------------|----|----------|-------|-----------------------------------|-------|
| C#1                                      | ON |          |       |                                   |       |
| GPA                                      |    | 0.498    | 0.127 | 0.302                             | 0.820 |
| Parameterization using Reference Class 1 |    |          |       |                                   |       |
| C#2                                      | ON |          |       |                                   |       |
| GPA                                      |    | 2.009    | 0.512 | 1.220                             | 3.310 |



## Chapter 17

# Automatic 3-Step Distal Only

---

### Data source:

This utilizes a dataset on undergraduate *Cheating* available from the poLCA package (Dayton, 1998): See documentation [here](#)

---

### 17.1 Load packages

```
library(MplusAutomation)
library(tidyverse) #collection of R packages designed for data science
library(here) #helps with filepaths
library(janitor) #clean_names
library(gt) # create tables
library(cowplot) # a ggplot theme
library(DiagrammeR) # create path diagrams
library(glue) # allows us to paste expressions into R code
library(data.table) # used for `melt()` function
library(poLCA)
library(reshape2)
```

---

## 17.2 Automated Three-Step

---

*Application: Undergraduate Cheating behavior*

“Dichotomous self-report responses by 319 undergraduates to four questions about cheating behavior” (poLCA, 2016).

---

Prepare data

```
data(cheating)

cheating <- cheating %>% clean_names()

df_cheat <- cheating %>%
  dplyr::select(1:4) %>%
  mutate_all(funs(.-1)) %>%
  mutate(gpa = cheating$gpa)

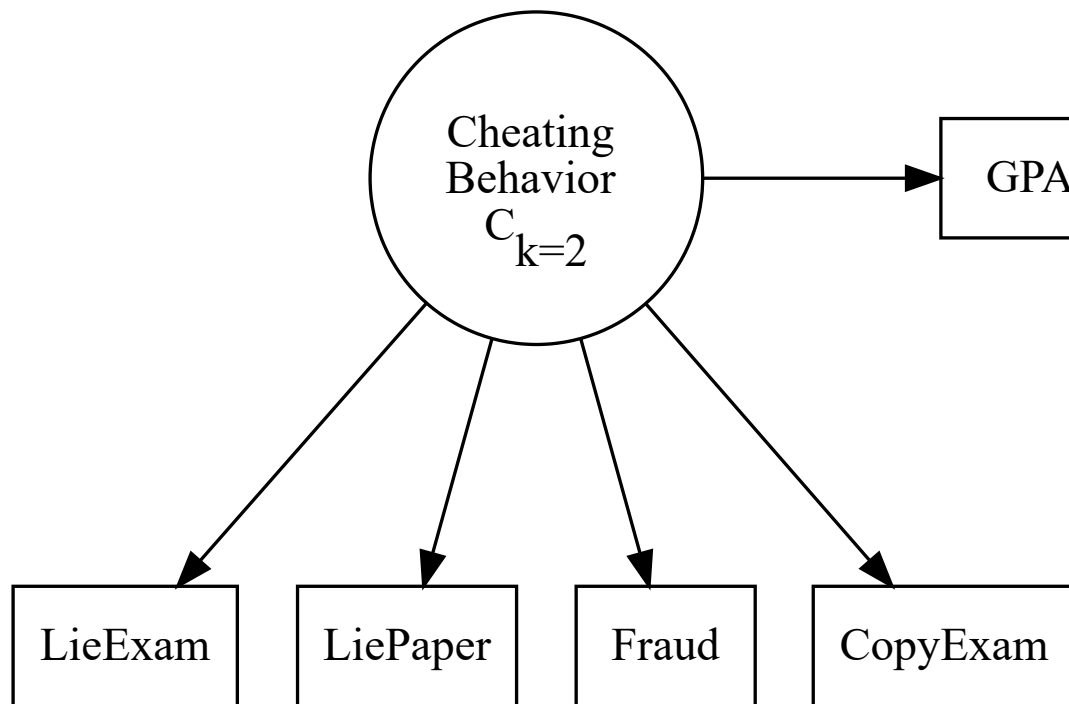
# Detaching packages that mask the dplyr functions
detach(package:poLCA, unload = TRUE)
detach(package:MASS, unload = TRUE)
```

---

### 17.2.1 DU3STEP

**DU3STEP** incorporates distal outcome variables (assumed to have unequal means and variances) with mixture models.

---



### 17.2.1.1 Run the DU3step model with gpa as distal outcome

---

```
m_stepdu <- mplusObject(
  TITLE = "DU3STEP - GPA as Distal",
  VARIABLE =
    "categorical = lieexam-copyexam;
    usevar = lieexam-copyexam;
    auxiliary = gpa (du3step);
    classes = c(2);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors = 10;",

  OUTPUT = "sampstat patterns tech11 tech14;",

  PLOT =
    "type = plot3;
    series = lieexam-copyexam(*)",

  usevariables = colnames(df_cheat),
  rdata = df_cheat)

m_stepdu_fit <- mplusModeler(m_stepdu,
  dataout=here("three_step", "auto_3step", "du3step.dat"),
  modelout=here("three_step", "auto_3step", "c2_du3step.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)
```

---

### 17.2.1.2 Plot Distal Outcome mean differences

```
modelParams <- readModels(here("three_step", "auto_3step", "c2_du3step.out"))

# Extract class size
c_size <- as.data.frame(modelParams[["class_counts"]][["modelEstimated"]][["proportion"]])
  rename("cs" = 1) %>%
```

```

mutate(cs = round(cs*100, 2))

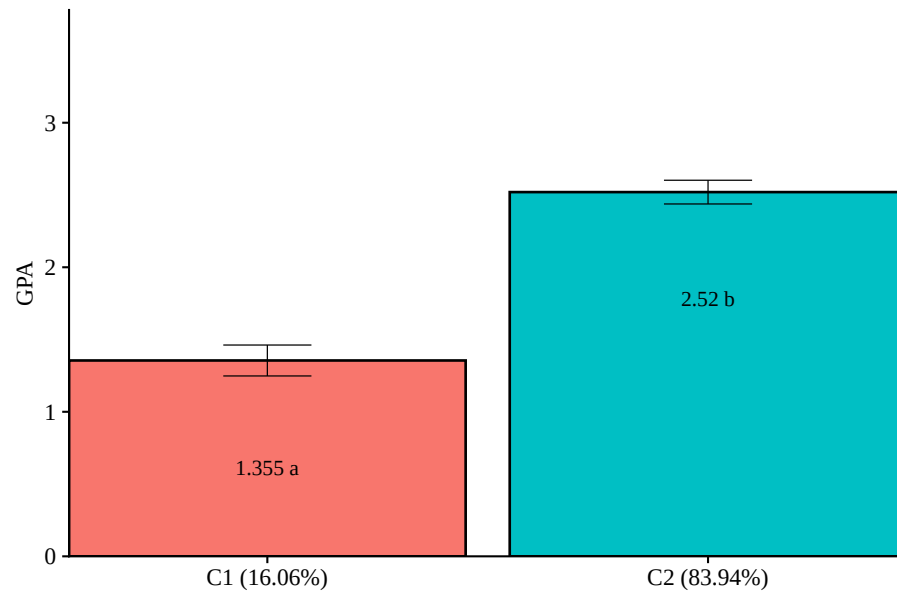
c_size_val <- paste0("C", 1:nrow(c_size), glue(" ({c_size[1:nrow(c_size),]}%)"))

# Extract information as data frame
estimates <- as.data.frame(modelParams[["lcCondMeans"]][["overall"]]) %>%
  reshape2::melt(id.vars = "var") %>%
  mutate(variable = as.character(variable),
         LatentClass = case_when(
           endsWith(variable, "1") ~ c_size_val[1],
           endsWith(variable, "2") ~ c_size_val[2])) %>% #Add to this based on the number of classes
  head(-3) %>%
  pivot_wider(names_from = variable, values_from = value) %>%
  unite("mean", contains("m"), na.rm = TRUE) %>%
  unite("se", contains("se"), na.rm = TRUE) %>%
  mutate(across(c(mean, se), as.numeric))

# Add labels (NOTE: You must change the labels to match the significance testing!!)
value_labels <- paste0(estimates$mean, c(" a", " b"))

# Plot bar graphs
estimates %>%
  ggplot(aes(fill = LatentClass, y = mean, x = LatentClass)) +
  geom_bar(position = "dodge", stat = "identity", color = "black") +
  geom_errorbar(aes(ymin=mean-se, ymax=mean+se),
               size=.3,
               width=.2,
               position=position_dodge(.9)) +
  geom_text(aes(y = mean, label = value_labels),
            family = "serif", size = 4,
            position=position_dodge(.9),
            vjust = 8) +
  #scale_fill_grey(start = .5, end = .7) +
  labs(y="GPA", x="") +
  theme_cowplot() +
  theme(text = element_text(family = "serif", size = 12),
        axis.text.x = element_text(size=12),
        legend.position="none") +
  coord_cartesian(expand = FALSE,
                  ylim=c(0,max(estimates$mean*1.5))) # Change ylim based on distal outcome range

```



```
# Save plot
ggsave(here("figures", "Du3STEP_plot.jpeg"),
       dpi=300, width=10, height = 7, units="in")
```



## **Part VII**

# **Latent Transition Analysis**



## Chapter 18

# Enumeration

---

**Data Source:** The data used to illustrate these analyses include elementary school student *Science Attitude* survey items collected during 7th and 10th grades from the **Longitudinal Study of American Youth** (LSAY; Miller, 2015).

---

To install package {rhdf5}

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

#BiocManager::install("rhdf5")
```

*Load packages*

```
library(MplusAutomation)
library(rhdf5)
library(tidyverse)
library(here)
library(glue)
library(janitor)
library(gt)
library(reshape2)
library(cowplot)
library(ggrepel)
```

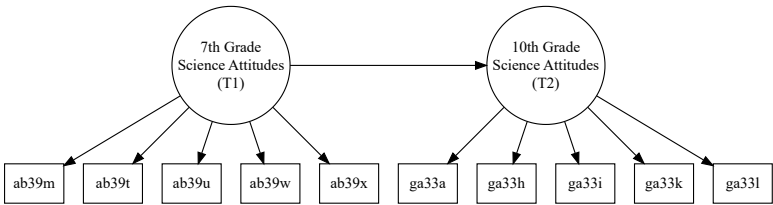
```
library(haven)
library(modelsummary)
library(corrplot)
library(DiagrammeR)
library(filesstrings)
library(PNWColors)
```

---

Read in LSAY data file, `lsay_new.csv`.

```
lsay_data <- read_csv(here("data", "lsay_lta.csv"), na = c("9999")) %>%
  mutate(across(everything(), as.numeric))
```

---



---

18.1 Descriptive Statistics

---

|          | Mean | SD   | Min  | Median | Max  | Histogram |
|----------|------|------|------|--------|------|-----------|
| female   | 0.48 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| minority | 0.23 | 0.42 | 0.00 | 0.00   | 1.00 |           |
| ab39m    | 0.61 | 0.49 | 0.00 | 1.00   | 1.00 |           |
| ab39t    | 0.40 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ab39u    | 0.49 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| ab39w    | 0.40 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ab39x    | 0.46 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| ga33a    | 0.58 | 0.49 | 0.00 | 1.00   | 1.00 |           |
| ga33h    | 0.43 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ga33i    | 0.51 | 0.50 | 0.00 | 1.00   | 1.00 |           |
| ga33k    | 0.42 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ga33l    | 0.42 | 0.49 | 0.00 | 0.00   | 1.00 |           |

### 18.1.1 Data Summary

```
data <- lsay_data

select_data <- data %>%
  select(female, minority, ab39m:ga33l)

f <- All(select_data) ~ Mean + SD + Min + Median + Max + Histogram
datasummary(f, data, output="markdown")
```

### 18.1.2 Correlation Table

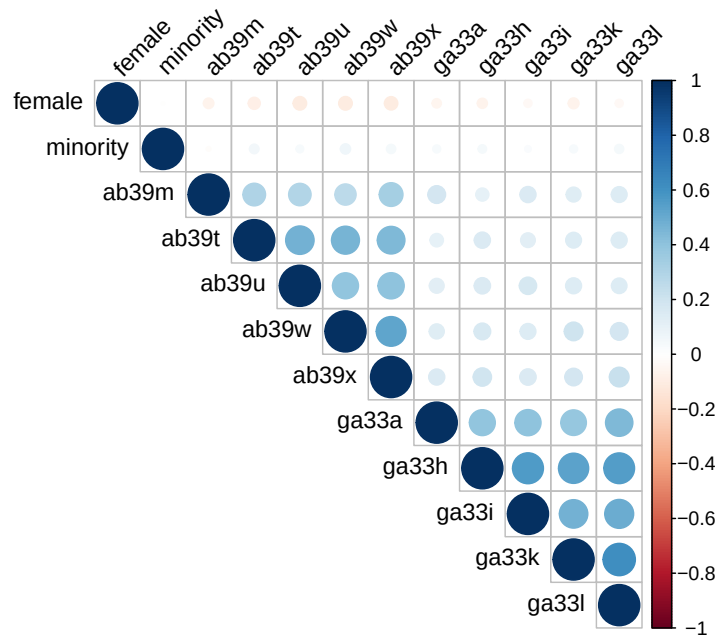
```
select_data %>%
  datasummary_correlation(output = "markdown")
```

|          | female | minority | ab39m | ab39t | ab39u | ab39w | ab39x | ga33a | ga33h | ga33i | ga33k | ga33l |
|----------|--------|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| female   | 1      | .        | .     | .     | .     | .     | .     | .     | .     | .     | .     | .     |
| minority | -.01   | 1        | .     | .     | .     | .     | .     | .     | .     | .     | .     | .     |
| ab39m    | -.06   | -.01     | 1     | .     | .     | .     | .     | .     | .     | .     | .     | .     |
| ab39t    | -.09   | .05      | .30   | 1     | .     | .     | .     | .     | .     | .     | .     | .     |
| ab39u    | -.11   | .03      | .30   | .48   | 1     | .     | .     | .     | .     | .     | .     | .     |
| ab39w    | -.11   | .06      | .27   | .46   | .40   | 1     | .     | .     | .     | .     | .     | .     |
| ab39x    | -.11   | .05      | .33   | .45   | .41   | .52   | 1     | .     | .     | .     | .     | .     |
| ga33a    | -.05   | .04      | .19   | .11   | .13   | .13   | .15   | 1     | .     | .     | .     | .     |
| ga33h    | -.06   | .04      | .10   | .15   | .16   | .16   | .20   | .40   | 1     | .     | .     | .     |
| ga33i    | -.04   | .02      | .15   | .13   | .18   | .15   | .16   | .40   | .57   | 1     | .     | .     |
| ga33k    | -.07   | .04      | .13   | .15   | .15   | .20   | .18   | .39   | .54   | .48   | 1     | .     |
| ga33l    | -.04   | .05      | .15   | .15   | .14   | .18   | .23   | .44   | .56   | .49   | .62   | 1     |

### 18.1.3 Correlation Plot

```
f_cor <- data %>%
  select(female, minority, ab39m:ga33l) %>%
  cor(use = "pairwise.complete.obs")

corrplot(f_cor,
  method = "circle",
  type = "upper",
  tl.col="black",
  tl.srt=45)
```



## 18.2 Enumeration

### 18.2.1 Enumerate Time Point 1 (7th grade)

```
# NOTE CHANGE: '1:6' indicates the number of k-class models to estimate
# User can change this number to fit research context
# In this example, the code loops or iterates over values 1 through 6 ( '{k}' )
#
t1_enum_k_16 <- lapply(1:6, function(k) {
  enum_t1 <- mplusObject(

# The 'glue' function inserts R code within a string or "quoted green text" using the .
#
  TITLE = glue("Class-{k}_Time1"),

  VARIABLE = glue(
    "!!! NOTE CHANGE: List of the five 7th grade science attitude indicators !!!
```



```

    categorical = ab39m-ab39x;
    usevar = ab39m-ab39x;

    classes = c({k});"),

ANALYSIS =
  "estimator = mlr;
  type = mixture;
  !!! NOTE CHANGE: The intial and final start values. Reduce to speed up estimation time. !!!
  starts = 500 100;
  processors=10;",

OUTPUT = "sampstat residual tech11 tech14;",

PLOT =
  "type = plot3;
  series = ab39m-ab39x(*);",

usevariables = colnames(lsay_data),
rdata = lsay_data)

# NOTE CHANGE: Fix to match appropriate sub-folder name
# See after `here` function (e.g., "enum_LCA_time1")
enum_t1_fit <- mplusModeler(enum_t1,
  dataout=here("lta", "enum_t1", "t1.dat"),
  modelout=glue(here("lta", "enum_t1", "c{k}_lca_enum_time1.inp")),
  check=TRUE, run = TRUE, hashfilename = FALSE)
})

```

**NEXT STEP** Check the output (.out) files to check for convergence warnings or syntax errors.

---

### 18.2.2 Enumerate Time Point 2 (10th grade)

```

t2_enum_k_16 <- lapply(1:6, function(k) {
  enum_t2 <- mplusObject(

    TITLE = glue("Class-{k}_Time2"),

    VARIABLE =

```

```

glue(
  "!!! NOTE CHANGE: List of the five 10th grade science attitude indicators !!!
  categorical = ga33a-ga33l;
  usevar = ga33a-ga33l;

  classes = c({k});"),

ANALYSIS =
  "estimator = mlr;
  type = mixture;
  starts = 500 100;
  processors=10;",

OUTPUT = "sampstat residual tech11 tech14;",

PLOT =
  "type = plot3;
  series = ga33a-ga33l(*);",

usevariables = colnames(lsay_data),
rdata = lsay_data)

enum_t2_fit <- mplusModeler(enum_t2,
  dataout=here("lta","enum_t2","t2.dat"),
  modelout=glue(here("lta","enum_t2","c{k}_lca_enum_time2.inp")),
  check=TRUE, run = TRUE, hashfilename = FALSE)
})

```

---

### 18.2.3 Create Model Fit Summary Table

---

Read all models for enumeration table

```

output_enum_t1 <- readModels(here("lta","enum_t1"), quiet = TRUE)
output_enum_t2 <- readModels(here("lta","enum_t2"), quiet = TRUE)

```

Extract model fit data

```

enum_extract1 <- LatexSummaryTable(output_enum_t1,
  keepCols=c("Title", "Parameters", "LL", "BIC", "aBIC",

```

```

      "BLRT_PValue", "T11_VLMR_PValue", "Observations"))
enum_extract2 <- LatexSummaryTable(output_enum_t2,
      keepCols=c("Title", "Parameters", "LL", "BIC", "aBIC",
      "BLRT_PValue", "T11_VLMR_PValue", "Observations"))

```

#### 18.2.4 Calculate Indices Derived from the Log Likelihood (LL)

```

allFit1 <- enum_extract1 %>%
  mutate(aBIC = -2*LL+Parameters*log((Observations+2)/24)) %>%
  mutate(CAIC = -2*LL+Parameters*(log(Observations)+1)) %>%
  mutate(AWE = -2*LL+2*Parameters*(log(Observations)+1.5)) %>%
  mutate(SIC = -.5*BIC) %>%
  mutate(expSIC = exp(SIC - max(SIC))) %>%
  mutate(BF = exp(SIC-lead(SIC))) %>%
  mutate(cmPk = expSIC/sum(expSIC)) %>%
  select(1:5,9:10,6:7,13,14) %>%
  arrange(Parameters)

allFit2 <- enum_extract2 %>%
  mutate(aBIC = -2*LL+Parameters*log((Observations+2)/24)) %>%
  mutate(CAIC = -2*LL+Parameters*(log(Observations)+1)) %>%
  mutate(AWE = -2*LL+2*Parameters*(log(Observations)+1.5)) %>%
  mutate(SIC = -.5*BIC) %>%
  mutate(expSIC = exp(SIC - max(SIC))) %>%
  mutate(BF = exp(SIC-lead(SIC))) %>%
  mutate(cmPk = expSIC/sum(expSIC)) %>%
  select(1:5,9:10,6:7,13,14) %>%
  arrange(Parameters)

allFit <- full_join(allFit1,allFit2)

```

---

### 18.3 Format Fit Table

```

rows_m1 <- 1:6
rows_m2 <- 7:12

```

```

allFit %>%
  mutate(Title = str_remove(Title, "_Time*")) %>%
  gt() %>%
  tab_header(
    title = md("***Model Fit Summary Table**") %>%
  cols_label(
    Title = "Classes",
    Parameters = md("Par"),
    LL = md("*LL*"),
    T11_VLMR_PValue = "VLMR",
    BLRT_PValue = "BLRT",
    BF = md("BF"),
    cmPk = md("*cmP_k*")) %>%
  tab_footnote(
    footnote = md(
      "*Note.* Par = Parameters; *LL* = model log likelihood;
      BIC = Bayesian information criterion;
      aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion;
      AWE = approximate weight of evidence criterion;
      BLRT = bootstrapped likelihood ratio test p-value;
      VLMR = Vuong-Lo-Mendell-Rubin adjusted likelihood ratio test p-value;
      cmPk = approximate correct model probability."),
    locations = cells_title()) %>%
  tab_options(column_labels.font.weight = "bold") %>%
  fmt_number(10, decimals = 2,
    drop_trailing_zeros=TRUE,
    suffixing = TRUE) %>%
  fmt_number(c(3:9,11),
    decimals = 2) %>%
  fmt_missing(1:11,
    missing_text = "--") %>%
  fmt(c(8:9,11),
    fns = function(x)
      ifelse(x<0.001, "<.001",
        scales::number(x, accuracy = 0.01))) %>%
  fmt(10, fns = function(x)
    ifelse(x>100, ">100",
      scales::number(x, accuracy = .1))) %>%
  tab_row_group(
    group = "Time-1",
    rows = 1:6) %>%
  tab_row_group(
    group = "Time-2",
    rows = 7:12) %>%

```

```

row_group_order(
  groups = c("Time-1", "Time-2")
) %>%
tab_style(
  style = list(
    cell_text(weight = "bold")
  ),
  locations = list(cells_body(
    columns = BIC,
    row = BIC == min(BIC[rows_m1]) # Model 1
  ),
  cells_body(
    columns = aBIC,
    row = aBIC == min(aBIC[rows_m1])
  ),
  cells_body(
    columns = CAIC,
    row = CAIC == min(CAIC[rows_m1])
  ),
  cells_body(
    columns = AWE,
    row = AWE == min(AWE[rows_m1])
  ),
  cells_body(
    columns = cmPk,
    row = cmPk == max(cmPk[rows_m1])
  ),
  cells_body(
    columns = BIC,
    row = BIC == min(BIC[rows_m2]) # Model 2
  ),
  cells_body(
    columns = aBIC,
    row = aBIC == min(aBIC[rows_m2])
  ),
  cells_body(
    columns = CAIC,
    row = CAIC == min(CAIC[rows_m2])
  ),
  cells_body(
    columns = AWE,
    row = AWE == min(AWE[rows_m2])
  ),
  cells_body(
    columns = cmPk,

```

```

    row = cmPk == max(cmPk[rows_m2])
  ),
  cells_body(
    columns = BF,
    row = BF > 10),
  cells_body(
    columns = BLRT_PValue,
    row = ifelse(BLRT_PValue < .05 & lead(BLRT_PValue) > .05, BLRT_PValue < .05, NA),
  cells_body(
    columns = T11_VLMR_PValue,
    row = ifelse(T11_VLMR_PValue < .05 & lead(T11_VLMR_PValue) > .05, T11_VLMR_PValue, NA)
  )
)

```

---

## 18.4 Compare Time 1 & Time 2 LCA Plots

---

Read models for plotting (4-class models)

```

model_t1_c4 <- output_enum_t1$c4_lca_enum_time1.out
model_t2_c4 <- output_enum_t2$c4_lca_enum_time2.out

```

### 18.4.1 Create a function `plot_lca_function` that requires 5 arguments:

- `model_name`: name of Mplus model object (e.g., `model_t1_c4`)
- `item_num`: the number of items in LCA measurement model (e.g., 5)
- `class_num`: the number of classes ( $k$ ) in LCA model (e.g., 4)
- `item_labels`: the item labels for x-axis (e.g., `c("Enjoy", "Useful", "Logical", "Job", "Adult")`)
- `plot_title`: include the title of the plot here (e.g., "Time 1 LCA Conditional Item Probability Plot")

```

plot_lca_function <- function(model_name, item_num, class_num, item_labels, plot_title){
  mplus_model <- as.data.frame(model_name$gh5$means_and_variances_data$estimated_probs$var)
  plot_t1 <- mplus_model[seq(2, 2*item_num, 2),]

```

Model Fit Summary Table<sup>1</sup>

| Classes  | Par | <i>LL</i>  | BIC              | aBIC             | CAIC             | AWE              | BLRT            | VLMR            |
|----------|-----|------------|------------------|------------------|------------------|------------------|-----------------|-----------------|
| Time-1   |     |            |                  |                  |                  |                  |                 |                 |
| Class-11 | 5   | -10,250.60 | 20,541.34        | 20,525.45        | 20,546.34        | 20,596.47        | –               | –               |
| Class-21 | 11  | -8,785.32  | 17,658.92        | 17,623.97        | 17,669.93        | 17,780.22        | <.001           | <.001           |
| Class-31 | 17  | -8,693.57  | 17,523.59        | 17,469.57        | 17,540.59        | <b>17,711.04</b> | <.001           | <.001           |
| Class-41 | 23  | -8,664.09  | <b>17,512.79</b> | <b>17,439.71</b> | <b>17,535.79</b> | 17,766.40        | <b>&lt;.001</b> | <b>&lt;.001</b> |
| Class-51 | 29  | -8,662.39  | 17,557.54        | 17,465.40        | 17,586.54        | 17,877.31        | 1.00            | 0.66            |
| Class-61 | 35  | -8,661.54  | 17,604.01        | 17,492.80        | 17,639.01        | 17,989.94        | 0.67            | 0.93            |
| Time-2   |     |            |                  |                  |                  |                  |                 |                 |
| Class-12 | 5   | -7,658.79  | 15,356.19        | 15,340.30        | 15,361.19        | 15,409.80        | –               | –               |
| Class-22 | 11  | -6,073.81  | 12,232.56        | 12,197.61        | 12,243.56        | 12,350.50        | <.001           | <.001           |
| Class-32 | 17  | -5,988.36  | 12,107.99        | 12,053.98        | <b>12,124.99</b> | <b>12,290.27</b> | <.001           | <.001           |
| Class-42 | 23  | -5,964.45  | <b>12,106.50</b> | <b>12,033.43</b> | 12,129.51        | 12,353.12        | <b>&lt;.001</b> | <b>0.00</b>     |
| Class-52 | 29  | -5,961.68  | 12,147.30        | 12,055.16        | 12,176.30        | 12,458.25        | 0.67            | 0.36            |
| Class-62 | 35  | -5,961.26  | 12,192.79        | 12,081.59        | 12,227.79        | 12,568.07        | 1.00            | 0.57            |

<sup>1</sup>Note. Par = Parameters; *LL* = model log likelihood; BIC = Bayesian information criterion; aBIC = sample size adjusted BIC; CAIC = consistent Akaike information criterion; AWE = approximate weight of evidence criterion; BLRT = bootstrapped likelihood ratio test p-value; VLMR = Vuong-Lo-Mendell-Rubin adjusted likelihood ratio test p-value; cmPk = approximate correct model probability.

```

c_size <- as.data.frame(model_name$class_counts$modelEstimated$proportion)
colnames(c_size) <- paste0("cs")
c_size <- c_size %>% mutate(cs = round(cs*100, 2))
colnames(plot_t1) <- paste0("C", 1:class_num, glue(" ({c_size[1:class_num,]}%)"))

plot_t1 <- cbind(Var = paste0("U", 1:item_num), plot_t1)
plot_t1$Var <- factor(plot_t1$Var,
  labels = item_labels)
plot_t1$Var <- fct_inorder(plot_t1$Var)
pd_long_t1 <- melt(plot_t1, id.vars = "Var")

p <- pd_long_t1 %>%

```

```

ggplot(aes(x = as.integer(Var), y = value,
shape = variable, colour = variable, lty = variable)) +
geom_point(size = 4) + geom_line() +
scale_x_continuous("", breaks = 1:5, labels = plot_t1$Var) +
scale_colour_grey() +
labs(title = plot_title, y = "Probability") +
theme_cowplot() +
theme(legend.title = element_blank(),
      legend.position = "top")

p
return(p)
}

```

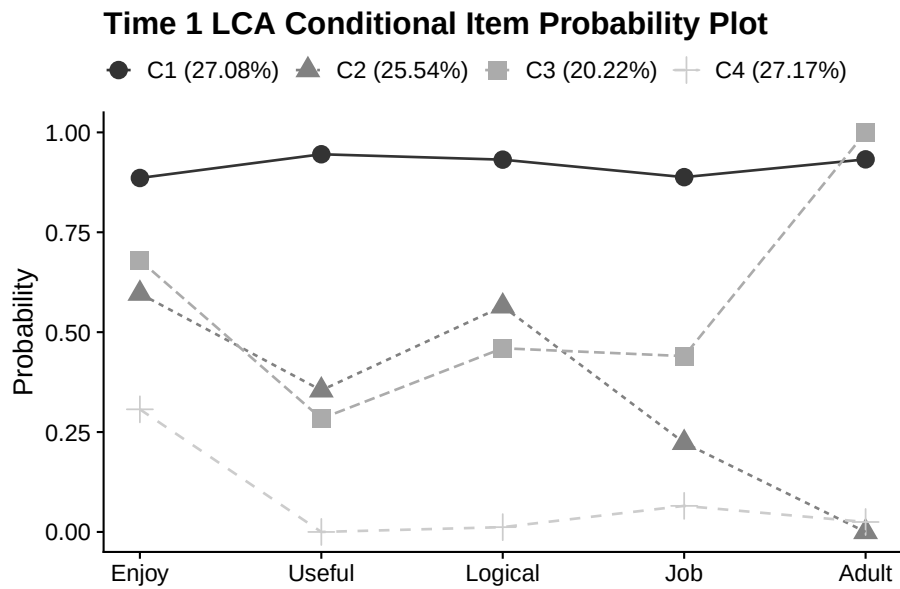
#### 18.4.2 Time 1 LCA - Conditional Item Probability Plot

```

plot_lca_function(
  model_name = model_t1_c4,
  item_num = 5,
  class_num = 4,
  item_labels = c("Enjoy", "Useful", "Logical", "Job", "Adult"),
  plot_title = "Time 1 LCA Conditional Item Probability Plot"
)

```

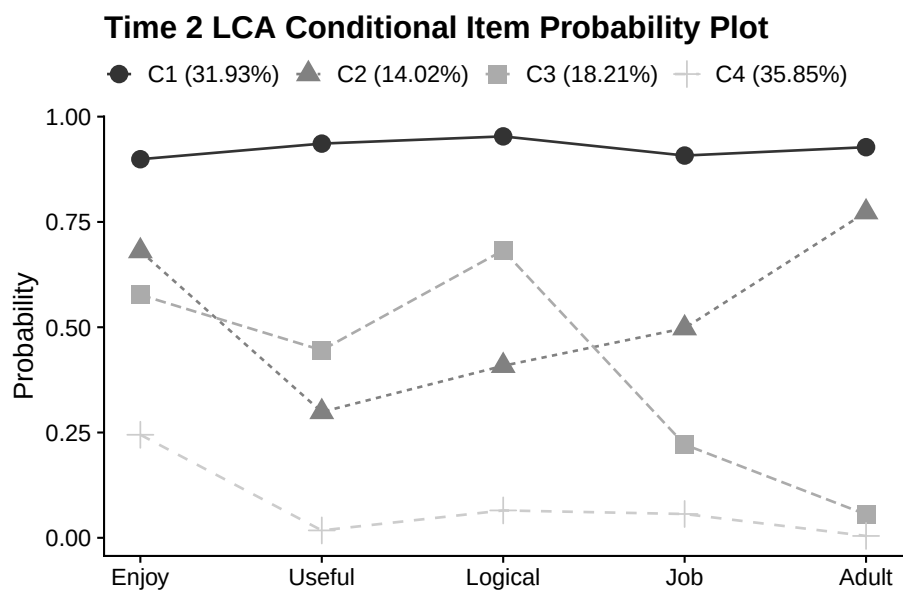




```
ggsave(here("figures", "t1_c4_lca_plot.png"), dpi=300, height=5, width=7, units="in")
```

### 18.4.3 Time 2 LCA - Conditional Item Probability Plot

```
plot_lca_function(
  model_name = model_t2_c4,
  item_num = 5,
  class_num = 4,
  item_labels = c("Enjoy", "Useful", "Logical", "Job", "Adult"),
  plot_title = "Time 2 LCA Conditional Item Probability Plot"
)
```



```
ggsave(here("figures", "t2_c4_lca_plot.png"), dpi=300, height=5, width=7, units="in")
```

---

## 18.5 Estimate Latent Transition Analysis (LTA) Model

---

### 18.5.1 Estimate Invariant LTA Model

```
lta_inv <- mplusObject(
  TITLE =
    "Invariant LTA",
  VARIABLE =
    "usev = ab39m ab39t ab39u ab39w ab39x ! 7th grade indicators
      ga33a ga33h ga33i ga33k ga33l; ! 10th grade indicators"
```

```

categorical = ab39m-ab39x ga33a-ga33l;

classes = c1(4) c2(4);",

ANALYSIS =
"estimator = mlr;
type = mixture;
starts = 500 100;
processors=10;",

MODEL =
"%overall%
c2 on c1;

MODEL c1:
%c1#1%
[AB39M$1-AB39X$1] (1-5); !!! labels that are repeated will constrain parameters to equality
%c1#2%
[AB39M$1-AB39X$1] (6-10);
%c1#3%
[AB39M$1-AB39X$1] (11-15);
%c1#4%
[AB39M$1-AB39X$1] (16-20);

MODEL c2:
%c2#1%
[GA33A$1-GA33L$1] (1-5);
%c2#2%
[GA33A$1-GA33L$1] (6-10);
%c2#3%
[GA33A$1-GA33L$1] (11-15);
%c2#4%
[GA33A$1-GA33L$1] (16-20);",

SAVEDATA =
"file = LTA_Inv_CPROBS.dat;
save = cprob;
missflag = 9999;",

OUTPUT = "tech1 tech15 svalues;",

usevariables = colnames(lsay_data),
rdata = lsay_data)

lta_inv_fit <- mplusModeler(lta_inv,

```

```
dataout=here("lta","lta_model","lta.dat"),
modelout=here("lta","lta_model","4-class-invariant.inp"),
check=TRUE, run = TRUE, hashfilename = FALSE)
```

---

### 18.5.2 Estimate Non-Invariant Estimated LTA Model

```
lta_non_inv <- mplusObject(

  TITLE =
    "Non-Invariant LTA",

  VARIABLE =
    "usev = ab39m ab39t ab39u ab39w ab39x ! 7th grade indicators
      ga33a ga33h ga33i ga33k ga33l; ! 10th grade indicators

    categorical = ab39m-ab39x ga33a-ga33l;

    classes = c1(4) c2(4);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors=10;",

  MODEL =
    "%overall%
    c2 on c1; !!! estimate all multinomial logistic regressions !!!

    !!! The above syntax can also be written as: !!!
      ! c2#1 on c1#1 c1#2 c1#3; !
      ! c2#2 on c1#1 c1#2 c1#3; !
      ! c2#3 on c1#1 c1#2 c1#3; !

    MODEL c1: !!! the following syntax will allow item thresholds to be estimated for

    %c1#1%
    [AB39M$1-AB39X$1];
    %c1#2%
    [AB39M$1-AB39X$1];
```

```

    %c1#3%
    [AB39M$1-AB39X$1];
    %c1#4%
    [AB39M$1-AB39X$1];

    MODEL c2:
    %c2#1%
    [GA33A$1-GA33L$1];
    %c2#2%
    [GA33A$1-GA33L$1];
    %c2#3%
    [GA33A$1-GA33L$1];
    %c2#4%
    [GA33A$1-GA33L$1];",

    OUTPUT = "tech1 tech15 svalues;",

    usevariables = colnames(lsay_data),
    rdata = lsay_data)

lta_non_inv_fit <- mplusModeler(lta_non_inv,
                                dataout=here("lta","lta_model","lta.dat"),
                                modelout=here("lta","lta_model","4-class-non-invariant.inp"),
                                check=TRUE, run = TRUE, hashfilename = FALSE)

```

### 18.5.3 Conduct Sattorra-Bentler adjusted Log Likelihood Ratio Difference Testing

- non-invariant (comparison): This model has **more** parameters.
- invariant (nested): This model has **less** parameters.

```

# *0 = null or nested model & *1 = comparison or parent model

lta_models <- readModels(here("lta","lta_model"), quiet = TRUE)

# Log Likelihood Values
L0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["LL"]]
L1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["LL"]]

# LRT equation

```

```

lr <- -2*(L0-L1)

# Parameters
p0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["Parameters"]]
p1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["Parameters"]]

# Scaling Correction Factors
c0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["LLCorrectionFactor"]]
c1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["LLCorrectionFactor"]]

# Difference Test Scaling correction
cd <- ((p0*c0)-(p1*c1))/(p0-p1)

# Chi-square difference test(TRd)
TRd <- (lr)/(cd)

# Degrees of freedom
df <- abs(p0 - p1)

# Significance test
(p_diff <- pchisq(TRd, df, lower.tail=FALSE))
#> [1] 0.6245173

```

**RESULT:** The Log Likelihood  $\chi^2$  difference test comparing the invariant and non-invariant LTA models was,  $\chi^2(20) = 21.542, p = .624$ .

Reference

---

Read invariance model and extract parameters (intercepts and multinomial regression coefficients)

```

lta_inv1 <- readModels(here("lta","lta_model","4-Class-Invariant.out" ), quiet = TRUE)

par <- as_tibble(lta_inv1[["parameters"]][["unstandardized"]]) %>%
  select(1:3) %>%
  filter(grepl('ON|Means', paramHeader)) %>%
  mutate(est = as.numeric(est))

```

Manual method to calculate transition probabilities:

Although possible to extract transition probabilities directly from the output the following code illustrates how the parameters are used to calculate each

transition. This is useful for conducting advanced LTA model specifications such as making specific constraints within or between transition matrices, or testing the equivalence of specific transition probabilities.

```
# Name each parameter individually to make the subsequent calculations more readable
a1 <- unlist(par[13,3]); a2 <- unlist(par[14,3]); a3 <- unlist(par[15,3]); b11 <- unlist(par[1,3])
b21 <- unlist(par[4,3]); b31 <- unlist(par[7,3]); b12 <- unlist(par[2,3]); b22 <- unlist(par[5,3])
b32 <- unlist(par[8,3]); b13 <- unlist(par[3,3]); b23 <- unlist(par[6,3]); b33 <- unlist(par[9,3])

# Calculate transition probabilities from the logit parameters
t11 <- exp(a1+b11)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t12 <- exp(a2+b21)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t13 <- exp(a3+b31)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t14 <- 1 - (t11 + t12 + t13)

t21 <- exp(a1+b12)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t22 <- exp(a2+b22)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t23 <- exp(a3+b32)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t24 <- 1 - (t21 + t22 + t23)

t31 <- exp(a1+b13)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t32 <- exp(a2+b23)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t33 <- exp(a3+b33)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t34 <- 1 - (t31 + t32 + t33)

t41 <- exp(a1)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t42 <- exp(a2)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t43 <- exp(a3)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t44 <- 1 - (t41 + t42 + t43)
```

#### 18.5.4 Create Transition Table

```
t_matrix <- tibble(
  "Time1" = c("C1=Anti-Science", "C1=Amb. w/ Elevated", "C1=Amb. w/ Minimal", "C1=Pro-Science"),
  "C2=Anti-Science" = c(t11, t21, t31, t41),
  "C2=Amb. w/ Elevated" = c(t12, t22, t32, t42),
  "C2=Amb. w/ Minimal" = c(t13, t23, t33, t43),
  "C2=Pro-Science" = c(t14, t24, t34, t44))

t_matrix %>%
  gt(rowname_col = "Time1") %>%
  tab_stubhead(label = "7th grade") %>%
  tab_header(
```

**Student transitions from 7th grade (rows) to 10th grade (columns)<sup>1</sup>**

| 7th grade           | 10th grade      |                     |                    |
|---------------------|-----------------|---------------------|--------------------|
|                     | C2=Anti-Science | C2=Amb. w/ Elevated | C2=Amb. w/ Minimal |
| C1=Anti-Science     | 0.27            | 0.27                |                    |
| C1=Amb. w/ Elevated | 0.09            | 0.56                |                    |
| C1=Amb. w/ Minimal  | 0.15            | 0.21                |                    |
| C1=Pro-Science      | 0.08            | 0.35                |                    |

<sup>1</sup>*Note.* Transition matrix values are the identical to Table 5, however Table 5 has the values rearranged by class for interpretation purposes. Classes may be arranged directly through Mplus syntax using start values.

```

title = md("**Student transitions from 7th grade (rows) to 10th grade (columns)**")
subtitle = md("&nbsp;") %>%
fmt_number(2:5,decimals = 2) %>%
tab_spanner(label = "10th grade",columns = 2:5) %>%
tab_footnote(
  footnote = md(
    "*Note.* Transition matrix values are the identical to Table 5, however Table 5
    has the values rearranged by class for interpretation purposes. Classes may be arr
    directly through Mplus syntax using start values."),
    locations = cells_title())

```



## Chapter 19

# Measurement Invariance

---

**Data Source:** The data used to illustrate these analyses include elementary school student *Science Attitude* survey items collected during 7th and 10th grades from the **Longitudinal Study of American Youth** (LSAY; Miller, 2015).

---

To install package {rhdf5}

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

#BiocManager::install("rhdf5")
```

*Load packages*

```
library(MplusAutomation)
library(rhdf5)
library(tidyverse)
library(here)
library(glue)
library(janitor)
library(gt)
library(reshape2)
library(cowplot)
library(ggrepel)
```

```
library(haven)
library(modelsummary)
library(corrplot)
library(DiagrammeR)
library(filesstrings)
library(PNWColors)
```

---

Read in LSAY data file, `lsay_new.csv`.

```
lsay_data <- read_csv(here("data", "lsay_lta.csv"), na = c("9999")) %>%
  mutate(across(everything(), as.numeric))
```

---

## 19.1 Estimate Latent Transition Analysis (LTA) Model

---

### 19.1.1 Estimate Invariant LTA Model

```
lta_inv <- mplusObject(

  TITLE =
    "Invariant LTA",

  VARIABLE =
    "usev = ab39m ab39t ab39u ab39w ab39x ! 7th grade indicators
      ga33a ga33h ga33i ga33k ga33l; ! 10th grade indicators

    categorical = ab39m-ab39x ga33a-ga33l;

    classes = c1(4) c2(4);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
```

```

starts = 500 100;
processors=10;";

MODEL =
"%overall%
c2 on c1;

MODEL c1:
%c1#1%
[AB39M$1-AB39X$1] (1-5);  !!! labels that are repeated will constrain parameters to equality
%c1#2%
[AB39M$1-AB39X$1] (6-10);
%c1#3%
[AB39M$1-AB39X$1] (11-15);
%c1#4%
[AB39M$1-AB39X$1] (16-20);

MODEL c2:
%c2#1%
[GA33A$1-GA33L$1] (1-5);
%c2#2%
[GA33A$1-GA33L$1] (6-10);
%c2#3%
[GA33A$1-GA33L$1] (11-15);
%c2#4%
[GA33A$1-GA33L$1] (16-20);",

SAVEDATA =
"file = LTA_Inv_CPROBS.dat;
save = cprob;
missflag = 9999;";

OUTPUT = "tech1 tech15 svalues;";

usevariables = colnames(lsay_data),
rdata = lsay_data)

lta_inv_fit <- mplusModeler(lta_inv,
  dataout=here("lta","lta_model","lta.dat"),
  modelout=here("lta","lta_model","4-class-invariant.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

## 19.1.2 Estimate Non-Invariant Estimated LTA Model

```

lta_non_inv <- mplusObject(

  TITLE =
    "Non-Invariant LTA",

  VARIABLE =
    "usev = ab39m ab39t ab39u ab39w ab39x ! 7th grade indicators
      ga33a ga33h ga33i ga33k ga33l; ! 10th grade indicators

    categorical = ab39m-ab39x ga33a-ga33l;

    classes = c1(4) c2(4);",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors=10;",

  MODEL =
    "%overall%
    c2 on c1; !!! estimate all multinomial logistic regressions !!!

    !!! The above syntax can also be written as: !!!
      ! c2#1 on c1#1 c1#2 c1#3; !
      ! c2#2 on c1#1 c1#2 c1#3; !
      ! c2#3 on c1#1 c1#2 c1#3; !

    MODEL c1: !!! the following syntax will allow item thresholds to be estimated for

    %c1#1%
    [AB39M$1-AB39X$1];
    %c1#2%
    [AB39M$1-AB39X$1];
    %c1#3%
    [AB39M$1-AB39X$1];
    %c1#4%
    [AB39M$1-AB39X$1];

    MODEL c2:
    %c2#1%
    [GA33A$1-GA33L$1];

```

```

      %c2#2%
      [GA33A$1-GA33L$1];
      %c2#3%
      [GA33A$1-GA33L$1];
      %c2#4%
      [GA33A$1-GA33L$1];",

  OUTPUT = "tech1 tech15 svalues;",

  usevariables = colnames(lsay_data),
  rdata = lsay_data)

lta_non_inv_fit <- mplusModeler(lta_non_inv,
                                dataout=here("lta","lta_model","lta.dat"),
                                modelout=here("lta","lta_model","4-class-non-invariant.inp"),
                                check=TRUE, run = TRUE, hashfilename = FALSE)

```

### 19.1.3 Conduct Sattorra-Bentler adjusted Log Likelihood Ratio Difference Testing

- non-invariant (comparison): This model has **more** parameters.
- invariant (nested): This model has **less** parameters.

```

# *0 = null or nested model & *1 = comparison or parent model

lta_models <- readModels(here("lta","lta_model"), quiet = TRUE)

# Log Likelihood Values
L0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["LL"]]
L1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["LL"]]

# LRT equation
lr <- -2*(L0-L1)

# Parameters
p0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["Parameters"]]
p1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["Parameters"]]

# Scaling Correction Factors
c0 <- lta_models[["X4.class.invariant.out"]][["summaries"]][["LLCorrectionFactor"]]

```

```

c1 <- lta_models[["X4.class.non.invariant.out"]][["summaries"]][["LLCorrectionFactor"]]

# Difference Test Scaling correction
cd <- ((p0*c0)-(p1*c1))/(p0-p1)

# Chi-square difference test(TRd)
TRd <- (lr)/(cd)

# Degrees of freedom
df <- abs(p0 - p1)

# Significance test
(p_diff <- pchisq(TRd, df, lower.tail=FALSE))
#> [1] 0.6245173

```

**RESULT:** The Log Likelihood  $\chi^2$  difference test comparing the invariant and non-invariant LTA models was,  $\chi^2(20) = 21.542, p = .624$ .

Reference

---

Read invariance model and extract parameters (intercepts and multinomial regression coefficients)

```

lta_inv1 <- readModels(here("lta","lta_model","4-Class-Invariant.out"), quiet = TRUE)

par <- as_tibble(lta_inv1[["parameters"]][["unstandardized"]]) %>%
  select(1:3) %>%
  filter(grepl('ON|Means', paramHeader)) %>%
  mutate(est = as.numeric(est))

```

Manual method to calculate transition probabilities:

Although possible to extract transition probabilities directly from the output the following code illustrates how the parameters are used to calculate each transition. This is useful for conducting advanced LTA model specifications such as making specific constraints within or between transition matrices, or testing the equivalence of specific transition probabilities.

```

# Name each parameter individually to make the subsequent calculations more readable
a1 <- unlist(par[13,3]); a2 <- unlist(par[14,3]); a3 <- unlist(par[15,3]); b11 <- unlist(par[16,3]); b12 <- unlist(par[17,3]); b13 <- unlist(par[18,3]); b21 <- unlist(par[4,3]); b31 <- unlist(par[7,3]); b12 <- unlist(par[2,3]); b22 <- unlist(par[5,3]); b32 <- unlist(par[8,3]); b23 <- unlist(par[6,3]); b33 <- unlist(par[9,3]);

```

```

b32 <- unlist(par[8,3]); b13 <- unlist(par[3,3]); b23 <- unlist(par[6,3]); b33 <- unlist(par[9,3])

# Calculate transition probabilities from the logit parameters
t11 <- exp(a1+b11)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t12 <- exp(a2+b21)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t13 <- exp(a3+b31)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0))
t14 <- 1 - (t11 + t12 + t13)

t21 <- exp(a1+b12)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t22 <- exp(a2+b22)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t23 <- exp(a3+b32)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0))
t24 <- 1 - (t21 + t22 + t23)

t31 <- exp(a1+b13)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t32 <- exp(a2+b23)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t33 <- exp(a3+b33)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0))
t34 <- 1 - (t31 + t32 + t33)

t41 <- exp(a1)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t42 <- exp(a2)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t43 <- exp(a3)/(exp(a1)+exp(a2)+exp(a3)+exp(0))
t44 <- 1 - (t41 + t42 + t43)

```

#### 19.1.4 Create Transition Table

```

t_matrix <- tibble(
  "Time1" = c("C1=Anti-Science", "C1=Amb. w/ Elevated", "C1=Amb. w/ Minimal", "C1=Pro-Science"),
  "C2=Anti-Science" = c(t11, t21, t31, t41),
  "C2=Amb. w/ Elevated" = c(t12, t22, t32, t42),
  "C2=Amb. w/ Minimal" = c(t13, t23, t33, t43),
  "C2=Pro-Science" = c(t14, t24, t34, t44))

t_matrix %>%
  gt(rowname_col = "Time1") %>%
  tab_stubhead(label = "7th grade") %>%
  tab_header(
    title = md("**Student transitions from 7th grade (rows) to 10th grade (columns)**"),
    subtitle = md("&nbsp;")) %>%
  fmt_number(2:5, decimals = 2) %>%
  tab_spanner(label = "10th grade", columns = 2:5) %>%
  tab_footnote(
    footnote = md(
      "*Note.* Transition matrix values are the identical to Table 5, however Table 5

```

Student transitions from 7th grade (rows) to 10th grade (columns)<sup>1</sup>

| 7th grade           | 10th grade      |                     |                    |
|---------------------|-----------------|---------------------|--------------------|
|                     | C2=Anti-Science | C2=Amb. w/ Elevated | C2=Amb. w/ Minimal |
| C1=Anti-Science     | 0.27            | 0.27                |                    |
| C1=Amb. w/ Elevated | 0.09            | 0.56                |                    |
| C1=Amb. w/ Minimal  | 0.15            | 0.21                |                    |
| C1=Pro-Science      | 0.08            | 0.35                |                    |

<sup>1</sup>Note. Transition matrix values are the identical to Table 5, however Table 5 has the values rearranged by class for interpretation purposes. Classes may be arranged directly through Mplus syntax using start values.

```
has the values rearranged by class for interpretation purposes. Classes may be arranged
directly through Mplus syntax using start values."),
locations = cells_title()
```



## Chapter 20

# Adding Covariates

---

**Data Source:** The data used to illustrate these analyses include elementary school student *Science Attitude* survey items collected during 7th and 10th grades from the **Longitudinal Study of American Youth** (LSAY; Miller, 2015).

---

To install package {rhdf5}

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

#BiocManager::install("rhdf5")
```

*Load packages*

```
library(MplusAutomation)
library(rhdf5)
library(tidyverse)
library(here)
library(glue)
library(janitor)
library(gt)
library(reshape2)
library(cowplot)
library(ggrepel)
```

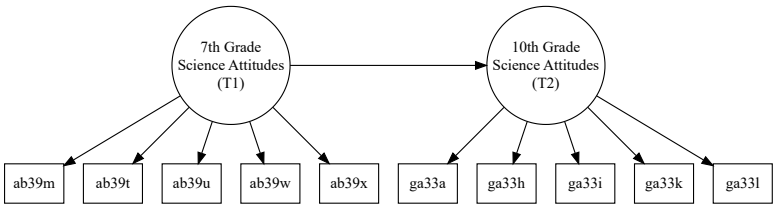
```
library(haven)
library(modelsummary)
library(corrplot)
library(DiagrammeR)
library(filesstrings)
library(PNWColors)
```

---

Read in LSAY data file, `lsay_new.csv`.

```
lsay_data <- read_csv(here("data", "lsay_lta.csv"), na = c("9999")) %>%
  mutate(across(everything(), as.numeric))
```

---



---

20.1 Descriptive Statistics

---

|          | Mean | SD   | Min  | Median | Max  | Histogram |
|----------|------|------|------|--------|------|-----------|
| female   | 0.48 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| minority | 0.23 | 0.42 | 0.00 | 0.00   | 1.00 |           |
| ab39m    | 0.61 | 0.49 | 0.00 | 1.00   | 1.00 |           |
| ab39t    | 0.40 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ab39u    | 0.49 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| ab39w    | 0.40 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ab39x    | 0.46 | 0.50 | 0.00 | 0.00   | 1.00 |           |
| ga33a    | 0.58 | 0.49 | 0.00 | 1.00   | 1.00 |           |
| ga33h    | 0.43 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ga33i    | 0.51 | 0.50 | 0.00 | 1.00   | 1.00 |           |
| ga33k    | 0.42 | 0.49 | 0.00 | 0.00   | 1.00 |           |
| ga33l    | 0.42 | 0.49 | 0.00 | 0.00   | 1.00 |           |

### 20.1.1 Data Summary

```
data <- lsay_data

select_data <- data %>%
  select(female, minority, ab39m:ga33l)

f <- All(select_data) ~ Mean + SD + Min + Median + Max + Histogram
datasummary(f, data, output="markdown")
```

## 20.2 Adding Covariates

Continuing from the previous section ??, we use the ML three-step method to estimate LTA models with predictors and distal outcomes). Estimate the unconditional model for each latent variable with the predictors included in the `auxiliary` option for at least one of the models.

Covariates

- `sci_issues7`: Interest in science issues (1 = Not at all interested, 2 = Moderately Interested, 3 = Very interested)

- `sci_irt7`: 7th Grade Science IRT Score (Continuous)
  - `female`: Gender (0 = Male, 1 = Female)
- 

### 20.2.1 Step 1 - Estimate Unconditional Model w/ Auxiliary Specification

*7th Grade*

```
step1 <- mplusObject(
  TITLE = "Step 1 - T1",
  VARIABLE =
    "usevar = ab39m ab39t ab39u ab39w ab39x;
    categorical = ab39m ab39t ab39u ab39w ab39x;

    classes = c(4);

    auxiliary = sci_issues7 sci_irt7 female;

    idvariable = casenum;",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;
    optseed = 534483;",

  SAVEDATA =
    "File=3step_t1.dat;
    Save=cprob;",

  OUTPUT = "residual tech11 tech14 svalues",

  PLOT =
    "type = plot3;
    series = ab39m-ab39x(*);",

  usevariables = colnames(lsay_data),
  rdata = lsay_data)

step1_fit <- mplusModeler(step1,
  dataout=here("lta","cov_model","t1.dat"),
```

```

modelout=here("lta","cov_model","one_T1.inp") ,
check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

### 10th Grade

```

step1 <- mplusObject(
  TITLE = "Step 1 - T1",
  VARIABLE =
    "usevar = ga33a ga33h ga33i ga33k ga33l;
    categorical = ga33a ga33h ga33i ga33k ga33l;

    classes = c(4);

    !auxiliary = sci_issues7 sci_irt7 female;

    idvariable = casenum;",

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;
    optseed = 392418;",

  SAVEDATA =
    "File=3step_t2.dat;
    Save=cprob;",

  OUTPUT = "residual tech11 tech14 svalues",

  PLOT =
    "type = plot3;
    series = ga33a-ga33l(*);",

  usevariables = colnames(lsay_data),
  rdata = lsay_data)

step1_fit <- mplusModeler(step1,
  dataout=here("lta","cov_model","t2.dat"),
  modelout=here("lta","cov_model","one_T2.inp") ,
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

`plot_lca_function` requires 5 arguments:

- `model_name`: name of Mplus model object (e.g., `model_t1_c4`)
- `item_num`: the number of items in LCA measurement model (e.g., 5)
- `class_num`: the number of classes ( $k$ ) in LCA model (e.g., 4)
- `item_labels`: the item labels for x-axis (e.g., `c("Enjoy", "Useful", "Logical", "Job", "Adult")`)
- `plot_title`: include the title of the plot here (e.g., "Time 1 LCA Conditional Item Probability Plot")

```
plot_lca_function <- function(model_name,item_num,class_num,item_labels,plot_title){

mplus_model <- as.data.frame(model_name$gh5$means_and_variances_data$estimated_probs$values)
plot_t1 <- mplus_model[seq(2, 2*item_num, 2),]

c_size <- as.data.frame(model_name$class_counts$modelEstimated$proportion)
colnames(c_size) <- paste0("cs")
c_size <- c_size %>% mutate(cs = round(cs*100, 2))
colnames(plot_t1) <- paste0("C", 1:class_num, glue(" ({c_size[1:class_num,]}%)"))

plot_t1 <- cbind(Var = paste0("U", 1:item_num), plot_t1)
plot_t1$Var <- factor(plot_t1$Var,
                      labels = item_labels)
plot_t1$Var <- fct_inorder(plot_t1$Var)
pd_long_t1 <- melt(plot_t1, id.vars = "Var")

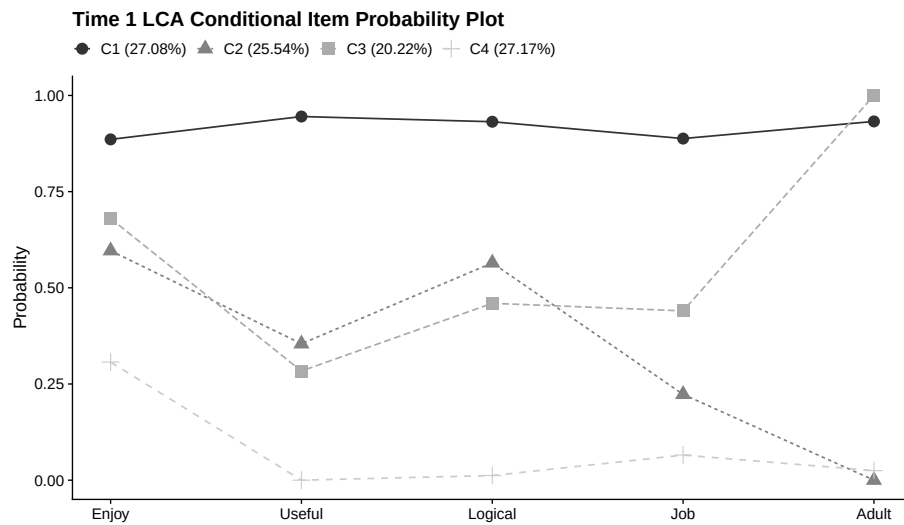
p <- pd_long_t1 %>%
  ggplot(aes(x = as.integer(Var), y = value,
             shape = variable, colour = variable, lty = variable)) +
  geom_point(size = 4) + geom_line() +
  scale_x_continuous("", breaks = 1:5, labels = plot_t1$Var) +
  scale_colour_grey() +
  labs(title = plot_title, y = "Probability") +
  theme_cowplot() +
  theme(legend.title = element_blank(),
        legend.position = "top")

p
return(p)
}
```

Plot Time 1

```
output_T1 <- readModels(here("lta", "cov_model", "one_T1.out"))
```

```
plot_lca_function(
  model_name = output_T1,
  item_num = 5,
  class_num = 4,
  item_labels = c("Enjoy", "Useful", "Logical", "Job", "Adult"),
  plot_title = "Time 1 LCA Conditional Item Probability Plot"
)
```

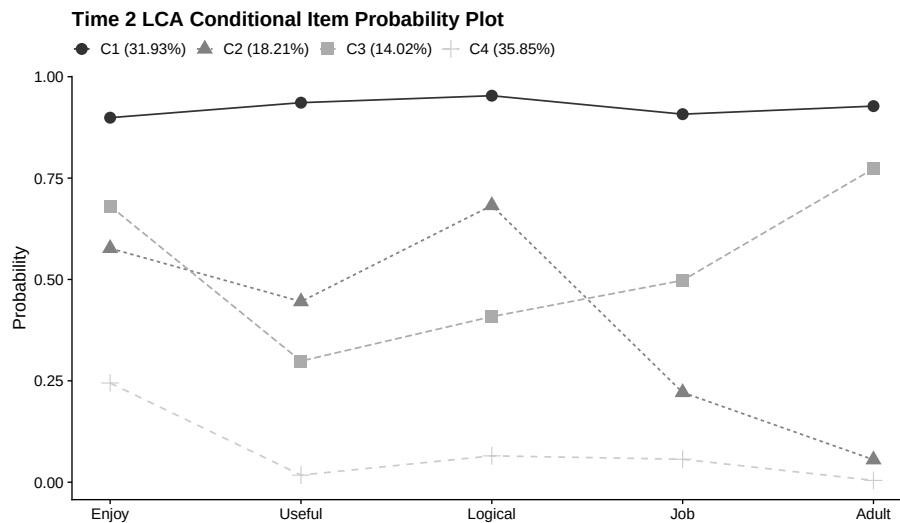


Plot Time 2

```
output_T2 <- readModels(here("lta", "cov_model", "one_T2.out"))

plot_lca_function(
  model_name = output_T2,
  item_num = 5,
  class_num = 4,
  item_labels = c("Enjoy", "Useful", "Logical", "Job", "Adult"),
  plot_title = "Time 2 LCA Conditional Item Probability Plot"
)
```





### 20.2.2 Step 2 - Determine Measurement Error

Extract logits for the classification probabilities for the most likely latent class:

```
logit_cprobs_T1 <- as.data.frame(output_T1[["class_counts"]]  
                                [["logitProbs.mostLikely"]])  
  
logit_cprobs_T2 <- as.data.frame(output_T2[["class_counts"]]  
                                [["logitProbs.mostLikely"]])
```

Extract saved dataset:

```
savedata_T1 <- as.data.frame(output_T1[["savedata"]])  
savedata_T2 <- as.data.frame(output_T2[["savedata"]])
```

Rename the column in savedata named "C" and change to "N"

```
colnames(savedata_T1)[colnames(savedata_T1)=="C"] <- "N_T1"  
colnames(savedata_T2)[colnames(savedata_T2)=="C"] <- "N_T2"  
  
savedata <- savedata_T1 %>%  
  full_join(savedata_T2, by = "CASENUM")
```

### 20.2.3 Step 3 - Add Auxiliary Variables

```

step3 <- mplusObject(
  TITLE = "ML Three Step LTA Model",

  VARIABLE =
    "nominal=N_T1 N_T2;
    usevar = N_T1 N_T2 SCI_IRT7 FEMALE;
    classes = c1(4) c2(4);" ,

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 0;",

  MODEL =
    glue(
    " %OVERALL%

      c2 on c1;
      c1 c2 on  SCI_IRT7 FEMALE;

    MODEL c1:

    %c1#1%
    [N_T1#1@{logit_cprobs_T1[1,1]}};
    [N_T1#2@{logit_cprobs_T1[1,2]}};
    [N_T1#3@{logit_cprobs_T1[1,3]}};

    %c1#2%
    [N_T1#1@{logit_cprobs_T1[2,1]}};
    [N_T1#2@{logit_cprobs_T1[2,2]}};
    [N_T1#3@{logit_cprobs_T1[2,3]}};

    %c1#3%
    [N_T1#1@{logit_cprobs_T1[3,1]}};
    [N_T1#2@{logit_cprobs_T1[3,2]}};
    [N_T1#3@{logit_cprobs_T1[3,3]}};

    %c1#4%
    [N_T1#1@{logit_cprobs_T1[4,1]}};
    [N_T1#2@{logit_cprobs_T1[4,2]}};
    [N_T1#3@{logit_cprobs_T1[4,3]}};

```

```

MODEL c2:

%c2#1%
[N_T2#1@{logit_cprobs_T2[1,1]}};
[N_T2#2@{logit_cprobs_T2[1,2]}};
[N_T2#3@{logit_cprobs_T2[1,3]}};

%c2#2%
[N_T2#1@{logit_cprobs_T2[2,1]}};
[N_T2#2@{logit_cprobs_T2[2,2]}};
[N_T2#3@{logit_cprobs_T2[2,3]}};

%c2#3%
[N_T2#1@{logit_cprobs_T2[3,1]}};
[N_T2#2@{logit_cprobs_T2[3,2]}};
[N_T2#3@{logit_cprobs_T2[3,3]}};

%c2#4%
[N_T2#1@{logit_cprobs_T2[4,1]}};
[N_T2#2@{logit_cprobs_T2[4,2]}};
[N_T2#3@{logit_cprobs_T2[4,3]}}];",

OUTPUT = "tech15;",

usevariables = colnames(savedata),
rdata = savedata)

step3_fit <- mplusModeler(step3,
  dataout=here("lta", "cov_model", "three.dat"),
  modelout=here("lta", "cov_model", "three.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

---

### 20.2.3.1 LTA Transition Plot

This code is adapted from the source code for the `plotLTA` function found in the [MplusAutomation](#) package

**NOTE:** The function found in `plot_transitions_function.R` is specific to a model with 2 time-points and 4-classes & must be updated to accommodate other models.

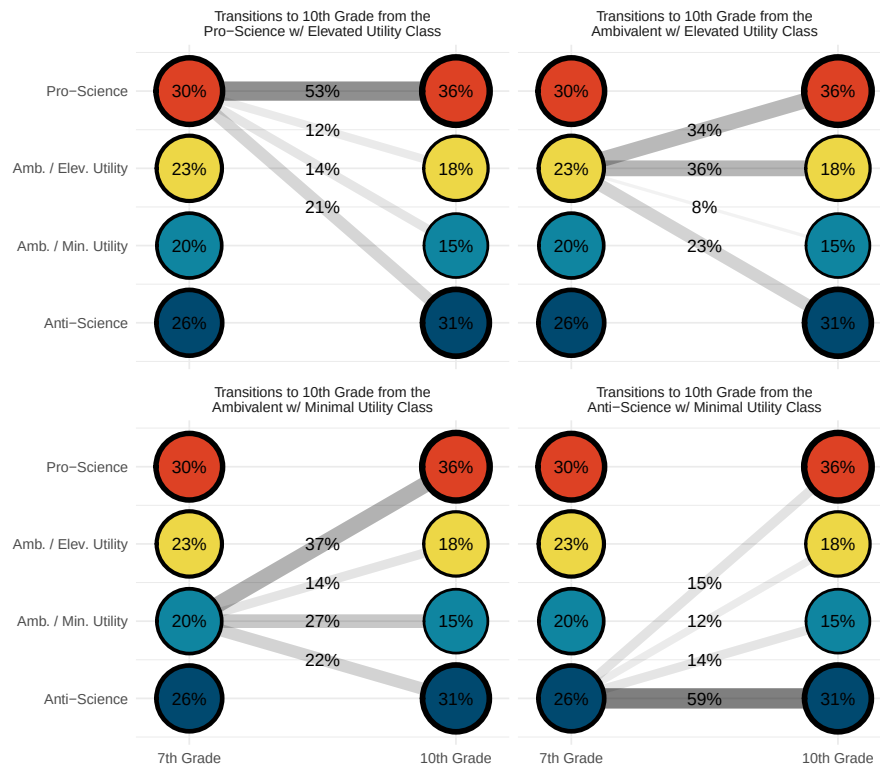
```

source(here("functions","plot_transitions_function.R"))

lta_model <- readModels(here("lta","cov_model","three.out"))

plot_transitions_function(
  model_name = lta_model,
  color_palette = pnw_palette("Bay", n=4, type = "discrete"),
  facet_labels = c(
    `1` = "Transitions to 10th Grade from the Pro-Science w/ Elevated Utility Class",
    `2` = "Transitions to 10th Grade from the Ambivalent w/ Elevated Utility Class",
    `3` = "Transitions to 10th Grade from the Ambivalent w/ Minimal Utility Class",
    `4` = "Transitions to 10th Grade from the Anti-Science w/ Minimal Utility Class"),
  timepoint_labels = c('1' = "7th Grade", '2' = "10th Grade"),
  class_labels = c(
    "Pro-Science",
    "Amb. / Elev. Utility",
    "Amb. / Min. Utility",
    "Anti-Science")
)

```



## Transition Probabilities

| 7th Grade            | 10th Grade  |                      |                     |              |  |
|----------------------|-------------|----------------------|---------------------|--------------|--|
|                      | Pro-Science | Amb. / Elev. Utility | Amb. / Min. Utility | Anti-Science |  |
| Pro-Science          | 0.53        | 0.12                 | 0.14                | 0.21         |  |
| Amb. / Elev. Utility | 0.34        | 0.35                 | 0.08                | 0.23         |  |
| Amb. / Min. Utility  | 0.37        | 0.14                 | 0.27                | 0.22         |  |
| Anti-Science         | 0.15        | 0.12                 | 0.14                | 0.59         |  |

Table:

```
lta_prob <- as.data.frame(lta_model$class_counts$transitionProbs$probability)

t_matrix <- tibble(
  "7th Grade" = c("Pro-Science", "Amb. / Elev. Utility", "Amb. / Min. Utility", "Anti-Science"),
  "Pro-Science" = c(lta_prob[1,1], lta_prob[2,1], lta_prob[3,1], lta_prob[4,1]),
  "Amb. / Elev. Utility" = c(lta_prob[5,1], lta_prob[6,1], lta_prob[7,1], lta_prob[8,1]),
  "Amb. / Min. Utility" = c(lta_prob[9,1], lta_prob[10,1], lta_prob[11,1], lta_prob[12,1]),
  "Anti-Science" = c(lta_prob[13,1], lta_prob[14,1], lta_prob[15,1], lta_prob[16,1]))

t_matrix %>%
  gt(rowname_col = "7th Grade") %>%
  tab_stubhead(label = "7th Grade") %>%
  tab_header(
    title = md("**Transition Probabilities**")) %>%
  fmt_number(2:5, decimals = 2) %>%
  tab_spanner(label = "10th Grade", columns = 2:5) #>%

#gtsave("matrix.docx")
```

## 20.2.3.2 Covariate Table

```
# REFERENCE CLASS 4
cov <- as.data.frame(lta_model[["parameters"]][["unstandardized"]]) %>%
  filter(param %in% c("SCI_IRT7", "FEMALE")) %>%
  mutate(param = case_when(
```

```

        param == "SCI_IRT7" ~ "Science IRT Score",
        param == "FEMALE" ~ "Gender"),
    se = paste0("(", format(round(se,2), nsmall =2), ")") %>%
    separate(paramHeader, into = c("Time", "Class"), sep = "#") %>%
    mutate(Class = case_when(
      Class == "1.ON" ~ "Pro-Science",
      Class == "2.ON" ~ "Amb. / Elev. Utility",
      Class == "3.ON" ~ "Amb. / Min. Utility"),
      Time = case_when(
        Time == "C1" ~ "7th Grade (T1)",
        Time == "C2" ~ "10th Grade (T2)",
      )
    ) %>%
    unite(estimate, est, se, sep = " ") %>%
    select(Time:pval, -est_se) %>%
    mutate(pval = ifelse(pval<0.001, paste0("<.001*"),
      ifelse(pval<0.05, paste0(scales::number(pval, accuracy = .001),
        scales::number(pval, accuracy = .001))))

# Create table

cov_m1 <- cov %>%
  group_by(param, Class) %>%
  gt() %>%
  tab_header(
    title = "Relations Between the Covariates and Latent Class") %>%
  tab_footnote(
    footnote = md(
      "Reference Group: Anti-Science"
    ),
  locations = cells_title()
  ) %>%
  cols_label(
    param = md("Covariate"),
    estimate = md("Estimate (*se*)"),
    pval = md("*p*-value") %>%
  sub_missing(1:3,
    missing_text = "") %>%
  sub_values(values = c(999.000), replacement = "-") %>%
  cols_align(align = "center") %>%
  opt_align_table_header(align = "left") %>%
  gt::tab_options(table.font.names = "serif")

cov_m1

```

### 20.2.3.3 Manually calculate transition probabilities by covariate

```

step3 <- mplusObject(
  TITLE = "LTA (invariant)",

  VARIABLE =
    "usevar = ab39m ab39t ab39u ab39w ab39x ! 7th grade indicators
      ga33a ga33h ga33i ga33k ga33l FEMALE;
    categorical = ab39m-ab39x ga33a-ga33l;
    classes = c1(4) c2(4);" ,

  ANALYSIS =
    "estimator = mlr;
    type = mixture;
    starts = 500 100;
    processors = 10;" ,

  MODEL =
    "%overall%

    c2 c1 on FEMALE;

    c2#1 on c1#1 (b11);
    c2#2 on c1#1 (b21);
    c2#3 on c1#1 (b31);

    c2#1 on c1#2 (b12);
    c2#2 on c1#2 (b22);
    c2#3 on c1#2 (b32);

    c2#1 on c1#3 (b13);
    c2#2 on c1#3 (b23);
    c2#3 on c1#3 (b33);

    [c2#1] (a1);
    [c2#2] (a2);
    [c2#3] (a3);

    c2#1 ON female (b212);
    c2#2 ON female (b222);
    c2#3 ON female (b232);
    c1#1 ON female (b112);

```

```

c1#2 ON female (b122);
c1#3 ON female (b132);

MODEL c1:
%c1#1%
[AB39M$1-AB39X$1] (1-5);  !!! labels that are repeated will constrain parameters
%c1#2%
[AB39M$1-AB39X$1] (6-10);
%c1#3%
[AB39M$1-AB39X$1] (11-15);
%c1#4%
[AB39M$1-AB39X$1] (16-20);

MODEL c2:
%c2#1%
[GA33A$1-GA33L$1] (1-5);
%c2#2%
[GA33A$1-GA33L$1] (6-10);
%c2#3%
[GA33A$1-GA33L$1] (11-15);
%c2#4%
[GA33A$1-GA33L$1] (16-20);",

OUTPUT = "tech1 tech15 svalues;",

MODELCONSTRAINT = " ! Compute joint and marginal probabilities:
  New(
    t11 t12 t13 t14
    t21 t22 t23 t24
    t31 t32 t33 t34
    t41 t42 t43 t44

    t11B t12B t13B t14B
    t21B t22B t23B t24B
    t31B t32B t33B t34B
    t41B t42B t43B t44B

    diff_11_22 x

  );

  t11 = exp(a1 +b11)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0));
  t12 = exp(a2 +b21)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0));
  t13 = exp(a3 +b31)/(exp(a1+b11)+exp(a2+b21)+exp(a3+b31)+exp(0));

```



```

t14 = 1 - (t11+t12+t13);

t21 = exp(a1 +b12)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0));
t22 = exp(a2 +b22)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0));
t23 = exp(a3 +b32)/(exp(a1+b12)+exp(a2+b22)+exp(a3+b32)+exp(0));
t24 = 1 - (t21+t22+t23);

t31 = exp(a1 +b13)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0));
t32 = exp(a2 +b23)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0));
t33 = exp(a3 +b33)/(exp(a1+b13)+exp(a2+b23)+exp(a3+b33)+exp(0));
t34 = 1 - (t31+t32+t33);

t41 = exp(a1)/(exp(a1)+exp(a2)+exp(a3)+exp(0));
t42 = exp(a2)/(exp(a1)+exp(a2)+exp(a3)+exp(0));
t43 = exp(a3)/(exp(a1)+exp(a2)+exp(a3)+exp(0));
t44 = 1 - (t41+t42+t43);

!c1#3 ON female (b132);

x= 1 ; ! x=1 is female, x=0 males

t11B = exp(a1 +b11+b212*x)/(exp(a1+b11+b212*x)+exp(a2+b21+b222*x)+exp(a3+b31+b232*x)
+exp(0));
t12B = exp(a2 +b21+b222*x)/(exp(a1+b11+b212*x)+exp(a2+b21+b222*x)+exp(a3+b31+b232*x)
+exp(0));
t13B = exp(a3 +b31+b232*x)/(exp(a1+b11+b212*x)+exp(a2+b21+b222*x)+exp(a3+b31+b232*x)
+exp(0));
t14B = 1 - (t11B+t12B+t13B);

t21B = exp(a1 +b12+b212*x)/(exp(a1+b12+b212*x)+exp(a2+b22+b222*x)+exp(a3+b32+b232*x)
+exp(0));
t22B = exp(a2 +b22+b222*x)/(exp(a1+b12+b212*x)+exp(a2+b22+b222*x)+exp(a3+b32+b232*x)
+exp(0));
t23B = exp(a3 +b32+b232*x)/(exp(a1+b12+b212*x)+exp(a2+b22+b222*x)+exp(a3+b32+b232*x)
+exp(0));
t24B = 1 - (t21B+t22B+t23B);

t31B = exp(a1 +b13+b212*x)/(exp(a1+b13+b212*x)+exp(a2+b23+b222*x)+exp(a3+b33+b232*x)
+exp(0));
t32B = exp(a2 +b23+b222*x)/(exp(a1+b13+b212*x)+exp(a2+b23+b222*x)+exp(a3+b33+b232*x)
+exp(0));
t33B = exp(a3 +b33+b232*x)/(exp(a1+b13+b212*x)+exp(a2+b23+b222*x)+exp(a3+b33+b232*x)
+exp(0));
t34B = 1 - (t31B+t32B+t33B);

```

```

t41B = exp(a1+b212*x)/(exp(a1+b212*x)+exp(a2+b222*x)+exp(a3+b232*x)+exp(0));
t42B = exp(a2+b222*x)/(exp(a1+b212*x)+exp(a2+b222*x)+exp(a3+b232*x)+exp(0));
t43B = exp(a3+b232*x)/(exp(a1+b212*x)+exp(a2+b222*x)+exp(a3+b232*x)+exp(0));
t44B = 1 - (t41B+t42B+t43B);

      diff_11_22= t11-t11B;",

  usevariables = colnames(savedata),
  rdata = savedata)

step3_fit <- mplusModeler(step3,
  dataout=here("lta", "cov_model", "calc_tran.dat"),
  modelout=here("lta", "cov_model", "calc_tran.inp"),
  check=TRUE, run = TRUE, hashfilename = FALSE)

```

Read invariance model and extract parameters (intercepts and multinomial regression coefficients)

```

lta_inv1 <- readModels(here("lta", "cov_model", "calc_tran.out" ), quiet = TRUE)

par <- as_tibble(lta_inv1[["parameters"]][["unstandardized"]]) %>%
  select(1:3) %>%
  filter(grepl('ON|Means|Intercept', paramHeader)) %>%
  mutate(est = as.numeric(est),
    label = c("b11", "b12", "b13", "b21", "b22", "b23", "b31", "b32", "b33", "b21"))

```

Manual method to calculate transition probabilities by covariate:

```

# Name each parameter individually to make the subsequent calculations more readable
a1 <- unlist(par[19,3]);
a2 <- unlist(par[20,3]);
a3 <- unlist(par[21,3]);

b11 <- unlist(par[1,3]);
b21 <- unlist(par[4,3]);
b31 <- unlist(par[7,3]);

b12 <- unlist(par[2,3]);
b22 <- unlist(par[5,3]);
b32 <- unlist(par[8,3]);

b13 <- unlist(par[3,3]);
b23 <- unlist(par[6,3]);
b33 <- unlist(par[9,3]);

```

```

b212 <- unlist(par[10,3]);
b222 <- unlist(par[11,3]);
b232 <- unlist(par[12,3]);

b112 <- unlist(par[13,3]);
b122 <- unlist(par[14,3]);
b132 <- unlist(par[15,3]);

x <- 0 # x=1 is female, x=0 males

# Calculate transition probabilities from the logit parameters
t11B <- exp(a1 + b11 + b212*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x))
t12B <- exp(a2 + b21 + b222*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x))
t13B <- exp(a3 + b31 + b232*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x))
t14B <- 1 - (t11B + t12B + t13B)

t21B <- exp(a1 + b12 + b212*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x))
t22B <- exp(a2 + b22 + b222*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x))
t23B <- exp(a3 + b32 + b232*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x))
t24B <- 1 - (t21B + t22B + t23B)

t31B <- exp(a1 + b13 + b212*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b33 + b232*x))
t32B <- exp(a2 + b23 + b222*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b33 + b232*x))
t33B <- exp(a3 + b33 + b232*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b33 + b232*x))
t34B <- 1 - (t31B + t32B + t33B)

t41B <- exp(a1 + b212*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x) + exp(0))
t42B <- exp(a2 + b222*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x) + exp(0))
t43B <- exp(a3 + b232*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x) + exp(0))
t44B <- 1 - (t41B + t42B + t43B)

x <- 1 # x=1 is female, x=0 males

# Calculate transition probabilities from the logit parameters
t11 <- exp(a1 + b11 + b212*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x) + exp(0))
t12 <- exp(a2 + b21 + b222*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x) + exp(0))
t13 <- exp(a3 + b31 + b232*x) / (exp(a1 + b11 + b212*x) + exp(a2 + b21 + b222*x) + exp(a3 + b31 + b232*x) + exp(0))
t14 <- 1 - (t11 + t12 + t13)

t21 <- exp(a1 + b12 + b212*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x) + exp(0))
t22 <- exp(a2 + b22 + b222*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x) + exp(0))
t23 <- exp(a3 + b32 + b232*x) / (exp(a1 + b12 + b212*x) + exp(a2 + b22 + b222*x) + exp(a3 + b32 + b232*x) + exp(0))
t24 <- 1 - (t21 + t22 + t23)

t31 <- exp(a1 + b13 + b212*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b33 + b232*x) + exp(0))

```

```

t32 <- exp(a2 + b23 + b222*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b232*x))
t33 <- exp(a3 + b33 + b232*x) / (exp(a1 + b13 + b212*x) + exp(a2 + b23 + b222*x) + exp(a3 + b232*x))
t34 <- 1 - (t31 + t32 + t33)

t41 <- exp(a1 + b212*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x))
t42 <- exp(a2 + b222*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x))
t43 <- exp(a3 + b232*x) / (exp(a1 + b212*x) + exp(a2 + b222*x) + exp(a3 + b232*x))
t44 <- 1 - (t41 + t42 + t43)

```

Create table

## 20.2.4 Create Transition Table

```

t_matrix <- tibble(
  "Time1" = c("C1=Anti-Science", "C1=Amb. w/ Elevated", "C1=Amb. w/ Minimal", "C1=Pro-Science"),
  "C2=Anti-Science" = c(t11, t21, t31, t41),
  "C2=Amb. w/ Elevated" = c(t12, t22, t32, t42),
  "C2=Amb. w/ Minimal" = c(t13, t23, t33, t43),
  "C2=Pro-Science" = c(t14, t24, t34, t44))

t_matrix %>%
  gt(rowname_col = "Time1") %>%
  tab_stubhead(label = "7th grade") %>%
  tab_header(
    title = md("**FEMALES: Student transitions from 7th grade (rows) to 10th grade (columns)"),
    fmt_number(2:5, decimals = 3) %>%
    tab_spanner(label = "10th grade", columns = 2:5) %>%
    tab_footnote(
      footnote = md(
        "*Note.* Transition matrix values are the identical to Table 5, however Table 5
        has the values rearranged by class for interpretation purposes. Classes may be arranged
        directly through Mplus syntax using start values."),
      locations = cells_title())

```

```

t_matrix <- tibble(
  "Time1" = c("C1=Anti-Science", "C1=Amb. w/ Elevated", "C1=Amb. w/ Minimal", "C1=Pro-Science"),
  "C2=Anti-Science" = c(t11B, t21B, t31B, t41B),
  "C2=Amb. w/ Elevated" = c(t12B, t22B, t32B, t42B),
  "C2=Amb. w/ Minimal" = c(t13B, t23B, t33B, t43B),

```

```

"C2=Pro-Science" = c(t14B,t24B,t34B,t44B))

t_matrix %>%
  gt(rowname_col = "Time1") %>%
  tab_stubhead(label = "7th grade") %>%
  tab_header(
    title = md("**MALES: Student transitions from 7th grade (rows) to 10th grade (columns)**")) %>%
  fmt_number(2:5, decimals = 3) %>%
  tab_spanner(label = "10th grade", columns = 2:5) %>%
  tab_footnote(
    footnote = md(
      "*Note.* Transition matrix values are the identical to Table 5, however Table 5
      has the values rearranged by class for interpretation purposes. Classes may be arranged
      directly through Mplus syntax using start values."),
    locations = cells_title())

```

Relations Between the Covariates and Latent Class<sup>1</sup>

| Time                                     | Estimate ( <i>se</i> ) | <i>p</i> -value |
|------------------------------------------|------------------------|-----------------|
| Science IRT Score - Pro-Science          |                        |                 |
| 7th Grade (T1)                           | 0.05 (0.01)            | <.001*          |
| 10th Grade (T2)                          | 0.056 (0.01)           | <.001*          |
| Gender - Pro-Science                     |                        |                 |
| 7th Grade (T1)                           | -1.001 (0.23)          | <.001*          |
| 10th Grade (T2)                          | 0.011 (0.22)           | 0.962           |
| Science IRT Score - Amb. / Elev. Utility |                        |                 |
| 7th Grade (T1)                           | -0.012 (0.01)          | 0.351           |
| 10th Grade (T2)                          | 0.049 (0.02)           | 0.002*          |
| Gender - Amb. / Elev. Utility            |                        |                 |
| 7th Grade (T1)                           | -0.515 (0.28)          | 0.071           |
| 10th Grade (T2)                          | -0.111 (0.33)          | 0.737           |
| Science IRT Score - Amb. / Min. Utility  |                        |                 |
| 7th Grade (T1)                           | 0.021 (0.01)           | 0.093           |
| 10th Grade (T2)                          | 0.022 (0.02)           | 0.202           |
| Gender - Amb. / Min. Utility             |                        |                 |
| 7th Grade (T1)                           | -0.761 (0.26)          | 0.003*          |
| 10th Grade (T2)                          | 0.075 (0.32)           | 0.815           |

<sup>1</sup>Reference Group: Anti-Science

**FEMALES: Student transitions from 7th grade (rows) to 10th grade (columns)<sup>1</sup>**

| 7th grade           | 10th grade      |                     |                    |                |
|---------------------|-----------------|---------------------|--------------------|----------------|
|                     | C2=Anti-Science | C2=Amb. w/ Elevated | C2=Amb. w/ Minimal | C2=Pro-Science |
| C1=Anti-Science     | 0.302           | 0.261               | 0.097              | 0.152          |
| C1=Amb. w/ Elevated | 0.129           | 0.510               | 0.152              | 0.273          |
| C1=Amb. w/ Minimal  | 0.164           | 0.308               | 0.273              | 0.094          |
| C1=Pro-Science      | 0.177           | 0.187               | 0.094              |                |

<sup>1</sup>*Note.* Transition matrix values are the identical to Table 5, however Table 5 has the values rearranged by class for interpretation purposes. Classes may be arranged directly through Mplus syntax using start values.

**MALES: Student transitions from 7th grade (rows) to 10th grade (columns)<sup>1</sup>**

| 7th grade           | 10th grade      |                     |                    |                |
|---------------------|-----------------|---------------------|--------------------|----------------|
|                     | C2=Anti-Science | C2=Amb. w/ Elevated | C2=Amb. w/ Minimal | C2=Pro-Science |
| C1=Anti-Science     | 0.306           | 0.273               | 0.085              | 0.133          |
| C1=Amb. w/ Elevated | 0.131           | 0.532               | 0.133              | 0.245          |
| C1=Amb. w/ Minimal  | 0.170           | 0.329               | 0.245              | 0.083          |
| C1=Pro-Science      | 0.181           | 0.198               | 0.083              |                |

<sup>1</sup>*Note.* Transition matrix values are the identical to Table 5, however Table 5 has the values rearranged by class for interpretation purposes. Classes may be arranged directly through Mplus syntax using start values.





## Part VIII

# Extensions of Latent Class Models



# Chapter 21

## Moderation

---

*Example: Longitudinal Study of American Youth*

**Data source:** : See documentation [here](#)

---

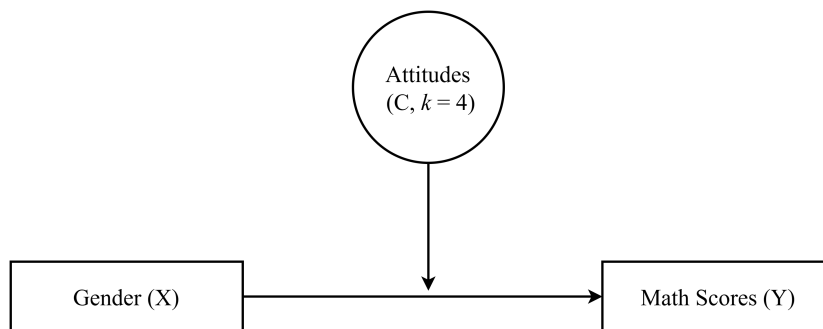
### 21.1 Load Packages

```
library(MplusAutomation)
library(tidyverse)
library(here)
library(glue)
library(gt)
library(cowplot)
library(kableExtra)
library(psych)
library(float)
library(janitor)
```

### LCA Indicators & Auxiliary Variables: Math Attitudes Example

| Name                | Variable Description                             |
|---------------------|--------------------------------------------------|
| enjoy               | I enjoy math.                                    |
| useful              | Math is useful in everyday problems.             |
| logical             | Math helps a person think logically.             |
| job                 | It is important to know math to get a good job.  |
| adult               | I will use math in many ways as an adult.        |
| Auxiliary Variables |                                                  |
| female              | Self-reported student gender (0=Male, 1=Female). |
| math_irt            | Standardized IRT math test score - 12th grade.   |

## 21.2 Moderation Path Diagram



Read in LSAY dataset

```
data <- read_csv(here("data", "lsay_subset.csv")) %>%
  clean_names() %>% # make variable names lowercase
  mutate(female = recode(gender, `1` = 0, `2` = 1)) # relabel values from 1,2 to 0,1
```

### LCA Indicator Proportions

| Variables | Proportion | Count |
|-----------|------------|-------|
| adult     | 0.702      | 1858  |
| enjoy     | 0.669      | 1784  |
| job       | 0.743      | 1947  |
| logical   | 0.640      | 1686  |
| useful    | 0.695      | 1835  |

## 21.3 Descriptive Statistics

### 21.3.1 Descriptive Statistics using R:

Quick view of all the relevant variables:

```
data %>%
  select(enjoy, useful, logical, job, adult, female, math_irt) %>%
  describe()
```

Proportion of indicators using R:

```
# Set up data to find proportions of binary indicators
ds <- data %>%
  pivot_longer(c(enjoy, useful, logical, job, adult), names_to = "Variable")

# Create table of variables and counts
tab <- table(ds$Variable, ds$value)

# Find proportions and round to 3 decimal places
prop <- prop.table(tab, margin = 1) %>%
  round(3)

# Combine everything to one table
dframe <- data.frame(Variables=rownames(tab), Proportion=prop[,2], Count=tab[,2])
#remove row names
row.names(dframe) <- NULL

gt(dframe) %>%
  tab_header(title = md("**LCA Indicator Proportions**"), subtitle = md("&nbsp;")) %>%
  tab_options(column_labels.font.weight = "bold", row_group.font.weight = "bold")
```

### 21.3.2 Descriptive Statistics using MplusAutomation:

```
basic_mplus <- mplusObject(  
  TITLE = "LSAL Descriptive Statistics;",  
  
  VARIABLE =  
    "usevar = enjoy, useful, logical, job, adult, female, math_irt;  
    categorical = enjoy, useful, logical, job, adult, female;",  
  
  ANALYSIS = "TYPE=basic;",  
  
  OUTPUT = "sampstat;",  
  
  usevariables = colnames(data),  
  rdata = data)  
  
basic_mplus_fit <- mplusModeler(basic_mplus,  
                                dataout = here("moderation", "LSAL_data.dat"),  
                                modelout = here("moderation", "basic.inp"),  
                                check = TRUE, run = TRUE, hashfilename = FALSE)
```

View the .out file:

## UNIVARIATE PROPORTIONS AND COUNTS FOR CATEGORICAL VARIABLES

## FEMALE

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.509 | 3026.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.491 | 2919.000 |
|------------|-------|----------|

## MOTHED

|            |       |         |
|------------|-------|---------|
| Category 1 | 0.147 | 854.000 |
|------------|-------|---------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.580 | 3362.000 |
|------------|-------|----------|

|            |       |         |
|------------|-------|---------|
| Category 3 | 0.103 | 597.000 |
|------------|-------|---------|

|            |       |         |
|------------|-------|---------|
| Category 4 | 0.118 | 684.000 |
|------------|-------|---------|

|            |       |         |
|------------|-------|---------|
| Category 5 | 0.052 | 300.000 |
|------------|-------|---------|

## URM

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.777 | 4313.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.223 | 1241.000 |
|------------|-------|----------|

## KA47A

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.466 | 1564.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.534 | 1793.000 |
|------------|-------|----------|

## KA47H

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.550 | 1837.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.450 | 1502.000 |
|------------|-------|----------|

## KA47I

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.452 | 1507.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.548 | 1825.000 |
|------------|-------|----------|

## KA47K

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.659 | 2200.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.341 | 1139.000 |
|------------|-------|----------|

## KA47L

|            |       |          |
|------------|-------|----------|
| Category 1 | 0.597 | 2003.000 |
|------------|-------|----------|

|            |       |          |
|------------|-------|----------|
| Category 2 | 0.403 | 1352.000 |
|------------|-------|----------|

Or, view of descriptive statistics using `get_sampstat()`:

```
# Using MplusAutomation
MplusAutomation::get_sampstat(basic_mplus_fit)

# Using base R
summary(data)
```

## 21.4 Enumeration

This code uses the `mplusObject` function in the `MplusAutomation` package and saves all model runs in the `mplus_enum` folder.

```
lca_enum_6 <- lapply(1:6, function(k) {
  lca_enum <- mplusObject(

    TITLE = glue("{k}-Class"),

    VARIABLE = glue(
      "categorical = enjoy, useful, logical, job, adult;
      usevar = enjoy, useful, logical, job, adult;
      classes = c({k});"),

    ANALYSIS =
      "estimator = mlr;
      type = mixture;
      processors = 12;
      starts = 500 100;",

    OUTPUT = "sampstat residual tech11 tech14;",

    usevariables = colnames(data),
    rdata = data)

  lca_enum_fit <- mplusModeler(lca_enum,
                              dataout=glue(here("moderation","enum", "LSAY_data.dat")),
                              modelout=glue(here("moderation","enum", "c{k}_lsal.inp"),
                              check=TRUE, run = TRUE, hashfilename = FALSE)
})
```

**IMPORTANT:** Before moving forward, make sure to examine each output